

RKDF University
Program Outcomes, Program Specific Outcomes and
Course Outcomes

Faculty of Engineering & Technology

Bachelors of Engineering (Civil Engineering)

Programme Outcome (POs)

- PO1** The graduates will have an ability to apply the basic knowledge of mathematics, science and engineering to real-life problems.
- PO2** The graduates will be able to understand problem identification, formulations, and to conduct experiments, by analysis and interpreting the data.
- PO3** Apply the knowledge of engineering specialization to the solution of complex engineering problems
- PO4** Design/development of solutions: design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO5** Create, select, and apply appropriate techniques, resources, and modern engineering and it tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO6** Environment and sustainability: understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
- PO7** Project management and finance: demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments

PROGRAM SPECIFIC OUTCOMES (PSOs)

- PSO1** Graduates will have an ability to describe, analyze and solve problems using mathematics and systematic problem solving technique
- PSO2** Graduates will have an ability to plan, execute, manage, maintain and rehabilitate civil engineering systems and processes.
- PSO3** Familiarize civil engineering components and systems.
- PSO4** Design and conduct experiments, analyze and interpret data

- PSO5** Solve problems in the structural, construction management, hydraulics, geotechnical, transportation and environmental disciplines of Civil Engineering
- PSO6** Function effectively in multi-disciplinary teams
- PSO7** Capably design and build civil engineering-based systems in the context of environmental, economical, and societal requirements and serve the community as ethical and responsible professionals.

Courses outcome:-

Semester - I

BE-1011 Engineering Chemistry

- CO1:** Gain knowledge about types of boiler problems, various physical and chemical techniques for water treatment and its analysis, desalination process used to produce potable water from brackish water
- CO2:** Differentiate between air and water pollution. Possess the knowledge about their adverse effect on the environment and their preventive measures.
- CO3:** Gain chemical knowledge on concepts of polymers, their structural properties and molding techniques required for solving interdisciplinary problems in polymer industries.
- CO4:** Gain basic knowledge about biomolecules, nanomaterial's, fullerenes, super conductors, and brass alloy, and also able to apply them in multi-disciplinary engineering branches.
- CO5:** Acquire knowledge on dyes and drugs, methods of dyeing, color theory, synthesis of antimalarial and antibiotic drugs.
- CO6:** Perform the experiments on pH-metry, Potentiometry, Conductometry, Colorimetry and chromatography as well as to analyze and interpret the data to address issues related to engineering problems
- CO7:** Acquire the knowledge of various types of Corrosion, their significance and preventive measures
- CO8:** Acquire the basics of non-conventional sources of energy and green chemistry.

BE-1021 Engineering Mathematics-I

- CO1:** Develops skill of higher derivative, expansion of functions in ascending power of variable & value of the function in neighborhood of some points.
- CO2:** Able to determine limits of indeterminate function. Applicable to already word problems & Engineering Problems.
- CO3:** Gain the knowledge to solve differential equation arising in different Engineering branch and able to form mathematical & physical interpretation of its solution which place important role in all branches of Engineering
- CO4:** Learn the evaluation policy of some special function like gamma & Beta function. & their relation which is helpful to evaluate some definite integral arising in various branch of Engineering
- CO5:** Able to calculate rank of matrix, characteristic equation & characteristic roots & use the applicability of Cayley Hamilton Theorem to find inverse of matrix which is very important in many engineering application

- CO6:** Develops the ability to trace the curve for a given equation of a curve & its nature
- CO7:** Gain knowledge to find radius of curvature & torsion of given curve which is helpful in civil Engineering , Mechanical Engineering & Rods and Building Construction & it is also useful in Research & development.

BE-1031 English

- CO1:** Ability to prepare and make small presentations.
- CO2:** Ability to write effective business letters, emails, CV and reports.
- CO3:** Comprehend answering strategies in group discussions and interviews.
- CO4:** Ability to voice opinion in discussions and convey ideas.
- CO5:** Comprehend different types of communication and importance of effective communication in a work place.

BE-1041 Basic Electrical & Electronics Engineering

- CO1:** Enhancement in understanding the basic concepts of Core Electrical Engineering.
- CO2:** Basic understanding of Electrical machines and power systems.
- CO3:** Analyze components associated with digital and analog electronic systems.
- CO4:** Demonstrate proficiency in the use of electronic equipment and devices.
- CO5:** Assist in the design, operation, and troubleshooting of electronic systems.
- CO6:** Analyzing electronics devices and circuits using computer simulations.
- CO7:** Solve electronic devices and systems using mathematical concepts.

BE-1051 Engineering Graphics & Design

- CO1:** To read, understand and apply the knowledge of orthographic projections (production related features and instructions) in manufacturing industry, process industry and other allied engineering application.
- CO2:** To communicate with the globally recognized engineers and the engineers of different discipline of engineering for research and development activities.
- CO3:** To get knowledge of projections of different solid objects.
- CO4:** To perceive the idea of sectional view and advantages of it
- CO5:** To apply the concept of intersections of solids for various engineering applications
- CO6:** To understand and apply the concept of surface development for fabricating and manufacturing industrial devices
- CO7:** To create the image of three dimensional figures with the help of isometric projections.

BE-1061 Workshop/ Manufacturing Practices

- CO1:** Acquire knowledge of the safety measures which are followed in workshop while using hand tools and general purpose machine tools.
- CO2:** Develop creativity, craftsmanship, approach to work and planning capabilities within students.
- CO3:** Given a drawing of a product/part such as carpentry job, fitting job, sheet metal job, assembly of system and pipe fitting, apply the various hand tools and general purpose machine tool to make or assemble the product/part
- CO4:** Select and use various measuring and gauging instrument which are required for different types of jobs.

BE-2011 Engineering Physics

- CO1:** Gain basic understanding of the combined effect of electric and magnetic fields their application for designing various electromagnetic and semiconductor devices.
- CO2:** Acquire fundamentals of Optics, especially wave nature of light (e.g., interference etc.) and its applications towards telescopes, microscopes, astronomy and fibre optics.
- CO3:** Develop basic knowledge on the historical development and time-to-time applications of quantum mechanics in electronic devices (e.g., Photovoltaic cell, Hall sensor etc.).
- CO4:** Obtain basic understanding of the particle nature of light (e.g., Photoelectric effect, Compton scattering etc.) and their applications.
- CO5:** Gain basic knowledge on the properties, production and applications of X-rays.
- CO6:** Understand the fundamentals of atomic structure and related theory & experiments.
- CO7:** Attain basic knowledge on different types of LASERs and their applications.
- CO8:** Develop an ability to conduct experiments, as well as to analyse and interpret data related to the Electromagnetism, Optics, Modern and Laser Physics.

BE-2021 Engineering Mathematics-II

- CO1:** Gain the knowledge to develop the concepts of surface $Z=f(x,y)$, its partial derivatives, Euler Theorem & modified Euler Theorem for homogenous function & deduce ability to solve problems related to partial derivatives.
- CO2:** Learn to expand any functions of two variables in the ascending power of variables and also develop error and approximation, extremum value of a given function related to engineering application.
- CO3:** Develop the ability to solve higher order & first degree linear non homogenous differential equation arising in various branch of engineering and related mathematical model develop arising to form mathematical modeling of Real World Problem with its physical interpretation.
- CO4:** Solve some differential equation which is not solvable in ordinary case but its series solution gives an idea of developing special function which has important role in some physical phenomena arising in engineering problems
- CO5:** Develop the concepts of Laplace transformation & inverse Laplace Transform with its property to solve partial Differential equation and Ordinary Differential Equation with given boundary conditions which is helpful in all engineering & research work.

BE-2031 Basic Mechanical Engineering

- CO1:** Gain knowledge about different conventional and non-conventional energy sources
- CO2:** Differentiate between different types of fuels and their calorific values and able to calculate the minimum mass (or volume) of air required for complete combustion of fuels
- CO3:** Gain knowledge about various types of boilers, the mountings and accessories and able to calculate the boiler efficiency and to design the chimney dimensions
- CO4:** Perform thermodynamic analysis of Otto, Diesel and Dual cycle models
- CO5:** Differentiate between the types and working of internal combustion engines: 2-stroke/4-stroke engines & SI and CI engines
- CO6:** Acquire the knowledge of the operation, construction and design of various components of thermal, hydro- and nuclear power plants
- CO7:** Operate the machine tools like lathe, shaper and drilling machine
- CO8:** Possess the knowledge about the principles of operation of various refrigeration and air conditioning systems for domestic as well as industrial purpose.

BE-2041 Basic Civil Engineering & Engineering Mechanics

- CO1:** A clear appreciation and understanding of the scope of environmental engineering and the types of problems and issues that are involved
- CO2:** An understanding of the interdisciplinary nature of problems associated with environmental engineering and the environment, and the broad range of skills and expertise that are required
- CO3:** The global climate system and human interactions of major biogeochemical cycles sufficiently to critically evaluate forecasts for global change
- CO4:** To describe and apply the fundamentals of air and water pollution to solve basic environmental engineering problems
- CO5:** The objectives of water and wastewater treatment and to the most important regulations for sustainable development
- CO6:** Solve for the resultants & moments of any force systems and determine equivalent force systems
- CO7:** Determine the internal forces in plane trusses and beams
- CO8:** Solve the mechanics problems associated with friction forces
- CO9:** Obtain the centroid, first moment and second moment of an area
- CO10:** Describe the motion of a particle in terms of its position, velocity and acceleration in different frames of reference and analyze the forces causing the motion of a particle
- CO11:** Apply work, energy, impulse and momentum relationships for a particle in motion
- CO12:** Understand free & forced vibration, single degree of freedom, concept of earthquake induced waves and its affect.

BE-2051 Programming for Problem Solving

- CO1:** Demonstrate a basic understanding of computer hardware and software
- CO2:** Develop proficiency in writing small to medium sized programs in a procedural programming language.
- CO3:** Apply problem-solving skills and knowledge of computing fundamentals to a wide variety of engineering, science and technology problems
- CO4:** Expose, diagnose, and fix errors in a program, using systematic testing and debugging techniques
- CO5:** Have developed interest in the field of computers to be able to adjust to the demands of current trends and technology

BE-2061 Language Laboratory

- CO1:** The student will acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

Semester – III**Engineering Mathematics-III-BE– 3011**

- CO1** Analyse the behavior of infinite series.
- CO2** Solve problems based on linear algebra.
- CO3** Acquire knowledge of Complex Functions.
- CO4** Understand methods to solve first order and higher order differential equations.
- CO5** Apply the concepts of differential equations to solve practical problems.

Transportation - Bridges and Tunnels-CE- 3021

- CO1** Know about railway track components, their materials, size, function and importance

- CO2 Carry out geometric design of railway track
- CO3 Know about various components in diverging, merging and crossings of railway tracks, stations, yards, signaling, interlocking and control systems.
- CO4 Know about requirements of railway track for high speed trains, safety aspects and maintenance.
- CO5 Understand about different types of bridges, their components, loads/stresses acting on bridges, requirement and function of the components, hydrological design, methods of erection, maintenance of bridges.
- CO6 Understand about importance, types, methods of construction, mucking, ventilation, lining and lighting in Tunnels.

Strength of Materials-CE- 3031

- CO1 Apply the linear laws of elasticity as related to stress and strain.
- CO2 Understand the concept of a complex stress system.
- CO3 Understand of the behavior of columns and struts under axial loading.
- CO4 Analyses the bending stress on different types of sections.
- CO5 Determine the effect of combined axial and bending stress.
- CO6 Demonstrate the use of critical thinking and problem solving technique as applied to mechanical and structural systems.

Building Design & Drawing -CE- 3041

- CO1- Visualize science in the form of technical graphics.
- CO2- Read and draft orthographic projections of points, lines, planes and solids as per the BIS codes.
- CO3- Understand and visualize section of solid, intersection and development of surfaces.
- CO4- Draw and read isometric and orthographic projections of geometric objects
- CO5- Imparting the knowledge of vastu for Home and other Commercial & Industrial Vastu applications

VALUE EDUCATION- CE- 3051

- CO1- Students will understand the importance of value based living.
- CO2- Students will gain deeper understanding about the purpose of their life.
- CO3- Students will understand and start applying the essential steps to become good leaders
- CO4- Students will emerge as responsible citizens with clear conviction to practice values and ethics in life

Computer Programming Lab-CE- 3061

- CO1- Get knowledge of Computer System, Window explorer, Control panel and Command prompt
- CO2- Work on Microsoft Word, Excel, PowerPoint and Mail Merging.
- CO3- Use concepts of C++ JAVA, JAVA virtual machine types, basic terminologies, operators, expressions, control structures.
- CO4- Implement programs using functions, arrays and strings.
- CO5- Understand the concept of OOP's, use of classes and objects and basics of file handling.

Semester – IV

Environmental Engineering-BE – 4011

- CO1- Convey a clear idea of the interdisciplinary nature of environmental and health risk

assessment.

- CO2- The ability to apply the fundamental knowledge of science and engineering to assess environmental and health risk.
- CO3- Ability to understand environmental laws and regulations to develop guidelines, procedures and processes for health and safety issues.
- CO4- Understand what are professional ethics and how do ethics affect the outcomes of environmental laws and regulations.

Concrete Technology- CE- 4021

- CO1- Identify Quality Control tests on concrete making materials
- CO2- Understand the behavior of fresh and hardened concrete
- CO3- Design concrete mixes as per IS and ACI codes
- CO4- Understand the durability requirements of concrete
- CO5- Understand the need for special concretes

Surveying & Geomatics- CE- 4031

- CO1- Surveying courses give basis knowledge of field survey work which includes linear and angular measurement
- CO2- It offers to under the basic requirement s work specific survey, which involves many parameters such as type of equipment, Precision, Time available, and Money etc.for a particular engineering application.
- CO3- It gives hands on practices to use various surveying equipment like chain Compass, Theodolite, Total station, Dump Levels etc., for linear and angular, Various type of leveling operation and Contouring
- CO4- Collection field data using various survey equipment leads them to prepare maps of suitable scale
- CO5- Apply the concept of Tacheometry in surveying difficult and hilly terrains to obtain the topographical map of area.
- CO6- Application and setting of the curves, control surveying and hydrographic surveying
- CO7- Application of advances surveying known as geomatics in field

Engineering Materials & Construction Techniques- CE- 4041

- CO1- Extend the knowledge about the characteristics, sources and defects in various materials.
- CO2- Design and test the materials either in the laboratory or in the field before their actual use at the site.
- CO3- Attain the knowledge of different components of building, their classification, materials and methods of construction and causes of their failures.
- CO4- Know the various services to be provided and the defects in the buildings along with the remedial measures for proper maintenance of the buildings.

Fluid Mechanics-1- CE- 4051

- CO1- Knowledge of the basic concepts and principles of fluid mechanics.
- CO2- Ability to analyze fluid flow problems with the application of momentum and energy equations.
- CO3- Ability to distinguish between various types of fluid flow.

Software Lab (AutoCAD)- CE- 4061

- CO1- Ability to create fully constrained solid models that can be quickly modified using standard software tools.
- CO2- Ability to use, identify and explain standard features in solid modeling including protrusions, revolutions, cutouts, and patterns
- CO3- Ability to use standard software tools to create engineering drawings, or other documents, to fully describe the geometries and dimensions of parts, as well as to document assemblies according to standard practice
- CO4- Ability to use standard software tools to create part assemblies and check for clearances.

Semester – V**Transportation Engineering-I- CE.- 5011**

- CO1- Understand concept of Geometric design of roads and various aspects of traffic engineering.
- CO2- Understand various types of bridges and their design aspects.
- CO3- Understand various traffic characteristics and analysis and use the data for road design.
- CO4- Understand various Highway materials and their suitability under different conditions.
- CO5- Understand various aspect of airport planning runway and taxiways. It also covers requirement airport obstruction, lightning and traffic control in airport.

Structure Design Drawing-I(RCC)- CE.- 5021

- CO1- To recognize the design philosophies of reinforced concrete structures
- CO2- To apply the principles, procedures and current code requirements to the analysis and design of reinforced concrete beams.
- CO3- To identify the behavior of reinforced concrete members in bond, anchorage, shear and torsion.
- CO4- To identify and design of slab
- CO5- To analyze and design reinforced concrete compression members and the load on the structure such as footings and staircases.

Water Resource and Irrigation Engineering -CE.- 5031(A)

- CO1- Demonstrated understanding of physical processes in the context of flood hydrology, including the hydrological cycle in general, and rainfall, loss and groundwater transport mechanisms in particular. It gives understanding of the methods that can be used to measure rainfall and flow, as well as their relative advantages and disadvantages. It gives ability to obtain design rainfall intensities and hyetographs which helps in select appropriate methods of determining design flows and hydrographs in urban and rural areas.
- CO2- Demonstrated ability to solve engineering design problems in the context of flood hydrology and hydrogeology
- CO3- Demonstrated the requirement of water resources planning and management for the various project
- CO4- Creating Plan an Irrigation System which also included analyzing irrigation canals and canal network for design. Creating Plan and design of diversion head works for analyzing design irrigation canal structures. Also included design water wells

Marine Construction- CE.- 5031(B)

- CO1- To understand various national level policies for building and development of marine construction
- CO2- To understand basic requirement of harbor planning and natural phenomena occurred under marine structure
- CO3- To understand the basic requirement for design aspect of marine structure which also include Docks and Locks.

Urban & Town Planning- CE.- 5031(C)

- CO1- To understand the definition, classification and other various aspects of urban planning.
- CO2- To understand the basing problem encountered in urban plan implementation.
- CO3- To understand the basic Act and bye-laws of town and country planning.
- CO4- To understand the basic problem of traffic transportation system in urban areas
- CO5- To understand the various plan required and implication of these plan is required in urban planning and development.

Fluid Mechanics-II- CE.- 5041(A)

- CO1- Students will be able to apply their knowledge of fluid mechanics in addressing problems in open channels.
- CO2- They will possess the skills to solve problems in uniform, gradually and rapidly varied flows in steady conditions.
- CO3- Problems pertain to design, construction as well as efficient working of various types of hydraulics structures and machines is considerably simplified by using dimensional analysis and model studies.
- CO4- Impact of Jet on vanes which is a base for analysis and design of turbo machines.
- CO5- They will have knowledge in hydraulic machines(pumps and turbines)

Remote Sensing & GIS -CE.- 5041(B)

- CO1- Students should be able to learn the basic principles of remote sensing. Describe the process of data acquisition of satellite images and their characteristics.
- CO2- To Retrieve the information content of remotely sensed data and Analyses the energy interactions in the atmosphere and earth surface features. Interpret the images for preparation of thematic maps
- CO3- Students able to explain the concepts and fundamentals of GIS
- CO4- Gain knowledge in modern image interpretation and recent analysis techniques.
- CO5- Compute knowledge of remote sensing and GIS in different civil engineering applications.

Renewable Energy Sources- CE.- 5041(C)

- CO1- To understand the growing requirement of renewable energy system with the growing requirement of population.
- CO2- To understand the design requirement of wind energy system
- CO3- To understand the requirement of solar energy system. How this can be effectively and efficiently. It gives insight of biomass energy system for energy generation and its use.
- CO4 - To understand how energy from oceans can be used form energy generation.
- CO5- To understand to conservation, audit and measurement of electric energy for efficient use of it.

Material Testing Lab- CE.- 5051

- CO1. To determine the normal consistency of cement.
- CO2. To determine the initial and final setting time of cement .
- CO3. To determine compressive strength of cement.
- CO4. To determine the soundness of cement.
- CO5. To determine the fineness modulus of fine aggregate & coarse aggregate.
- CO6. Mix design of concrete by IS code Method.
- CO7. Slump test for determining workability of concrete.
- CO8. Compressing strength of concrete cube.
- CO9. To determine the flexure strength of concrete.

Non-Destructive Testing -CE.- 5061

- CO1 Determine Compressive Strength of concrete by using Manual Rebound Schmidt Hammer.
- CO2 Determine Compressive Strength of concrete by using DIGISCHMIDT 2000 Concrete test hammer.
- CO3 Determine the Pulse Velocity, Crack Depth and Compressive Strength of concrete by using PUNDIT – lab ultrasonic instrument.
- CO4 Determine the concrete ability to resist chloride ion penetration.
- CO5 Determine the Dampness in Building Material by using James Instrument Aquameter.
- CO6 Determine Location of Rebars, measurement of concrete cover and bar diameter by using Rebar detector.
- CO7 Determine the Humidity of Concrete and other structure by using Vaisala Structural Humidity Measurement Kit SHM40.
- CO8 Determine the Rate of Corrosion in concrete structure by using CORMAP II.
- CO9 Determine Resistivity of Concrete by using Resipod.

Semester – VI

Environmental Engineering-II -CE.- 6011

- CO1- Perform common environmental experiments relating to water and wastewater quality, and know which tests are appropriate for given environmental problems.
- CO2- Statistically analyze and interpret laboratorial results.
- CO3- Apply the laboratorial results to problem identification, quantification, and basic environmental design and technical solutions.
- CO4- Understand and use the water and wastewater sampling procedures and sample preservations.
- CO5- Obtain the necessary background for subsequent courses in environmental engineering.
- CO6- Demonstrate the ability to write clear technical laboratorial reports.

Geo Tech Engineering- CE.- 6021

- CO1- To understand the origin of soil and to identify different types of soil.
- CO2- To understand the various physical and engineering characteristics of different types of soil.
- CO3- To understand the concept of slope stability.
- CO4- To appreciate the use of modern technology in the field of geotechnical engineering

Quantity Surveying & Costing CE.- 6031(A)

- CO1- To know the importance of preparing the types of estimates under different conditions.

- CO2- To apply logical thoughts and prepare the rate analysis and bills.
- CO3- To comprehend detailed report on estimation and valuation process
- CO4- To analyze and synthesize cost effective approach for civil engineering projects.
- CO5- To evaluate the cost of expenditure and prepare a detailed rate analysis report

Entrepreneurship Development and Management- CE.- 6031(B)

- CO1- Understanding basic concepts in the area of entrepreneurship,
- CO2- Understanding the role and importance of entrepreneurship for economic development, developing personal creativity and entrepreneurial initiative, adopting of the key steps in the elaboration of business idea.
- CO3- Understanding the stages of the entrepreneurial process and the resources needed for the successful development of entrepreneurial ventures.

Cost-Effective & Eco-Friendly Construction- CE.- 6031(C)

- CO1- It gives the proper understand of utilization of concepts of energy efficient and environment friendly materials and techniques.
- CO2- There are various applications of cost effective techniques in construction such as building, sanitation and low cost road.
- CO3- Cost analysis and comparison with conversional construction material and techniques is made.

Environmental Impact Assessment (EIA) -CE.- 6041(A)

- CO1- The student will develop of major problems in Environmental impact and control, regulations.
- CO2- The students will be able to familiar with regulations pertinent to environmental problems.
- CO3- The students will be able to describe general environmental impact problems, meteorological definitions.
- CO4- The students will be able to present the results as a report in the record notebook.
- CO5- The students will develop the ability to learn from the mistakes ethically and increase the quality of design.

Structural Design & Drawing-II (R.C.C)- CE.- 6031(B)

- CO1- To understand basic requirement for design of multistory building
- CO2- To understand the design requirement of Earth retaining structures
- CO3- To understand the design requirement of overhead water tanks
- CO4- To understand the design requirement of storage such as silos and bunkers.
- CO5- To understand the design required T-beams and slab bridges. It also includes Prestressing concept and other requirement.

Operational Research -CE.- 6041(C)

- CO1- Operational research given knowledge of various tool and techniques for optimization of various aspect of work.
- CO2- Linear models help to identify the co-relationship between various parameters.
- CO3- Transportation model helps to identify optimization of cost of distribution and network model (CPM & PERT) helps to optimization of route.
- CO4- An inventory model helps to maintain the inventory for required project.
- CO5- Queueing models helps to analysis and check on entire process carried.
- CO6- Decision models help in making decision by various input given at various stages of

work.

Quantity Surveying and Cost Lab-CE.- 6051

- CO1 Preparation of detailed estimate.
- CO2. Detailed estimate for services of plumbing and water supply or Electrification work.
- CO3. Detailed estimate for earth work for the road construction or arched culvert.
- CO4. Rate analysis for at least 8 items of construction.
- CO5. Preparation of DPR of Civil Engineering Project.

Prestressed Concrete Structure Lab- CE.- 6061

- CO1. Fabrication, casting and testing of simply supported prestressed concrete beam/slab (pres-tensioned or post-tensioned) for strength and deflection behaviour.
- CO2. Fabrication, casting and testing of beam/slab (pres-tensioned or post-tensioned) with different layout of cables for strength and deflection behaviour
- CO3. Fabrication, casting and testing of various prestressed structures as per contents given in the syllabus
- CO4. Minimum 15 problems from above topics along with cross checking using any opensource / professional software.
- CO5. Modeling and analysis of at least one real-life structure using open-source/ professional software

Semester – VII

Advanced Structural Design(Steel)- CE- 7011

- CO1 To introduce steel structures and its basic components and structural steel fasteners like welding and bolting
- CO2 To design tension members, compression members, beams and beam-columns
- CO3 To design column splices and bases.
- CO4 Able to understand properties of steel under loading conditions.
- CO5 Able to determine the ultimate bending moment capacity of steel members considering both yielding and lateral buckling.
- CO6 Ability to analyze railway bridge, footbridge, water tanks and Chimneys.

Computation Methods in Structure Engineering -CE- 7021(A)

- CO1- Understood different computational methods and their applicability for the solution of engineering problems.
- CO2- Learned different differentiation and integration techniques used in numerical methods.
- CO3- Learned different interpolation techniques and their application.
- CO4- Capable in selection and use of computational techniques for modelling of various engineering problems.

Design of Hydraulic Structure- CE- 7021(B)

- CO1- Knowledge about various type of hydraulic structures and their function
- CO2- Knowledge about Design principles of gravity and earth dams
- CO3- Knowledge about Components of diversion head work and their How do the energy balance is associated with a hydraulic jump
- CO4- Energy balance. What does critical flow and the specific energy concept mean

- CO5- How the flow pattern change under different influences
- CO6- How is the flow downstream a spillway and through an opening
- CO7- How the river develops various water lines
- CO8- Which pipeline is best for the given hydraulic condition.

Semester – VII

Internet of Things CE- 7031(A)

- CO1- Understand in depth about Internet of things.
- CO2- Establish secure communication for his network for his devices connected in IOT.
- CO3- Store his data securely on cloud and access it when required
- CO4- Design web based application using various internet protocols and services
- CO5- Use sensor technology and RFID and wireless networking for maintaining privacy and security concern in smart city and housing environmental considerations.

Construction Planning and Management- CE- 7031(B)

- CO1- Student will demonstrate basic knowledge of L.D.E., Vector, P.D.E., F.T. & Probability.
- CO2- Student will show the understanding of impact of Engineering Mathematics in Civil.
- CO3- Student will demonstrate their understanding of mathematical ideas from multiple perspectives, such as by
- CO4- Using the internal connections between geometry, algebra, and numerical computation,
- CO5- Applying the connections between theory and applications, or
- CO6- Distinguishing between a formal proof and a less formal arguments and understanding the different roles these play in mathematics.

Integrated Waste Management CE- 7031(C)

- CO1 Review the components of solid waste management system as per need of particular locality, town or city.
- CO2 Be aware of the significance of recycling, reuse and reduction and recovery of solid wastes.
- CO3 Develop an insight into the collection, transfer, and transport of municipal solid waste.
- CO4 Understand the importance and operation of a resource recovery facilities like waste-to-energy Technologies-Biochemical and thermochemical.
- CO5 Understand the design and operation of a municipal solid waste composting and landfilling.

Advances Surveying Lab -CE- 7041

- CO1 Study and Install IDE of Arduino and different types of Arduino.
- CO2 Write program using Arduino IDE for Blink LED.
- CO3 Write Program for RGB LED using Arduino.
- CO4 Study the Temperature sensor and Write Program for monitor temperature using Arduino.
- CO5 Study and Implement RFID, NFC using Arduino.
- CO6 Study and Configure Raspberry Pi.
- CO7 WAP for LED blink using Raspberry Pi.
- CO8 Study and Implement Zigbee Protocol using Arduino / Raspberry Pi.
- CO9. Study and implement MQTT protocol using Arduino.
- CO10. Study and implement CoAP protocol using Arduino.

Semester – VII

Theory of Structure CE- 8011

- CO1 To understand the concept and analyses of determinate and indeterminate structures.
- CO2 To understand the principle of virtual work and the application of influence line diagrams in structural analysis problems.
- CO3 The course runs through a number of techniques which are used for the analysis of civil engineering structures.

Traffic Engineering- CE- 8021(A)

- CO1 to give a deep knowledge of growing requirements of city in terms of Traffic Characteristics,
- CO2 to give a deep knowledge of growing requirements of city in terms of Traffic studies,
- CO3 to give a deep knowledge of growing requirements of city in terms of Traffic Operations and Control, Street light
- CO4 to give a deep knowledge of growing requirements of city in terms of Accident Studies & Mass Transportation

Foundation Engineering -CE- 8021(B)

- CO1 To determine the selection of foundation and sub-soil exploration on various types of soil
- CO2- To determine various methods of analysis the bearing capacity of shallow foundation.
- CO3- To determine the selection and mechanics of pile foundation
- CO-4 To determine the important of foundations on problematic soil & Introduction to Geosynthetics
- CO5- To determine the lateral earth pressure on the retaining wall

Bridge Engineering- CE- 8021(C)

- CO1 To design various types of bridge super structures such as R.C and steel bridge.
- CO2 It also involves design of Pier
- CO3 To design various types of Abutment
- CO4 To design various types of Wing Walls
- CO5 To design various types of Foundation and Bearings.

Earthquake Resistant Design of Structures- CE- 8021(D)

- CO1- To under the behavior of Earthquake and its characteristics
- CO2- To determine the response spectrum in multi-storeyed building
- CO3- To determine the earthquake intensity and effect by earth quake modelling

Pavement Design -CE- 8031(A)

- CO1- Identify the pavement components and design bituminous mixes,
- CO2- Analyze and design flexible and rigid pavements
- CO3- Evaluate structural condition of pavement.

Air quality Monitoring and Control -CE- 8031(B)

- CO1- The course objectives has three components i.e., sources of air pollution, pathways (air pollutants transformation and transport) and receptors.

- CO2- Students would get an insight into the dispersion of air pollution in the atmosphere
- CO3- This life cycle of air pollution will enable the student to first identify the pollutants and their sources and then the transport mechanisms of the pollutants followed by the affected population and their control mechanisms.

Finite Element Method -CE- 8031(C)

- CO-1 To learn basic principles of finite element analysis procedure.
- CO2- To learn the theory and characteristics of finite elements that represent engineering structures.
- CO3- To learn and apply finite element solutions to structural, thermal, dynamic problem to develop the knowledge and skills needed to effectively evaluate finite element analyses.

Artificial Intelligence- CE- 8031(D)

- CO1- Learn about importance of AI techniques. Adoption of Artificial Intelligence (AI) technologies is widely expanding in our society. Applications of AI include: self-driving cars, personal assistants, surveillance systems, robotic manufacturing, machine translation, financial services, cyber security, web search, video games, and code analysis and product recommendations.
- CO2 Know the exact application of AI Techniques. Such applications use AI techniques to interpret information from a wide variety of sources and use it to enable intelligent, goal-directed behavior.
- CO3 understand the working of Modern AI based systems. It often involves self-learning systems that are trained on massive amounts of data, and/or interacting intelligent agents that perform distributed reasoning and computation.
- CO4 Know about sensors used in AI based systems. AI connects sensors with algorithms and human interfaces, and extends itself into large networks of smart devices.
- CO5 Know the opportunities after having knowledge of AI techniques. The knowledge of Artificial Intelligence opens career opportunities in companies that are building the next generation of intelligence and language understanding for their products: for example intelligent personal assistants opinion mining systems, customer support system, biomedical applications, computer games, smart adaptive devices, robots, smart planning systems.

Earthquake Resistant Lab- CE- 8041

- CO1 Lump mass with equal height:- To understand effect of frequency and present travel on structure having equal height
- CO2 Lump mass with variable height:- To understand effect of frequency and present travel on structure having variable height
- CO3 Lump mass with different material :-To understand effect of frequency and present travel on structure having different material
- CO4 Lump mass with different thickness:- To understand effect of frequency and present travel on structure having different thickness
- CO5 Single degree freedom system:-To understand behavior of multi degree of freedom system under lateral load
- CO6 Multi degree freedom system:-To understand behavior of single degree of freedom

- system under lateral load
- CO7 Building without structural control system:- To understand behavior of Uncontrolled structure under earthquake
 - CO8 Building with X-Bracing:-To understand behavior of building having X- Bracing under earthquake
 - CO9 Building with Shear wall:- To understand behavior of building having shear wall under earthquake
 - CO10 Pounding Effect:- To understand pounding effect of structure
 - CO11 Liquefaction of Soil:-To understand liquefaction of soil under dynamic loading

Bachelors of Engineering (Computer Science and Engineering)

Programme Outcome (POs)

- PO1-** Scientific knowledge: Apply the knowledge of mathematics, science, and computing to the solution of complex scientific problems.
- PO2-** Problem analysis: Identify, formulate, research literature, and analyze complex scientific problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and applied sciences.
- PO3-** Design/development of solutions: Design solutions for complex problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4-** Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5-** Modern tools usage: Create, select, and apply appropriate techniques, resources, and modern computing and IT tools including prediction and modeling to complex scientific activities with an understanding of the limitations.
- PO6-** The software engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional practice.
- PO7-** Environment and sustainability: Understand the impact of the professional software engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8-** Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the scientific practice.
- PO9-** Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Programmes specific Outcomes

- PSO1:** Demonstrate basic knowledge of computer applications and apply.
- PSO2:** Understand, Analyze and Develop computer programs for efficient design of computer-based systems of varying complexity.

- PSO3:** Demonstrate understanding of the principles and working of the hardware and software aspects of computer systems.
- PSO4:** Use professional engineering practices, strategies and tactics for the development, operation and intendance of software.
- PSO5:** Design and develop computer programs/computer-based systems in the areas related to algorithms, networking, web design, cloud computing, IoT and data analytics of varying complexity
- PSO6:** Acquaint with the contemporary trends in industrial/research settings and “thereby innovate novel solutions to existing problems

SEMESTER-III

Engineering Mathematics-III-CSE -3011

- CO1-** Analyze and solve engineering problems using Laplace Series
- CO2-** Analyze and solve engineering problems using Fourier Series
- CO3-** To apply partial differential techniques to solve the physical engineering problems
- CO4-** Develop problem-solving techniques needed to accurately calculate probabilities. Apply problem-solving techniques to solving real-world events

Data structure & Algorithms-CSE -3021

- CO1-** Understand the concept of Dynamic memory management, data types, algorithms, Big O notation.
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queues
- CO3-** Describe the hash function and concepts of collision and its resolution methods.
- CO4-** Solve problem involving graphs, trees and heaps
- CO5-** Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data

Computer Organization and Architecture-CSE -3031

- CO1-** Explain the organization of basic computer, its design and the design of control unit.
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queues
- CO3-** Describe the operations and language of the register transfer, micro operations and input-output organization.
- CO4-** Understand the organization of memory and memory management hardware.

Operating System-CSE -3041

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems.
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queueDescribe the various CPU scheduling algorithms and remove deadlocks.
- CO3-** Explain various memory management techniques and concept of thrashing.
- CO4-** Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5-** Recognize file system interface, protection and security mechanisms.

Value Education-CSE -3051

- CO1-** To learn the philosophy of Life, personal value, social value, mind cultural value and personal health.
- CO2-** To learn professional ethical values, codes of ethics, responsibilities, safety, rights and related global issues. Understand the basics of operating systems like kernel, shell, types and views of operating systems

Software Lab 1 (C++)-CSE -3061

- CO1-** Introduces Object Oriented Programming concepts using the C++ language.
- CO2-** Introduces the principles of data abstraction, inheritance and polymorphism
- CO3-** Introduces the principles of virtual functions and polymorphism
- CO4-** Introduces handling formatted I/O and unformatted I/
- CO5-** Introduces exception handling.

Semester – IV

Environmental Engineering-CS -4011

- CO1-** Analyze and solve engineering problems using Laplace Series
- CO2-** Analyze and solve engineering problems using Fourier Series.
- CO3-** To apply partial differential techniques to solve the physical engineering problems
- CO4-** Develop problem-solving techniques needed to accurately calculate probabilities.
- CO5-** Apply problem-solving techniques to solving real-world events

Database Management System-CS -4021

- CO1-** Understand the concept of Dynamic memory management, data types, algorithms, Big O notation.
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queues
- CO3-** Describe the hash function and concepts of collision and its resolution methods.
- CO4-** Solve problem involving graphs, trees and heaps
- CO5-** Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data

Software Engineering-CS -4031

- CO1-** Explain the organization of basic computer, its design and the design of control unit.
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queues
- CO3-** Describe the operations and language of the register transfer, micro operations and input-output organization.
- CO4-** Understand the organization of memory and memory management hardware.
- CO5-** Elaborate advanced concepts of computer architecture, Parallel Processing, Interprocessor communication and synchronization

Design and Analysis of Algorithm-CS -4041

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queueDescribe the various CPU scheduling algorithms and remove deadlocks
- CO3-** Explain various memory management techniques and concept of thrashing
- CO4-** Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5-** Recognize file system interface, protection and security mechanisms.

Computer Network-CSE -4051

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queue. Describe the various CPU scheduling algorithms and remove deadlocks
- CO3-** Explain various memory management techniques and concept of thrashing
- CO4-** Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5-** Recognize file system interface, protection and security mechanisms.

Software Lab II (Java-CSE -4061

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queue. Describe the various CPU scheduling algorithms and remove deadlocks
- CO3-** Explain various memory management techniques and concept of thrashing
- CO4-** Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5-** Recognize file system interface, protection and security mechanisms.

Semester –V**Digital Circuit and System CS-511**

- CO1-** Develop a digital logic and apply it to solve real life problems.
- CO2-** Analyze, design and implement combinational logic circuits
- CO3-** Classify different semiconductor memories.
- CO4-** Analyze, design and implement sequential logic circuits

Object Oriented Programming-CS-521

- CO1-** Articulate the principles of object-oriented problem solving and programming.
- CO2-** Apply the concepts of class, method, constructor, instance, data abstraction, function abstraction, inheritance, overriding, overloading, and polymorphism
- CO3-** Program using objects and data abstraction, class, and methods in function abstraction
- CO4-** Analyze, write, debug, and test basic C++ codes using the approaches introduced in the course
- CO5-** Analyze problems and implement simple C++ applications using an object-oriented software engineering approach.

Discrete Structure CS -5031(A)

- CO1-** Write an argument using logical notation and determine if the argument is or is not valid
- CO2-** Demonstrate the ability to write and evaluate a proof or outline the basic structure of and give examples of each proof technique described.
- CO3-** Understand the basic principles of sets and operations in sets
- CO4-** Demonstrate an understanding of relations and functions and be able to determine their properties
- CO5-** Model problems in Computer Science using graphs and trees

Cyber Security CS -5031(B)

- CO1-** Be able to Analyze and evaluate the cyber security needs of an organization

- CO2-** Be able to Determine and analyze software vulnerabilities and security solutions to reduce the risk of exploitation
- CO3-** Be able to Measure the performance and troubleshoot cyber security systems.
- CO4-** Be able to Measure the performance and troubleshoot cyber security systems.
- CO5-** Be able to implement cyber security solutions and use of cyber security, information assurance, and cyber/computer forensics software/tools.
- CO6-** Be able to Design and develop security architecture for an organization

Data Analytics-CS -5031(C)

- CO1-** Understand the basic of data analytics using concept of statistics and probability.
- CO2-** Understand the needs of data processing techniques.
- CO3-** Implement the data analytics techniques using R, MATLAB and Python.
- CO4-** Apply the data analytics techniques in real life applications.

Theory of Computation CSE -5041(A)

- CO1-** Students will demonstrate knowledge of basic mathematical models of computation and describe how they relate to formal languages
- CO2-** Students will understand that there are limitations on what computers can do, and learn examples of unsolvable problems.
- CO3-** Students will learn that certain problems do not admit efficient algorithms, and identify such problems
- CO4-** will apply knowledge of computing and mathematics appropriate to the discipline
- CO5-** will function effectively as a member of a team in order to accomplish a common goal.

Internet & Web Technology CSE -5041(B)

- CO1-** Describe the concepts of WWW including browser and HTTP protocol
- CO2-** Develop modern web pages using the HTML and CSS features with different layouts as per the need of application
- CO3-** Use the javascript to develop the dynamic web pages.
- CO4-** Develop the modern web application using the client and server side technologies and web design fundamentals
- CO5-** User server side scripting with PHP to generate the web pages dynamically using the database connectivity.

Production to DBMS- CSE -5041(C)

- CO1-** Describe the characteristics of database, architecture and languages of Database system.
- CO2-** Implement Entity-Relationship diagrams.
- CO3-** Apply relational model concepts and constraints.
- CO4-** Apply normalization techniques for relational databases and familiarize with transaction processing.
- CO5-** Develop programs using PL/SQL.

Lab(Linux)-CSE -5051

- CO1-** Explain the fundamental concepts of open-source operating system Linux
- CO2-** Understand the basic set of commands and editors in Linux operating system.
- CO3-** Discuss shell programming in Linux operating system

Lab(Python)-CSE -5061

CO1- Write, Test and Debug Python programs

CO2- Use functions and represent Compound data using Lists, Tuples and Dictionaries

CO3- Use functions and represents Compound data using Lists, Tuples and Dictionaries

Semester –VI**Cloud Computing-CS -6011**

CO1- Understand the concepts, characteristics, delivery models and benefits of cloud computing

CO2- Understand the key security and compliance challenges of cloud computing

CO3- Understand the key technical and organizational challenges

CO4- Understand the different characteristics of public, private and hybrid cloud deployment models.

Compiler Design-CS -6021

CO1- Understand the different phases of compiler.

CO2- Design lexical analyzer for a sample language.

CO3- Apply different parsing algorithms to develop the parsers for a given grammar.

CO4- Understand syntax-directed translation and run-time environment.

CO5- Learn to implement code optimization techniques and a simple code generator.

Artificial Intelligence-CS -6031(A)

CO1- To acquire the knowledge of basic artificial intelligence

CO2- To acquire the knowledge of searching strategies of AI

CO3- To acquire the knowledge of game playing techniques and planning of AI

CO4- To understand the concept of statistical methods for reasoning and logic

CO5- To acquire the knowledge of expert systems and other learning methods of AI

Computer Graphics & Multimedia-CS -6031(B)

CO1- Understand the basics of computer graphics, different graphics systems and applications of computer graphics.

CO2- Render projected objects to naturalize the scene in 2D view and use of illumination models for this

CO3- To implement various algorithms to scan, convert the basic geometrical primitives, transformations, Area filling, clipping.

CO4- To define the fundamentals of animation, virtual reality and its related technologies.

CO5- To design an application with the principles of virtual reality

Advanced Computer Architecture-CS -6031(C)

CO1- Understand the performance and efficiency in advanced multiple-issue processors.

CO2- Understand symmetric shared-memory architectures and their performance.

CO3- Understand multiprocessor cache coherence using the directory based and snooping class of protocols.

CO4- Understand the various models to achieve memory consistency.

CO5- Understand the several advanced optimizations to achieve cache performance.

Pattern Recognition-CSE -6041(A)

CO1- Understand basic concepts in pattern recognition

- CO2-** Gain knowledge about state-of-the-art algorithms used in pattern recognition research
- CO3-** Understand pattern recognition theories, such as Bayes classifier, linear discriminant analysis.
- CO4-** Apply pattern recognition techniques to real-world problems such as document analysis and recognition.
- CO5-** Implement simple pattern classifiers, classifier combinations, and structural pattern recognizers.

Internet Of Things-CSE -6041(B)

- CO1-** Understand the vision of IoT from a global context.
- CO2-** Understand the application of IoT.
- CO3-** Determine the Market perspective of IoT.
- CO4-** Use of Devices, Gateways and Data Management in IoT.
- CO5-** Building state of the art architecture in IoT.

Skill Development Lab-CSE -6061

- CO1-** Understand the basics of software as a Product.
- CO2-** Understand the current requirements of industries
- CO3-** Implement the software as a product using different design patterns.
- CO4-** Apply the software development techniques in real life applications.

Semester –VII

Software Architecture -CS -7011

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queue. Describe the various CPU scheduling algorithms and remove deadlocks
- CO3-** Explain various memory management techniques and concept of thrashing
- CO4-** Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5-** Recognize file system interface, protection and security mechanisms.

Computational Intelligence -CS -7021(A)

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queue. Describe the various CPU scheduling algorithms and remove deadlocks
- CO3-** Explain various memory management techniques and concept of thrashing
- CO4-** Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5-** Recognize file system interface, protection and security mechanisms.

Deep and Reinforcement Learning- CS -7021(B)

- CO1-** Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2-** Understand basic data structures such as arrays, linked lists, stacks and queue. Describe the various CPU scheduling algorithms and remove deadlocks
- CO3-** Explain various memory management techniques and concept of thrashing

CO4- Use disk management and disk scheduling algorithms for better utilization of external memory.

CO5- Recognize file system interface, protection and security mechanisms.

Cryptography and Information Security CS -7031(A)

CO1- Understanding of the basics of Cryptography and Network Security and working knowledge of Mathematics used in Cryptology.

CO2- Understanding of previous attacks on cryptosystems to prevent future attacks from securing a message over an insecure channel by various means.

CO3- Knowledge about how to maintain the Confidentiality, Integrity and Availability of a data

CO4- Understanding of various protocols for network security to protect against the network threats.

CO5- Getting hands-on experience of various Information Security Tools.

Data Mining and Warehousing-CS -7031(B)

CO1- Understand the need of designing Enterprise data warehouses and will be enabled to approach business problems analytically by identifying opportunities to derive business.

CO2- Compare and contrast, various methods for storing & retrieving data from different data sources/repository.

CO3- Ascertain the application of data mining in various areas and Preprocess the given data and visualize it for a given application or data exploration/mining task

Disaster Management CS -7031(C)

CO1- To gain a preliminary understanding of approaches of Disaster Risk Reduction (DRR)

CO2- To enhance awareness of institutional processes in the country and

CO3- To develop rudimentary ability to respond to their surroundings with potential disaster response in areas where they live, with due sensitivity

Semester –VIII

Mobile Application Development -CS -8011

CO1- Apply essential Android Programming concepts

CO2- Develop various Android applications related to layouts & rich user interfaces

CO3- Develop Android applications related to mobile related server-less database

High Performance Computing-CS -8021(A)

CO1- Able to design, formulate, solve and implement high performance versions of standard single threaded algorithms

CO2- Use HPC platforms and parallel programming models .

CO3- Able to measure, analyse and assess the performance of HPC applications and their supporting hardware.

CO4- Able to administration, scheduling, code portability and data management in an HPC environment, with particular reference to Grid Computing

Block Chain Technologies- CS -8021(B)

- CO1-** Explain cryptographic building blocks and reason about their security
- CO2-** Learn how the individual components of the Bitcoin protocol make the whole system works: transactions, script, blocks, and the peer-to-peer network
- CO3-** Define how Bitcoin mining works
- CO4-** Exploit applications of Blockchain in real world sceneries

Object Oriented Software Engineering- CS -8021(C)

- CO1-** Identify domain objects, their properties and relationships among them
- CO2-** Identify and model/represent domain constraints on the objects and (or) on their relationships
- CO3-** Develop design solutions for problems on various O-O concepts
- CO4-** To model different perspectives of object-oriented software design (UML)

Image Processing and Computer Vision- CS -8031(A)

- CO1-** Design and implement various algorithms for digital image processing and computer vision.
- CO2-** Understanding the fundamental concepts and the mathematical tools used in digital image processing and computer vision
- CO3-** Identify basic concepts, terminology, theories, models and methods in the field of computer vision
- CO4-** Describe basic methods of computer vision related to multi-scale representation, edge detection and detection of other primitives, stereo, motion and object recognition
- CO5-** Assess which methods to use for solving a given problem, and analyse the accuracy of the methods

Game Theory with Engineering Applications -CS -8031(B)

- CO1-** Formulate mathematically multi-agent decision-making problems arising in engineering applications as games
- CO2-** Analyze using mathematical theory the equilibria of the games and compute them using optimization theory, analyze mathematically the outcome of the game.
- CO3-** Understand selected models and concepts of game theory
- CO4-** Produce simple economic models with basic game theory

Managing Innovation and Entrepreneurship- CS -8031(C)

- CO1-** Able to assess the commercial viability of new technologies, business opportunities and existing companies
- CO2-** Plan, organize, and execute a project or new venture with the goal of bringing new products and service to the market
- CO3-** Carry out scientific research in the field of entrepreneurship

(Electronics & CommunicationEngineering)

Programme Outcome (POs)

- PO1-** Engineering Graduates will successfully engage themselves in practice of engineering and achieve excellence in their professional career.
- PO2-** It will inculcate social and moral values within the students in today's technology driven society.
- PO3-** They will pursue wide-spectrum careers with professional and managerial capability.
- PO4-** Graduates develop themselves as technologists, innovators, managers & entrepreneurs and will advance in their profession.

Program Specific Outcomes (PSOs)

- PSO1-** To build up skills to analyze the requirements of electronics, understand the technical specifications, design and provide novel engineering solutions and efficient product design. The Department of Electronics and Communication Engineering has a strong focus on providing students with a strong background in mathematics, science and engineering. The department provides students with adequate practical training by way of laboratory sessions, design and problem based learning.
- PSO2-** To develop the ability among the students to understand the concept of Mathematics, Physics and core electronics subjects which will facilitate understanding of new technology. Through the work experience gained from summer/winter training programs conducted by the expert faculties from the department and at various reputed organizations in India, students can have up to date knowledge of the specific field and can pursue career in that field.
- PSO3-** To provide the student with a strong foundation in the engineering fundamentals necessary to formulate, solve and analyze engineering problems and to prepare them for graduate studies, R&D, consultancy and higher learning. Students shall participate and succeed in competitive examinations such as GATE,GRE and TOEFL,PSUs and may admit to various programs like Master of Science (Abroad/India),Master of Technology (India) and Doctor of Philosophy (India/Abroad).
- PSO4:** To give exposure to emerging edge technologies, adequate training and opportunities to work as team on multi disciplinary projects with effective communication skills and leadership traits. Students will be able to realize their ideas with the help of" Incubation Centre" established within college premises and will be able to participate in National/ International level project (Design/Coding) competitions organized by Industry/Institutes. With the help of technical and entrepreneur skills students will be able to employ appropriate techniques using hardware and software engineering tools for modern engineering applications and will demonstrate an ability to apply their knowledge of advanced mathematics and electronics engineering principles towards creating new technologies that helps students to establish their own industries/ventures.
- PSO5-** To prepare graduates who possess the necessary foundation required to take up gainful employment in core sector and allied sector or prepare them for a successful career and

work professionally to meet the technical requirement of Indian and multinational companies.

Course Outcomes

Semester- III

Electronic Devices-EC 3021

- CO1-** Get a comprehensive introduction of electronic properties of semiconductors,
- CO2-** Be familiar with electronic devices, and their applications to circuits.
- CO3-** Will get knowledge of biasing and other characteristics with circuit operation.
- CO4-** Helps to understand the basics of solid state devices.
- CO5-** Be able to analyze performance of various types of amplifiers.
- CO6-** Get Thorough Knowledge of Power Supplies.

Digital Electronics- EC - 3031

- CO1-** Understand the concepts of Boolean algebra and Logic gates and realize logic functions by utilizing these fundamentals.
- CO2-** Design complex digital systems and apply them in real world scenario.
- CO3-** Understand the concepts of combinational and sequential circuits and implement them using the minimization techniques (K-Map, Quine-McCluskey algorithm).
- CO4-** Identify and analyze the synchronous and asynchronous logic circuits
- CO5-** Understand concepts of memory, programmable logic and digital integrated circuits
- CO6-** Get thorough knowledge of registers and counters and apply them in logic circuits.

Network Analysis-EC - 3041

- CO1-** Understand basics of electrical circuits with nodal and mesh analysis. Understand network laws like KCL and KVL.
- CO2-** Learn the concept of electrical network theorems.
- CO3-** Apply Laplace Transform for steady state and transient analysis.
- CO4-** Get knowledge of Network Graph theory and determine different network functions.
- CO5** Realize the frequency domain techniques and understand relationship of two port networks.
- CO6-** Get a comprehensive knowledge of linear network theory, frequency response and steady state analysis.

Value Education-EC - 3051

- CO1-** Understand the Concepts of Values, need for Education in values for character development and Challenges for Value adoption.
- CO2-** Get knowledge of Personal Values, Social Values, Spiritual values and its Benefits for self and society.
- CO3-** Learn Business and Professional Ethics and Entrepreneurship and develop Corporate Social Responsibility.
- CO4-** Will understand the self anatomy, its cyclic processes within self and learn the states of the awareness.
- CO5-** Understand Quality of Life and know self esteem and understand its components.

Software Lab --EC - 3061

- CO1-** Get familiar with circuit simulation/ PCB designing software
- CO2** Study and learn circuit simulation software's
- CO3** Get Knowledge of the software applications of the software in the field of Electronic Devices, Electronic Instrumentation and Network Analysis
- CO4-** Will understand Designing and simulation of Basic Electronics
- CO5-** Understanding of software in optimization, designing and fabrication of PCBs of different circuits.

Semester IV**Electronic Instrumentation and Measurement-EC – 4021**

- CO1-** Understand the errors in measurements and their rectification.
- CO2-** Understand working of CRO and apply the knowledge to select and identify specific sensors(or complete instruments) for controlling machines and processes.
- CO3** Designing and validation of DC and AC bridges, get knowledge of various transducers, thermocouples, etc and learn their concepts in designing measurement systems.
- CO4-** Understand the operating principles of a range of DAC and ADC instruments and study the dynamic response and the calibration of the instruments.
- CO5-** Designing and Analysis of signal and sweep generator systems.
- CO6-** Learn about various measurement devices, their characteristics, their operation and their limitations

Analog Communication-EC-4031

- CO1-** Understand Signal Analysis. spectral density and types of systems
- CO2-** Get Knowledge of modulation Techniques and their analysis.
- CO3-** Get in-depth knowledge of modulation and different types of detectors
- CO4-** Learn about various transmitters and receivers.
- CO5-** Learn and Understand various sources of noise and its types

Analog Circuits-EC - 4041

- CO1-** Understand the basic principles and operations of amplifiers at low and high frequencies.
- CO2-** Understand the nature and characteristics of devices such as Operational Amplifiers and Oscillators.
- CO3-** Analysis and modeling of amplifiers such as Power amplifier and Tuned Amplifiers along with their applications.
- CO4-** Learn about various cascade and cascade amplifiers with their characteristics.
- CO5-** Study and analyze the differential amplifiers with its characteristics and its analysis.

Control Systems- EC 4051

- CO1-** Study about the modeling, characteristics and classification of types of control systems.
- CO2-** Understand about the concept of stability of time domain and time response of the systems.
- CO3-** Analyze the systems in time and frequency domains and use the concept of feedback to improve the system performance.
- CO4-** Study about the different approaches of system designs

- CO5-** Learn about state space modeling of system and its analysis and understand about the concept of state, state variables and state model.

Software Lab -II-EC - 4061

- CO1-** Learn Advanced Simulation Software's and its analysis package.
CO2- Study of the key features and applications of the software.
CO3- Learn about the applications of the software in the field of Electronic Circuits, Digital Electronics and Analog Communication
CO4- Designing, Optimization, simulation and verification of digital electronic systems
CO5- Realization of various signals and communication link etc.

Semester – V

Digital Communication - EC 5011

- CO1-** Understand basics of Digital Communication and Information Theory.
CO2- Learn the concept of Sampling Theory.
CO3- Understand the concept of Quantization, Quantization Noise, A-Law, etc
CO4- Get knowledge of various Digital Modulation Techniques
CO5- Learn about Digital Multiplexing Schemes and its concepts
CO6- Get a comprehensive knowledge of Polar, Bipolar Coding's, ISI, Nyquist Criterion, etc.

Electromagnetic Theory-EC-5021

- CO1-** Learn about different waveguides.
CO2- Understand about the concept of electromagnetism.
CO3- Analyze the systems and need for impedance matching and different impedance matching techniques
CO4- Study and learn about the various concepts of electric and magnetic fields.
CO5- Understand about field and circuit theory and understand about the concept of Maxwell's equations and time varying fields.

Professional Elective-EC – 5031Microprocessor and Controller(μ P and μ C)-EC 5031(A)

- CO1-** Understand the concept of 8086/8088 μ Processor and its Architecture
CO2- Get Introduction to the idea of instruction set and programming of 8086/8088 μ Processor
CO3- Knowledge of segmentation of memory and Interfacing peripherals
CO4- Knowledge of interfacing devices like DMA Controller, PCI, etc
CO5- Familiarity with Multimedia Extension (MMX) and Knowledge of 8051 Microcontroller and its Programming

Voice Communication (VC)- EC 5031(B)

- CO1-** Understand the concept of Basic Telephony, Modes of Data Transmission
CO2- Get Introduction to the idea of Line Coding, PSTN Network.
CO3- Knowledge of different types of Digital Telephony.
CO4- Knowledge of various sorts of Digital Communication.
CO5- Familiarity with Digital Switching and its protocols.

Communication Networks and Transmission Lines (CNTL)-EC 5031(C)

- CO1-** Understand the concept of Two Ports Networks
CO2- Get Introduction of various types of filters.
CO3- Knowledge of different types Real Functions.

CO4- Knowledge of various sorts of Transmission Line Fundamentals.

CO5- Familiarity with Radio frequencies and its basics

Open Elective-EC – 5041

Industrial Electronics-EC 5041(A)

CO1- Know concepts of power electronic devices as switches

CO2- Understand operation and applications of converters

CO3- Comprehend operation of inverters and commutation methods

CO4- Learn techniques of PWM inverters

CO5- Understand operation of choppers

CO6- Understand the working of Electronic Control of AC and DC motors.

Computer Organization - EC 5041(B)

CO1- Understand Computer Basics and its Fundamentals.

CO2- Get Knowledge of Control Unit Organization.

CO3- Get in-depth knowledge of different types of I/O Organization.

CO4- Learn about various memory organizations their specifications.

CO5- Learn and Understand about multiprocessors and their various parameters.

Data Communication -EC 5041(C)

CO1- Understand the concept of Basic Telephony, Modes of Data Transmission

CO2- Get Introduction to the idea of Line Coding, PSTN Network and OSI Models.

CO3- Knowledge of different types of LAN and WAN protocols.

CO4- Knowledge of various sorts of errors of Data Communication.

CO5- Familiarity with Networking Devices and its protocols.

CNTL Lab - EC 5051

CO1. Study of standing waves formation on a transmission line

CO2. Learn to measure various parameters of transmission line.

CO3. Get Knowledge of reactive loads and physical dimensions of Transmission line.

CO4. Get Knowledge of Balun Transformer and various filters.

CO5. Learn to measure input and output impedance of any RF circuits.

Subject code: Software Lab-III-EC 5061

CO1. Study of simulation softwares like MATLAB.

CO2. Introduction to MATLAB and its basics.

CO3. Modeling, Designing and development of Programs in MATLAB

CO4. Application of the software in the field of Control Systems, Data Communications and Communication Systems.

CO5. Developing Programs related to simulations of CRC, LRC, VRC, hamming codes, line encoding techniques.

Semester – VI

Wireless Communication-EC-6011

CO1- Understand generations of wireless mobile communication technologies

CO2 Be conversant with how various decisions were made while evolving the mobilecommunication system standards and get Knowledge of Handoff's and Channel Assignments.

- CO3-** Get the Knowledge of functioning of mobile communication network
- CO4-** Understand the concept of Digital Modulation Techniques for Mobile Communications.
- CO5-** To Understand features of wireless LAN technologies
- CO6-** Learn principles of Communication Protocols and its specifications.

Antenna & Wave Propagation-EC-6021

- CO1-** Understand Antenna Fundamentals and Vector Potentials.
- CO2-** Get Knowledge of various forms of Antenna Arrays.
- CO3-** Get in-depth knowledge of different types of Antennas and their features.
- CO4-** Learn about various antenna measurements and their specifications.
- CO5-** Learn and Understand about modes of propagation and their various parameters.

Departmental Elective-II-EC – 6031

VLSI Fabrication and Technology - EC 6031(A)

- CO1-** Get Knowledge of fundamental principles of VLSI (Very Large Scale Integrated) Circuit Design and Layout.
- CO2-** Have an overview of CMOS fabrication technologies, physical VLSI design issues (bottom-up design), Thin Film Fabrication.
- CO3-** Learn about Lithography and Different Types of Lithography.
- CO4-** Learn basic concepts of VLSI process techniques and Integration.
- CO5-** Get Knowledge of building blocks of large-scale CMOS digital integrated circuits
- CO6-** Have hands-on design experience using a professional IC design platform

Digital Signal Processing - EC 6031(B)

- CO1-** To learn and implement basic digital signal processing techniques like convolution, correlation and Fourier transform.
- CO2-** To learn and understand realization of Discrete Time Systems.
- CO3-** To design and implement different FIR filters.
- CO4-** To design and implement different IIR filters.
- CO5-** To characterize stochastic process.
- CO6-** To implement basic digital processing algorithms in dedicated DSP platforms.

Digital Image Processing- EC 6031(C)

- CO1** To learn and implement basic digital image processing techniques and understand its fundamentals.
- CO2** To learn and understand realization of 2D Discrete Fourier Transforms.
- CO3** To learn and understand Image Enhancement Spatial domain methods.
- CO4** To understand Image Restoration Image degradation/restoration methods.
- CO5** To Learn Image Compression Fundamentals of data compression.
- CO6** To get knowledge of Digital image watermarking and its parameters.

Software Lab- IVEC - 6051

- CO1-** Learn VHDL, identifiers, data objects, data types, operators, behavioral modeling, data flow modeling, structural modeling.
- CO2-** Get overview of digital design with Verilog, Hierarchical Modeling: basic concepts, models and ports, gate level modeling, data flow modeling and behavioral modeling

- CO3-** Learn about Design and simulation of following Logic Gates Verilog/ VHD
- CO4-** Logic gates: NAND, NOR, XOR, XNOR.
- CO5-** Get knowledge of Half adder, full adder, subtractor, latches, multiplexers- 2:1, 4:1, 8:1, comparators, decoders- 2:4, 3:8, 4:16.
- CO6** Learn the concept of 4-bit ripple carry full adder, 4-bit Ripple carry counter, parity generator, up/down counters.

DSP and Embedded System Lab-EC – 6061

- CO1-** Learn designing of different Filters.
- CO2-** Get overview of various Transforms of Digital Signal Processing
- CO3-** Learn about Design and simulation of 8051 microcontroller.
- CO4-** Get knowledge of 8051 Timers and Counters.
- CO5-** Learn the concept of LCD and its parameters.

Semester – VII

VLSI Design- EC 7011

- CO1-** To understand the fabrication process of CMOS technology.
- CO2-** To teach fundamentals of VLSI circuit design and implementation using circuit
- CO3-** simulators and layout editors. To study various problems due to VLSI technology advancement.
- CO4-** To study digital circuits using various logic methods and their limitations.
- CO5-** To highlight the circuit design issues in the context of VLSI technology.
- CO6** Upon successful completion of this course, the student will be able to: Demonstrate a clear understanding of CMOS fabrication flow and technology scaling.
 - Design MOSFET based logic circuit
 - Draw layout of a given logic circuit
 - Demonstrate an understanding of working principle of operation of different types of
 - memories
 - Demonstrate an understanding of working principles of clocking, power reduction and
 - Distribution

Departmental Elective III - EC 7021

Subject Name: - Microwave Engineering- EC 7021(A)

- CO1-** Identify of various types of Microwave electronic components and systems.
- CO2-** Understand different modes of operation of various RF and Microwave circuits.
- CO3-** Design and analyze of high frequency circuits and systems.
- CO4-** Solving complex RF & Microwave communication network design problems

Information Theory and Coding- EC 7021(B)

- CO1-** Acquire the knowledge in measurement of information and errors.
- CO2-** Know the application of coding theorem for efficient utilization of communication resources.
- CO3-** Understand the utilization of various communication channels for communication system.
- CO4-** Design the block and cyclic codes for error correction and detection in communication systems
- CO5-** Know the significance of source and channel codes in various applications.

NANO Electronics- EC 7021(C)

- CO1-** Acquire the knowledge in Semiconductor physics.
- CO2-** Know the application of Lithography.
- CO3-** Understand the utilization of various Tunnel Junction and its applications.
- CO4-** Understand designing of MOSFETs
- CO5-** Know the significance of scaling in various applications.

Open Elective III-EC – 7031

I.O.T.-EC 7031(A)

- CO1-** . Understand in depth about Internet of things. 2.. 3. 4. 5.
- CO2-** Establish secure communication for his network for his devices connected in IOT
- CO3-** Store his data securely on cloud and access it when required
- CO4-** Design web based application using various internet protocols and service
- CO5-** Use sensor technology and RFID and wireless networking for maintaining privacy and security concern in smart city and housing environmental considerations.
- CO6-** The course provides basic knowledge of how to connect various devices through Internet and control them remotely. It will provide methods for different types of networking and data storage. The course aims at providing communication overview and protocols for safe and secure data access and transfer and maintain confidentiality and integrity.

Operation Research-EC 7031(B)

- CO1-** Understand basics of Resource Scheduling and Network Analysis.
- CO2-** Learn the concept of Replacement and Game Theory.
- CO3-** Get knowledge of Inventory Control.
- CO4-** Get a comprehensive knowledge of Linear Programming.
- CO5-** Learn Advanced Linear Programming Problems.

Human Resource and Management- EC 7031(C)

- CO1-** Importance of Human Resource and Management and its Functions.
- CO2-** Understand the need and basics of Job Procurement and Manpower Planning.
- CO3-** Understand the Utility of Training and Development and get knowledge of factors affecting compensation Policy.
- CO4-** Get Knowledge of Industrial Relations.
- CO5-** Understand the concept of Employee Safety and Retirement Policy.
- CO6-** Learn the concepts of International HRM and its business.

Departmental Elective Lab-EC-7041

Microwave Lab- EC 7041

- CO1-** Study of Microwave electronic components and systems.
- CO2-** Understand different modes of operation of various RF and Microwave circuits.
- CO3-** Get Knowledge of reactive loads.
- CO4-** Design and analyze of high frequency circuits and systems.
- CO5-** Get understanding of Solving complex RF & Microwave communication network design problems.

Open Elective Lab-EC–7051 IOT Lab- EC 7051

- CO1-** Understand LAB INDEX Design
- CO2-** Developed and implement following using Arduino
- CO3-** Learn about Raspberry Pi compiler
- CO4-** Get knowledge of Python language in Linux/Windows environment.
- CO5-** Study applications of IOT

Semester – VIII

TV and Radar-EC 8011

- CO1-** Fundamental principles of TV Transmitter and Receiver.
- CO2-** Understand the need and basics of Three color Theory.
- CO3-** Become familiar with types of color Television Systems.
- CO4-** Understand the Utility of Radar Systems and learn Radar Equations and their applications.
- CO5-** Understand the concept of MTI and Pulse Doppler Radar.
- CO6-** Learn the Utility of Radar Transmitter and Receiver.

Departmental Elective IV-EC – 8021

Satellite Communication -EC 8021(A)

- CO1-** Importance of Satellite Systems and its Functions.
- CO2-** Understand the need and basics of Geostationary Orbits.
- CO3-** Understand the Utility of the Space Segments.
- CO4-** Get Knowledge of Space Links.
- CO5-** Understand the concept of Satellite Services

AI and Signal Processing - EC 8021(B)

- CO1-** Ability to develop a basic understanding of AI building blocks presented in intelligent agents.
- CO2-** Ability to choose an appropriate problem-solving method and knowledge representation technique.
- CO3-** Ability to analyze the strength and weaknesses of AI approaches to knowledge-intensive problem-solving.
- CO4-** Understand real time applications of Fourier transform.
- CO5-** Describe discrete time systems in terms of difference equations.
- CO6-** To impart knowledge about Artificial Intelligence and to give understanding of the main abstractions and reasoning for intelligent systems and signal processing

Wireless Network - EC 8021(C)

- CO1-** Review the concepts of wireless and mobile communication
- CO2-** Understand LTE and OFDM technologies for mobile telephony
- CO3-** Understand the basic concepts of wireless sensor network
- CO4-** Understand mobile networking and compare transport layer protocols for mobile and traditional networks
- CO5-** Understand the technology and standards of IoT, ZigBee

Open Elective III-EC–8031

Cellular Mobile Communication- EC 8031(A)

- CO1-** Understand various communication standards and multiple accesses
- CO2-** Cellular concepts and basics and design
- CO3-** Frequency allocation, Reuse, and antennae for mobile communication
- CO4-** Various interferences and reduction techniques
- CO5-** Various cellular generation, standards and trends

5G Technology-EC 8031(B)

- CO1-** Describe 5G Technology advances and their benefits

- CO2-** Distinguish the key RF, PHY, MAC and air interface changes required to support 5G
- CO3-** Demonstrate Device to device communication and millimeter wave communication
- CO4-** Implementation options for 5G
- CO5-** Modeling of MIMO system

Speech Processing-EC 8031(C)

- CO1-** Model speech production system and describe the fundamentals of speech.
- CO2-** Extract and compare different speech parameters.
- CO3-** Choose an appropriate statistical speech model for a given application.
- CO4-** Design a speech recognition system.
- CO5-** Use different speech synthesis techniques.

Departmental Elective Lab-EC – 8041Advanced Communication Engineering La- EC 8041

- CO1-** Understand ASK, FSK**CO2-** Study modulation Schemes**CO3-** Learn about Multiplexing and De multiplexing. **CO4-** Get knowledge of various types of Antennas**CO5-** Study applications of Directional Couplers

Bachelors of Engineering (Electrical & ElectronicsEngineering)

Programme Outcome (POs)

- PO1-** Apply the knowledge of mathematics,science, engineering fundamentals and an engineering specialization to the solution of the complex engineering problems
- PO2-** Identify formulate, review research literature, and analyze complex engineering problems and reaching sustained conclusion using the principle of mathematics , natural sciences and engineering sciences
- PO3-** Design the solutions for complex engineering problem and design system components or processes that meet the specified needs with appropriate considerations for the public health and safety and cultural and societal and environmental considerations
- PO4** Use research based knowledge and research method including design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions
- PO5-** Create , select and apply appropriate techniques, resources , modern engineering and IT tools including prediction and modeling to complex engineering activities with understanding of the limitations
- PO6-** Apply reasoning informed by contextual knowledge, to assess societal health, safety, legal and cultural issues and consequent responsibility relevant to professional engineering practices.
- PO7-** Understand the impact of professional engineering solution in societal and environmental context and demonstrate the knowledge of, and need for sustainable development.
- PO8-** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practices.
- PO9-** Function effectively as an individual and as a member or leader in diversity and multi-disciplinary settings.

- PO10 Communicate effectively on complex engineering activities with the engineering community and with society at large such as being able to comprehend and write effective reports and design documentations, make effective presentations and view and receive clear instructions.
- PO11 Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team to manage project and in multi-disciplinary environments.
- PO12 Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Program Specific outcomes-

- PSO1- Use logical & technical skills to model, simulate and analyse electrical components and systems.
- PSO2- Integrate the knowledge of fundamental electronics, power electronics and embedded systems for the controllability, reliability and sustainability of electrical systems.
- PSO3- Contribute for the development of smart power grid and integrating green energy on it to meet the increasing demand of the society.
- PSO4 Graduates will be able to solve real life problems of Power system and Power Electronics using Mi Power, PSPICE and MATLAB software tools and hardware.
- PSO5- Graduates will be able to develop and support systems based on renewable and sustainable Energy sources.

Semester – III

Environmental Engineering- EX- 3011

- CO1 Convey a clear idea of the interdisciplinary nature of environmental and health risk assessment. Understand how each discipline of CEE maps into the society and the world market in the realm of environmental issues
- CO2 Understand how each discipline of CEE maps into the society and the world market in the realm of environmental issues.
- CO3 The ability to apply the fundamental knowledge of science and engineering to assess environmental and health risk.
- CO4 Understand what is life-time learning and how does it contribute to advancements of career.
- CO5 Ability to understand environmental laws and regulations to develop guidelines, procedures and processes for health and safety issues.
- CO6 Understand what are professional ethics and how do ethics affect the outcomes of environmental laws and regulations.
- CO7 Understand professionalism, ethics, and environmental laws and regulations.
- CO8 Understand about life-long learning, and current global and contemporary issues in environmental and health risk assessment.

Electromagnetic Fields-EX – 3021

- CO1 Explain fundamental laws governing electromagnetic fields and evaluate the physical quantities of electromagnetic fields (Field intensity, Flux density etc.) in different media using the fundamental laws.
- CO2 Define and recognize different co-ordinate systems to describe the spatial variations of the physical quantities dealt in electromagnetic field theory as they are functions of

- space and time. Apply different techniques of vector calculus to understand different concepts of electromagnetic field theory.
- CO3 Solve Laplace's & Poisson's equations, solution of Laplace's equation.
 - CO4 Determine the electromagnetic force exerted on charged particles, current elements, working principle of various electric and electromagnetic energy conversion devices are based on this force.
 - CO5 Design electromagnetic energy storage devices like capacitor, inductor which are frequently used in electrical systems and choose suitable materials required to assemble such electromagnetic energy storage devices..
 - CO6 Deduce and justify the concepts of electromagnetic waves, means of transporting energy or information, in the form of radio waves, TV signals, radar beams and light rays.
 - CO7 Generalize the concepts of guided structures like transmission line, means of transporting energy or information, commonly used in power distribution and communication.
 - CO8 Explain Biot-Savart's law, Faraday's Law

Electrical Measurement and Measuring Instruments-EX- 3031

- CO1 Understand about different instruments that are used for measurement purpose.
- CO2 Analyze the Performance characteristics of each instrument.
- CO3 Understand about different types of signal generators and recorders.
- CO4 Understand how different physical parameters and their co-relation with measuring Instrument.
- CO5 Internal and general repairing of instruments and problem solving capacity.
- CO6 Perform the experiments on pH-meter, Potentiometry, Conductometry, Colorimetry and chromatography as well as to analyze and interpret the data to address issues related to engineering problems.
- CO7 Knowledge about Dynamometer type wattmeter – induction type energy meter- 1 phase & 3 phase.
- CO8 Acquire the Principle of operation and construction of PMMC, MI, Dynamometer.

Network Analysis-EX – 3041

- CO1 Introduction to LLBP circuit elements R,L,C
- CO2 Select a suitable measuring instrument for a given electrical machine.
- CO3 Conduct experimental investigation and gain knowledge of Filter circuit.
- CO4 Solve the Laplace Transform and Inverse Laplace Transform.
- CO5 Analyze the response of Step, Ramp, Impulse, Sinusoidal, Cosinusoidal, Exponential, and Gate signals.
- CO6 Perform the experiments on Network Theorems – Thevenin's & Norton's theorem, superposition, reciprocity, compensation, maximum power transfer and Millman's theorem
- CO7 Solve the two port network analysis.
- CO8 Concept of phasor & vector, impedance & admittance. Node & mesh analysis of RL, RC and RLC networks with sinusoidal and other driving sources.

Value Education -EX - 3051

- CO1 To teach and inculcate the importance of value based living
- CO2 To give students a deeper understanding about the purpose of life.

- CO3 To teach and inculcate the essential qualities to become a good leader.
- CO4 understand the importance of value based living
- CO5 understand and start applying the essential steps to become good leaders.
- CO6 emerge as responsible citizens with clear conviction to practice values and ethics in life.
- CO7 To teach the students on being tolerant and to adjust with family, friends and society through case studies.
- CO8 To motivate students to prove that the character building is important for value based living through Debates / Testimonies.

Semester – IV

Engineering Mathematics-III-E.E.- 4011

- CO1 Find Laplace Transform of periodic functions, Dirac delta function and solve differential equations, integral equations and integro- differential equations and can analyze the solution. Expand the function in Fourier series and half range. Also Find Fourier transform and Fourier Integral and solve Integral equations.
- CO2 Analyze solutions of partial differential equations and solve the initial value and boundary value problems. Extremis the functional and also solve Isoperimetric problems.
- CO3 Understand Principle of operation and characteristics, biasing arrangements of Field effect transistors and MOSFETs and types of oscillator.
- CO4 Evaluate conversion of numbers from one code to other code, logic gates and truth table of digital circuits.

ELECTRICAL MACHINE-I-EX- 4021

- CO1 Formulate and then analyze the working of any electrical machine using mathematical model under loaded and unloaded conditions.
- CO2 Understand and explain the principle of operation and performance of Induction Machine, Synchronous Machines and Fractional kW Motors
- CO3 Analyze the response of Induction Machine, Synchronous Machines and Fractional kW Motors.
- CO4 Troubleshoot the operation of Induction Machine, Synchronous Machines and Fractional kW Motors
- CO5 Analyze given require specification of electrical machine and select a suitable measuring instrument for a given application
- CO6 Select the suitable specification of machine for different purpose
- CO7 To expose the students to the concepts of various types of electrical machines and applications of electrical machines in electric power engineering and electric drives.
- CO8 Requirements, power capability, efficiency, operating characteristics, control requirements and electrical demands of these machines

Digital Electronics EE - 4031

- CO1 Convert different type of codes and number systems which are used in digital communication and computer systems.
- CO2 Employ the codes and number systems converting circuits and Compare different types

of logic families which are the basic unit of different types of logic gates in the domain of economy, performance and efficiency.

- CO3 Analyze different types of digital electronic circuit using various mapping and logical tools and know the techniques to prepare the most simplified circuit using various mapping and mathematical methods.
- CO4 Design different types of with and without memory element digital electronic circuits for particular operation, within the realm of economic, performance, efficiency, user friendly and environmental constraints.

Electrical Engineering Drawing-EX- 4041

- CO1 Electrical Drawing Is An Important Engineering Subject Taught To Electrical / Electronic Engineering Students Both At Degree And Diploma Level Institutions. The Course Content Generally Covers Assembly And Working Drawings Of Electrical Machines And Machine Parts, Drawing Of Electrical Circuits,
- CO2 Learning how to draw the shapes, angles and lines and others which is essential for engineer.
- CO3 The ability to read and understand information contained on drawings is essential to perform most engineering-related jobs
- CO4 Engineering drawings are the industry's means of communicating detailed and accurate information on how to fabricate, assemble, troubleshoot, repair, and operate a piece of equipment or a system
- CO5 Importance of Engineering Drawing Technical drawing allows efficient communication among engineers and can be kept as a record of the planning process. Since a picture is worth a thousand word
- CO6 Learning the main idea from assembly and detail drawing.
- CO7 Technical drawing is essential for communicating ideas in industry and engineering. To make the drawings easier to understand, people use familiar symbols, perspectives, units of measurement, notation systems, visual styles, and page layout. Technical drawings are understood to have one intended meaning.
- CO8 Get information about the important tools for engineering drawing. This will give student basic knowledge of technical drawings professions and means of communications to others.

POWER SYSTEM-I-EX-4051

- CO1 Gain the knowledge on Steady State, Transient and Voltage Stability aspects. Acquire in-depth advance knowledge in the domain of modern and industrial oriental power systems.
- CO2 Analyze the given power system network with respect to stability point of view, Critically analyze various power systems components, models and their operation, optimization of cost criteria
- CO3 Fundamentals and concepts to analyze, formulate and solve complex problems of electrical power systems and its components and control of frequency and voltages
- CO4 Use advanced techniques, skills and modern scientific and engineering tools for professional practice for power system to enhanced power quality, reliability, security and load ability.
- CO5 To prepare graduates who have the ability to identify and address current and future

- problems in the domain of power systems, power electronics and electrical machines
- CO6 Power Systems in Restructured Environment; Distributed and Dispersed Generation; Environment Aspects of Electric Power Generation
 - CO7 Reactive Power Sensitivity and Voltage Control; Load Compensation with Capacitor Banks; Line Compensation with Reactors; Shunt and Series Compensation; Fixed Series Capacitors; Thyristor Controlled Series Capacitors
 - CO8 Impart the knowledge of generation of electricity based on conventional and non-conventional energy sources.

JAVA PROGRAMMING- EX-4061

- CO1 Write, compile, and execute Java programs that may include basic data types and control flow constructs using J2SE or other Integrated Development Environments (IDEs) such as Eclipse, Net Beans, and Developer.
- CO2 Write, compile and execute Java programs using object oriented class structures with parameters, constructors, and utility and calculations methods, including inheritance, test classes and exception handling.
- CO3 Write, compile, and execute Java programs using arrays and recursion.
- CO4 Write, compile, and execute Java programs manipulating Strings and text documents.
- CO5 Write, compile, execute Java programs that include GUIs and event driven programming.
- CO6 Write a final project that may be selected from among the following: applets for inclusion in web pages; applets to access enterprise data bases in robust, enterprise three level applications; secure communications over the internet; or an approved project chosen by the student.

Semester – V

Electrical Machine-II-EX- 5011

- CO1 Acquire knowledge about the constructional details and principle of operation of alternator
- CO2 Acquire knowledge about the working of synchronous machines as generators and motors.
- CO3 Acquire knowledge about testing and applications of synchronous machines.
- CO4 Acquire knowledge about the constructional details and principle of operation of three phase and single phase induction motors.
- CO5 Acquire knowledge about the starting and speed control of induction motors
- CO6 Acquire knowledge about testing and applications of induction motors.
- CO7 Acquire hands on experience of conducting various tests on dc machines and obtaining their performance indices using standard analytical as Well as graphical methods.
- CO8 Acquire hands on experience of conducting various tests on transformers and obtaining their performance indices using standard analytical as well as graphical methods

Switchgear and Protection-EX-5021

- CO1 Acquire the knowledge of various abnormal conditions that could occur in power system.
- CO2 Ability to design various protective devices in power system for protecting equipment and personnel.
- CO3 Knowledge of various types of existing circuit breakers, their design and constructional details.

- CO4 Knowledge of various conventional relays, their design and latest developments.
- CO5 Knowledge of standards and specifications related to switchgear and protection.
- CO6 Select suitable method and mathematical model for short circuit and load flow studies
- CO7 Design and develop different protection schemes
- CO8 Understand
 - various types of faults
 - Protective schemes
 - Power system protective equipment
 - significance of relay testing and co-ordination

Energy Conservation Management-EX- 5031

- CO1 Gain knowledge on non-renewable sources like solar, biomass, wind energies, Realize solar energy applications using photo voltaic cell.
- CO2 Acquire the knowledge of basic principles of energy auditing, types and objectives, Instruments used .
- CO3 To learn energy efficient control methods and schemes for improvement of starting efficiency in electrical motors.
- CO4 Understand efficient control strategies, Optimal selection, sizing, operation of variable speed drives like pumps and fans.
- CO5 Acquire the knowledge of analysis of Transformer loading and Feeder loss evaluation, Methods, scheme for reactive power management, energy efficient illumination system.
- CO6 Acquire the knowledge of different types and schemes of cogeneration. Analyses biogas performance and testing
- CO7 To learn the Energy conservation measures and optimal operation methods for electric load like air conditioning, refrigeration, geysers-solar.
- CO8 Acquire the knowledge of fundamentals of economic operation of an electrical system. Water heaters, compressors, electrolytic process

CONTROL SYSTEM- EX- 5041

- CO1 Learn the characteristics of nonlinear systems and common types of nonlinearities
- CO2 Analyze the behavior of nonlinear systems using both phase plane and describing function methods
- CO3 Test the stability of nonlinear systems
- CO4 Design full order and reduced order observers
- CO5 Design state feedback controllers
- CO6 Learn microprocessor/ microcontroller/ DSP based control programmable logic controllers
- CO7 To design a Lead, lag and Lag-Lead compensator sand to obtain the characteristics by experiment and simulation using MATLAB
- CO8 Set up a system for closed loop voltage regulation for a dc separately excited generator using amplidyne and obtain its characteristics

ELECTRICAL SIMULATION LAB-I- EX- 5051

- CO1- At the end At the end of the course, the student should have the
- CO2- Ability to understand the concept of MATLAB programming in solving power systems problems.
- CO3. Ability to understand power system planning and operational studies.
- CO4. Ability to acquire knowledge on Formation of Bus Admittance and Impedance Matrices

and Solution of Networks.

- CO5. Ability to analyze the power flow using GS and NR method.
- CO6. Ability to find Symmetric and Unsymmetrical fault.
- CO7. Ability to understand the economic dispatch.
- CO8. Ability to analyze the electromagnetic transients.
- CO9. Ability to acquire knowledge on power system analysis methods.

Semester – VI

Microprocessors and Microcontrollers-EX-6011

- CO1 Design the VLSI circuit and study the Introduction of Intel 8085 architecture.
- CO2 Understand the Assembly language program.
- CO3 Analyze the Interrupts and methods of data transfer
- CO4 Understand the External peripheral devices.

Power System-II EX-6021

- CO1 To comprehend the different issues related to overhead lines and underground cables.
- CO2 To train the students with a solid foundation in power system concepts required to solve engineering problems.
- CO3 To provide the knowledge about the system transients, sag and various issues related to cables and transmission lines.
- CO4 The student is capable of using different methods of power system analysis and design in sufficient depth for short circuit calculations and stability.

Departmental Electives-EX-6031

Utilization of Electrical Energy- EX – 6031(A)

- CO1 Illumination engineering is a field that spans many topics, including the development of sources, design of luminaires and light pipes, measurement of lighting conditions, and machine vision
- CO2 To learn the process of Heating&Welding and understand the process of Electrolysis.
- CO3 To understand the action of drawing a body, vehicle, train, or the like, along a surface, as a road, track, railroad, or waterway.
- CO4 Understand starting, speed control and braking of electric system
- CO5 Learn and analysis the working of drives & Hybrid system used in traction.
- CO6 Learn and analysis the working of drives & Hybrid system used in traction.

Electromagnetic Waves-EX 6031 (B)

- CO1- Apply vector calculus to static electric-magnetic fields in different engineering situations.
- CO2- Analyze Maxwell's equation in different forms (differential and integral) and apply them to diverse engineering problems
- CO3- Examine the phenomena of wave propagation in different media and its interfaces and in applications of microwave engineering.
- CO4- Analyze the nature of electromagnetic wave propagation in guided medium which are used in microwave applications.

Digital Signal Processing -EX 6031 (C)

- CO1- Use concepts of trigonometry, complex algebra, Fourier transform, z-transform to

- analyze the operations on signals and acquire knowledge about Systems
- CO2- Select proper tools for analog-to-digital and digital-to-analog conversion. Also select proper tools for time domain and frequency domain implementation
 - CO3- Design, implementation, analysis and comparison of digital filters for processing of discrete time signals
 - CO4- Integrate computer-based tools for engineering applications
 - CO5- Employ signal processing strategies at multidisciplinary team activities
 - CO6- Assess the techniques, skills, and modern engineering tools necessary for analysis of different electrical signals and filtering out noise signals in engineering practice. Also develop creative and innovative designs that achieve desired performance criteria within specified objectives and constraints, understand the need for lifelong learning and continuing professional education

Open Electives-EX-6041

Electronic Instrumentation-EX – 6041(A)

- CO1 Explain basic concepts and definitions in measurement, bridge configurations and their applications
- CO2 Elaborate discussion about the importance of signal generators and analyzers in measurement
- CO3 Explain the basic features of oscilloscope and different types of oscilloscopes
- CO4 Apply the complete knowledge of various electronics instruments/transducers to measure the physical quantities in the field of science, engineering and technology.

Internet of Things (IoT)- EX 604 (B)

- CO1- Able to understand the application areas of IOT
- CO2- Able to realize the revolution of Internet in Mobile Devices, Cloud & Sensor Networks
- CO3- Able to understand building blocks of Internet of Things and characteristics.

Analog & Digital Communication- EX 604 (C)

- CO1- Explain the concept of amplitude and angle modulations
- CO2- Compare different types of shift keying techniques
- CO3- Describe various data and pulse transmission schemes
- CO4- Study the error in the communication system using error Control coding techniques.
- CO5- Describe about multi user communication concepts, radio communication and wireless technology.

Electrical Simulation Lab – II-EX 605

- CO1- Understand the main features and importance of the MATLAB/ SCI LAB mathematical programming environment.
- CO2- Apply working knowledge of MATLAB/ SCI LAB package to simulate and solve Electrical, Electronic circuits and Applications.
- CO3- Solve, Simulate and Analyse various DC circuits.
- CO4- Solve, Simulate and Analyse various AC circuits.
- CO5- Solve, Simulate and Analyse various Analog and Digital Electronics circuits
- CO6- Solve, Simulate and Analyse simple Transformer and DC Generator circuits.

Semester-VII

Power Electronics-EX - 7011

- CO1 Learn basic operation of various power semiconductor devices & understand basic principle of switching circuits.
- CO2 Analyze and design an AC/DC rectifier circuit & DC/DC converter circuits
- CO3 Analyze DC/AC inverter circuit
- CO4 Understand the role power electronics in the improvement of energy usage efficiency and demonstrate the development of renewable energy technologies.

Departmental Electives

Power System Analysis & Control - EX-7021(A)

- CO1- Analyze power system operation and stability control.
- CO2- Able to calculate steady state voltages and bus angles given the load and generation using load flow calculation methods i.e. GSLF, NRLF, FDLF
- CO3- Able to analyze hydro-thermal coordination problems
- CO4- Able to carryout contingency analysis and ranking

Electrical Machines – III EX-7021(B)

- CO1 Gets the knowledge of synchronous machine construction, operation and load characteristics.
- CO2 To attain the knowledge of synchronous machine construction, operation
- CO3 This course deals with mainly definition, construction, principle of operation, types, efficiency, regulation of the synchronous generator -different types of tests to study the characteristics and regulation of synchronous generators, construction, principle of operation of synchronous motor
- CO4 knowledge of construction and operation of single phase induction motors.

Power System Planning & Reliability- EX-7021(C)

- CO1- To list the objectives of load forecasting and to apply the AI technique for load forecasting
- CO2- To determine the reliability of ISO and interconnected generation systems.
- CO3- To analyze the transmission system reliability
- CO4- To explain the expansion planning and capacitor placement problem in transmission system and radial distributions system
- CO5- To design the primary and secondary distribution system and to explain distribution system protective scheme and its coordination.

Open Elective-EX-7031

Power System Protection-EX-7031(A)

- CO1- Knowledge of electrical protection importance's and role in electrical power system networks as well as apparatuses.
- CO2- Know of protection types and operation principle
- CO3- Knowledge to make settings and learning the different types of settings schemes.

Semiconductor Devices -EX-7031(B)

- CO1- Describe the properties of materials and application of semiconductor electronics
- CO2- Apply the knowledge of semiconductors to illustrate the functioning of basic electronic devices.

- CO3- Demonstrate the switching and amplification application of the semiconductor devices
- CO4- Demonstrate the control applications using semiconductor devices
- CO5- Identify the fabrication methods of integrated circuits
- CO6- Classify and describe the semiconductor devices for special applications

Electrical Machine-II-EX-7031(C)

- CO1- Relate spatially displaced armature windings for the generation of various magnetic fields in AC machines
- CO2- Identify different types of armature windings and winding factors for calculating induced EMF
- CO3- Illustrate the electromagnetic laws for the operation of three phase synchronous and asynchronous machines
- CO4- Describe different tests for calculating the performance parameters of three phase induction motors.
- CO5- Explain the fundamental control practices like starting, reversing and speed control strategies for different applications.
- CO6- Describe the different methods for the computation of voltage regulation of an alternator.

Energy Modelling, Economics and Project Management-EX-8011

- CO1 Energy modeling allows the design team to optimize building design and prioritize investment in those areas that will return the best value from the building's energy source.
- CO2 Understand various terminologies related to market like demand, elasticity of demand, factors of production and advertising elasticity.
- CO3 Understand the functions of management – planning, organizing, Directing, Controlling and Communicating.
- CO4 Comprehend monopoly, oligopoly, depreciation and methods to evaluate depreciation.

Electrical Drive-EX-8021

- CO1 Understand starting, speed control and braking and their numerical
- CO2 Define and distinguish heating and cooling of motors, flywheel and solve their numerical
- CO3 Demonstrate PLC's, contactors and relays in the Industry
- CO4 Learn and analysis the working of drives used in traction and the drives used in the Industry

Open Electives-EX-8031

High Voltage Engineering-EX-8031(A)

- CO1- Acquire the knowledge of necessity and methods of testing various apparatus in power system.
- CO2- Knowledge of various circuits for generating high voltages for testing various apparatus and their measurement method.
- CO3- Knowledge of the various reasons of overvoltage in power system and protection methods against them.
- CO4- Knowledge of insulation coordination and design of insulation levels of various parts of power system

Power System Economics- EX-8031(B)

- CO1- Identify and explain the different methods of generation, distribution, control and compensation involved in the operation of power systems.

- CO2- Design the mathematical models of the mechanical and electrical components involved in the operation of power systems and demonstrate the understanding of the open loop and closed loop control practices associated with the voltage and frequency control of single area or interconnected multi area power systems.
- CO3- Specify the equivalent electrical parameters of transmission line to prepare and analyze models to predict the range and ratings of the equipment's to be used, the protection required against line transients and determine the appropriate methods of compensation required for operational stability
- CO4- Solve the problems related to the economic dispatch of power, plant scheduling, unit commitment and formulate strategies to minimize transmission line losses and penalties incurred

Advanced Communication System- EX-8031(C)

- CO1- Analyze the design parameters of a single and multi-carrier communication system.
- CO2- Use mathematical tools to analyze the performance of communication systems.
- CO3- Use probability theory and stochastic processes in communication system application
- CO4- Learn synchronization and adaptive equalization techniques.

Renewable & Non Conventional Energy System-EX-8201

- CO1 Learn fundamentals of solar radiation geometry, application of solar energy
- CO2 Understand selection of sites for wind farm, different types of wind generators
- CO3 Understand the basic of small hydro, ocean & wave energy.
- CO4 Understand basics of small hydro power generation & bio mass

MAJOR PROJECT-EX - 8041

- CO1- Identify, analyze & define the problem.
- CO2- Generate alternative solutions to the problem identified
- CO3- Compare & select feasible solutions from alternatives generated.
- CO4- Design, develop, manufacture & operate equipment/program.
- CO5- Acquire higher-level technical knowledge by studying recent development in Engineering field.
- CO6- Compare machines/devices/apparatus for performance practices.
- CO7- Work effectively in a team.

Bachelors of Engineering (Information Technology)

Programme Outcome (POs)

- PO1- An ability to design hardware and software systems, components and processes to meet desired needs within realistic constraints such as economic, environmental, social, political,

ethical, health and safety, manufacturability and sustainability.

PO2- An ability to work individually or as a team with responsibility to function on multi-disciplinary teams.

PO3- An ability to identify, formulate and solve computing problems, accounting for the interaction between hardware and software.

PO4- An understanding of professional, legal, and ethical issues and responsibilities

PO5- An ability to communicate effectively in speech and in writing including documentation of hardware and software systems.

PO6- An ability to design and conduct experiments, as well as to organize, analyze and interprets data to produce meaningful conclusions and recommendations.

PO7- An ability to understand the impact of engineering solutions on the environment, economy and social issues both in the local and global perspective.

PO8- Demonstrate an ability to acquire new knowledge in the computing discipline and to engage in life-long learning

Program Specific outcomes-

PSO1 An ability to specify, analyze& design a usable computing system that efficiently utilizes system software and hardware to cover current user requirement in a socially and economically acceptable form.

PSO2 An ability to state, design and implement a secure and reliable information communication system by using concepts of computer networks, network security, information theory and parallel algorithm.

PSO3 An ability to state, design and implement knowledge based discovery and machine based learning in computer system by using the various concepts of soft computing, neural networks, image processing, digital system design and artificial intelligence.

PSO4 An ability to analyse and design an efficient information management system which uses the concepts & tools of databases, data mining and information coding and deliver technological solutions for the end users.

Course Outcomes:

Semester – III

Engineering Mathematics-III-IT -3011

CO1 Analyze and solve engineering problems using Laplace Series.

CO2 Analyze and solve engineering problems using Fourier Series.

CO3 To apply partial differential techniques to solve the physical engineering problems.

CO4 Develop problem-solving techniques needed to accurately calculate probabilities.

CO5 Apply problem-solving techniques to solving real-world events.

Data structure & Algorithms-IT -3021

- CO1 Understand the concept of Dynamic memory management, data types, algorithms, Big O notation.
- CO2 Understand basic data structures such as arrays, linked lists, stacks and queues
- CO3 Describe the hash function and concepts of collision and its resolution methods.
- CO4 Solve problem involving graphs, trees and heaps.
- CO5 Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data

Computer Organization and Architecture IT -3031

- CO1 Explain the organization of basic computer, its design and the design of control unit
- CO2 Understand basic data structures such as arrays, linked lists, stacks and queues
- CO3 Describe the operations and language of the register transfer, micro operations and input-output organization.
- CO4 Understand the organization of memory and memory management hardware
- CO5 Elaborate advanced concepts of computer architecture, Parallel Processing, inter processor communication and synchronization

Operating System IT -3041

- CO1 Understand the basics of operating systems like kernel, shell, types and views of operating systems
- CO2 Understand basic data structures such as arrays, linked lists, stacks and queue Describe the various CPU scheduling algorithms and remove deadlocks
- CO3 Explain various memory management techniques and concept of thrashing.
- CO4 Use disk management and disk scheduling algorithms for better utilization of external memory.
- CO5 Recognize file system interface, protection and security mechanisms.

Value Education IT -3051

- CO1 To learn the philosophy of Life, personal value, social value, mind cultural value and personal health.
- CO2 To learn professional ethical values, codes of ethics, responsibilities, safety, rights and related global issues.
- CO3 Understand the basics of operating systems like kernel, shell, types and views of operating systems

Software Lab 1 (C++) IT -3061

- CO1 Introduces Object Oriented Programming concepts using the C++ language.
- CO2 Introduces the principles of data abstraction, inheritance and polymorphism
- CO3 Introduces the principles of virtual functions and polymorphism
- CO4 Introduces handling formatted I/O and unformatted I/O
- CO5 Introduces exception handling

Semester – IV**Environmental Engineering IT -4011**

- CO1 Understand key current environmental problems

- CO2 Be able to identify and value the effect of the pollutants on the environment: atmosphere, water and soil
- CO3 Be able to plan strategies to control, reduce and monitor pollution
- CO4 Be able to select the most appropriate technique to purify and/or control the emission of pollutants
- CO5 Be able to apply the basis of an Environmental Management System (EMS) to an industrial activity

Database Management System IT -4021

- CO1 Understand database concepts and structures and query language
- CO2 Understand the E R model and relational model
- CO3 To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.
- CO4 Understand Functional Dependency and Functional Decomposition.
- CO5 Apply various Normalization techniques

Software Engineering IT -4031

- CO1 Students will be able to decompose the given project in various phases of a lifecycle
- CO2 Students will be able to choose appropriate process model depending on the user requirements.
- CO3 Students will be able perform various life cycle activities like Analysis, Design, Implementation, Testing and Maintenance
- CO4 Students will be able to know various processes used in all the phases of the product.
- CO5 Students can apply the knowledge, techniques, and skills in the development of a software product

Analog and Digital Communication IT -4041

- CO1 Explain the differences between analog and digital communication systems and compare their respective advantages and disadvantages
- CO2 Understand signal multiplexing, modulation and demodulation; bandwidth requirements; signal power spectrum requirements for analog and digital communication systems.
- CO3 Apply signal and system analytical tools in both the time and frequency domains; including Fourier transforms, frequency response functions, time duration versus bandwidth, and convolution
- CO4 Understand amplitude, phase and frequency modulation and demodulation and how they compare.
- CO5 Application of the Sampling theorem to analog-to-digital conversion and understand the limitations of practical sampling, quantization and encoding.

Computer Network IT -4051

- CO1 Understand and describe the layered protocol model.
- CO2 Understand and explain Data Communications System and its components
- CO3 Identify the different types of network topologies and protocols
- CO4 Identify the different types of network devices and their functions within a network
- CO5 Understand and building the skills of subnetting and routing mechanisms

Software Lab II (Java) IT -4061

- CO1 Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity
- CO2 Identify classes, objects, members of a class and the relationships among them needed for finding the solution to specific problem
- CO3 Demonstrates how to achieve reusability using inheritance, interfaces and packages and describes faster application development can be achieved.
- CO4 Demonstrate understanding and use of different exception handling mechanisms and concept of multithreading for robust faster and efficient application development
- CO5 Identify and describe common abstract user interface components to design GUI in Java using Applet & AWT along with response to events

Semester –V

Object Oriented Analysis and Design IT -5021

- CO1 Specify, analyze and design the use case driven requirements for a particular system
- CO2 Model the event driven state of object and transform them into implementation specific layouts
- CO3 Identify, Analyze the subsystems, various components and collaborate them interchangeably
- CO4 Understand the difference between writing programs for the software and doing analysis and design

Theory of Computation IT -5031(A)

- CO1 Students will demonstrate knowledge of basic mathematical models of computation and describe how they relate to formal languages
- CO2 Students will understand that there are limitations on what computers can do, and learn examples of unsolvable problems.
- CO3 Students will learn that certain problems do not admit efficient algorithms, and identify such problems
- CO4 will apply knowledge of computing and mathematics appropriate to the discipline.
- CO5 will function effectively as a member of a team in order to accomplish a common goal.

Microprocessor & Interfacing -IT -5031(B)

- CO1 Understand the basic concepts of microcomputer systems.
- CO2 Understand the architecture and software aspects of microprocessor 8086
- CO3 Write Assembly language program in 8086
- CO4 Interface peripherals for 8086
- CO5 Design elementary aspect of microprocessor based system..

Numerical Method and Statistics IT -5031(C)

- CO1 Ability to flowchart and pseudo code logic for problem solving
- CO2 Solve root finding problems using several methods
- CO3 Solve systems of linear algebraic equations using Gauss elimination and LU decomposition.
- CO4 Perform regression and interpolation on datasets.

Internet & Web Technology IT -5041(A)

- CO1 Create web pages using PHP
- CO2 Identify the difference between the HTML PHP and XML documents

- CO3 3Identify the engineering structural design of XML and parse tree
- CO4 4Understand the concept of JAVA SCRIPTS
- CO5 Understand the JSP and Servlet concepts.

E-Commerce and Governance IT -5041(B)

- CO1 the basicconcepts and technologies used in the field of management information systems
- CO2 the processes of developing and implementing information Systems. Be aware of the ethical, social, and sec urity issues of information systems
- CO3 the role of information systems in organizations, the strategic management processes, and the implications for the management
- CO4 how various information systems work together to accomplish the information objectives of an organization

Java Programming IT -5041(C)

- CO1 the knowledge of the structure and model of the Java programming language
- CO2 use the Java programming language for various programming tasks
- CO3 develop software in the Java programming language
- CO4 evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements
- CO5 propose the use of certain technologies by implementing them in the Java programming language to solve the given problem

Lab(Linux) IT -5051

- CO1 Explain the fundamental concepts of open-source operating system Linux
- CO2 Understand the basic set of commands and editors in Linux operating system.
- CO3 Discuss shell programming in Linux operating system

Lab(Python) IT -5061

- CO1 Write, Test and Debug Python rograms
- CO2 Implement Conditionals and Loops for Python programs
- CO3 Use functions and represent Compound data using Lists, Tuples and Dictionaries

Semester –VI

Wireless and Mobile Computing IT -6011

- CO1 Explain the basic concepts of wireless network and wireless generations
- CO2 Demonstrate the different wireless technologies such as CDMA, GSM, GPRS etc
- CO3 Explain the design considerations for deploying the wireless network infrastructure
- CO4 Differentiate and support the security measures, standards. Services and layer wise securityconsiderations

Compiler Design IT -6021

- CO1 Understand the different phases of compiler.
- CO2 Design alexical analyzer for a sample language.
- CO3 Apply differentparsing algorithms to develop the parsers for a given grammar.
- CO4 Understand syntax-directed translation and run-time environment.
- CO5 Learn to implement code optimization techniques and a simple code generator.

Artificial Intelligence IT -6031(B)

- CO1 To acquire the knowledge of basic artificial intelligence
- CO2 To acquire the knowledge of searching strategies of AI
- CO3 To acquire the knowledge of game playing techniques and planning of AI
- CO4 To understand the concept of statistical methods for reasoning and logic
- CO5 To acquire the knowledge of expert systems and other learning methods of AI

Computer Graphics & Multimedia IT -6031(A)

- CO1 Understand the basics of computer graphics, different graphics systems and applications of computer graphics.
- CO2 Render projected objects to naturalize the scene in 2D view and use of illumination models for this
- CO3 To implement various algorithms to scan, convert the basic geometrical primitives, transformations, Area filling, clipping.
- CO4 To define the fundamentals of animation, virtual reality and its related technologies.
- CO5 To design an application with the principles of virtual reality

Embedded System IT -6031(C)

- CO1 Explain the embedded system concepts and architecture of embedded systems
- CO2 Describe the architecture of 8051 microcontroller and write embedded program for 8051 microcontroller
- CO3 Select elements for an embedded systems tool
- CO4 Understand the memory types used in embedded systems
- CO5 Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability

Pattern Recognition IT -6041(A)

- CO1 Have knowledge of some existing applications of wireless sensor actuator networks
- CO2 Learn the various hardware, software platforms that exist for sensor networks
- CO3 Have knowledge of the various protocols for sensor networks
- CO4 Analyze modeling and simulation of sensor networks
- CO5 Understand what research problems sensor networks pose in disciplines such as signal processing, wireless communications and even control systems

Internet of Things IT -6041(B)

- CO1 Understand the vision of IoT from a global context.
- CO2 Understand the application of IoT.
- CO3 Determine the Market perspective of IoT.
- CO4 Use of Devices, Gateways and Data Management in IoT.
- CO5 Building state of the art architecture in IoT.

Cloud Computing- IT -6041(C)

- CO1 Understand the concepts, characteristics, delivery models and benefits of cloud computing
- CO2 Understand the key security and compliance challenges of cloud computing
- CO3 Understand the key technical and organisational challenges
- CO4 Understand the different characteristics of public, private and hybrid cloud deployment models.

Data Analytics Lab IT -6051

- CO1 Understand the basics of software as a product..
- CO2 List the components of Hadoop and Hadoop Eco-System
- CO3 Access and Process Data on Distributed File System
- CO4 Manage Job Execution in Hadoop Environment
- CO5 Develop Big Data Solutions using Hadoop Eco System

Skill Development Lab IT -6061

- CO1 Understand the basics of software as a Product
- CO2 Understand the current requirements of industries
- CO3 Implement the software as a product using different design patterns.
- CO4 Apply the software development techniques in real life applications.

Semester –VII**Soft Computing IT -7011**

- CO1 Understand concept of ANN and explain the XOR problem
- CO2 Use supervised neural networks to classify given inputs
- CO3 Understand unsupervised neural networks for clustering data .
- CO4 Build Fuzzy inference system using concepts of fuzzy logic.

Simulation &Modelling IT -7021(A)

- CO1 Define, describe and apply basic concepts related to modeling, identification and simulation
- CO2 Classify various simulation models and give practical examples for each category.
- CO3 Demonstrate the ability to apply knowledge of probability and statistics for simulation & modeling
- CO4 Construct a model for a given set of data and motivate its validity.

Deep and Reinforcement Learning IT -7021(B)

- CO1 Describe in-depth about theories, models and algorithms in machine learning.
- CO2 Compare and contrast different learning algorithms with parameters.
- CO3 Examine the nature of a problem at hand and find the appropriate learning algorithms and it's parameters that can solve it efficiently enough.
- CO4 Design and implement of deep and reinforcement learning approaches for solving real-life problems.

Data Science IT -7021(C)

- CO1 Ability to identify the characteristics of datasets and compare the trivial data and big data for various applications
- CO2 Ability to select and implement machine learning techniques and computing environment that are suitable for the applications under consideration.
- CO3 Ability to understand and apply scaling up machine learning techniques and associated computing techniques and technologies.
- CO4 Ability to recognize and implement various ways of selecting suitable model parameters for different machine learning techniques..

Cyber Laws and Forensics IT -7031(A)

- CO1 Become aware of various cyber crimes and cyber laws
- CO2 Underline the need of digital forensic and role of digital evidences
- CO3 Understand different types of digital evidences that can be presented to support

investigations

CO4 List the methods to generate legal evidence and supporting investigation reports

CO5 Use various digital forensic tools

Digital Image Processing- IT -7031(B)

CO1 Explain basic concepts of image processing

CO2 Have knowledge of techniques employed for the enhancement of images

CO3 Interpret image segmentation and representation techniques.

Component Based Technology- IT -7031(C)

CO1 Familiarization with Component Based Systems, their Purpose and Scope.

CO2 Analyse Software Engineering Practices related to CBD.

CO3 Apply design Of Software Component Infrastructures

Data Science Lab IT -7041

CO1 Understand the R Programming Language

CO2 Exposure on solving of data science problems.

CO3 Understand the classification and Regression Model.

Case Tools Lab IT -7051

CO1 Understand the R Programming Language

CO2 Exposure on solving of data science problems

CO3 Understand the classification and Regression Model.

Semester –VIII

Information Security IT -8011

CO1 Understand key terms and concepts in information security and Cryptography and evaluate the cyber security needs of an organization.

CO2 Acquire knowledge to secure computer systems, protect personal data, and secure computer networks in an organization

CO3 Design operational and strategic information security strategies and policies.

Engineering Economics & Management IT -8021(A)

CO1 To impart knowledge, with respect to concepts, principles and practical applications of Economics, which govern the functioning of a firm/organization under different market conditions.

CO2 To help the students to understand the fundamental concepts and principles of management; the basic roles, skills, functions of management, various organizational structures and basic knowledge of marketing.

Machine Learning IT -8021(B)

CO1 Recognize the characteristics of machine learning strategies.

CO2 Apply various supervised learning methods to appropriate problems.

CO3 Identify and integrate more than one technique to enhance the performance of learning.

CO4 Create probabilistic and unsupervised learning models for handling unknown pattern.

Robotics IT -8021(C)

CO1 Understand robot mechanism

CO2 Perform kinematic and dynamic analyses with simulation

CO3 Design control laws for a robot

CO4 Integrate mechanical and electrical hardware for a real prototype of robotic device

Parallel Computing IT -8031(A)

- CO1 To develop an understanding of various basic concepts associated with parallel computing environments
- CO2 Understand, appreciate and apply parallel and distributed algorithms in problem solving
- CO3 Acquire skills to measure the performance of parallel and distributed programs
- CO4 Design parallel programs to enhance machine performance in parallel hardware environment
- CO5 Design and implement parallel programs in modern environments such as CUDA, OpenMP, etc

Game Theory with Engineering Applications IT -8031(B)

- CO1 formulate mathematically multi-agent decision-making problems arising in engineering applications as games
- CO2 analyze using mathematical theory the equilibria of the games and compute them using optimization theory, analyze mathematically the outcome of the game.
- CO3 Understand selected models and concepts of game theory
- CO4 Produce simple economic models with basic game theory

Block Chain Technology IT -8031(C)

- CO1 Understand block chain technology
- CO2 Acquire knowledge of cryptocurrencies
- CO3 Develop block chain based solutions and write smart contract using Hyperledger Fabric and Ethereum frameworks

Bachelors of Engineering (Mechanical Engineering)

Programme Outcome (POs)

- PO1- UG student has the proficiency in area of Mechanical Engineering and has the ability to distinguish, estimate, scrutinize and creations of solutions for various Mechanical Engineering problems.
- PO2- UG student has the ability to apply the fundamental knowledge to generate solution of complex engineering problems. The student has the capability to apply the basic research principles to conduct research in wider spectrum of theoretical and practical aspect of the problem.
- PO3- UG student has the ability to conceptualize the problem and present a broad spectrum of solutions. The student has the ability to arrive at a feasible and optimal solution after considering public at large and environmental effects.
- PO4 UG student has the ability to apply the simulation and optimization techniques to Industrial problems. Student has the ability to conduct experiments, logically analyse and interpret the experimental or theoretical outcomes. He has ability to develop the scientific theory or technical knowledge in the domain of the problem
- PO5- UG student has the ability to create or apply the modern IT tools and techniques to find out fundamental solution of the problems. Student has the ability to model and simulate the Mechanical Engineering problems with the help of advanced tools and techniques.
- PO6- Student has the attitude to think innovatively for developing solutions to the societal problems and develop cost effective solutions to the society at large.

- PO7- UG student has the ability to communicate effectively with society at large, related to complex engineering activities. Student has the ability to write effective reports and design documentation by adhering to suitable standards, make effectual presentation and give and receive clear instructions.

Program Specific outcomes-

- PSO1- Our graduate engineers will apply all the basic principles of mechanical engineering required in both private and public sector organizations. They can contribute to all national level research projects viz DST, SERB, CSIR, DRDO etc.
- PSO2- We produce graduate engineers specialized in Thermal, Manufacturing and Design
- PSO3- Our students are well equipped with industrial management skills, and interdisciplinary technologies
- PSO4 Apply the fundamentals of mathematics, science and engineering knowledge to identify, formulate, design, investigate and solve complex engineering problems of machines & mechanisms, kinematics and dynamics, mechanical components & systems and realize the dream of India to establish world class manufacturing facility having computer-based design, analysis, simulation and fabrication with best quality practices.
- PSO5 Apply appropriate techniques and modern engineering hardware and software tools in the design and integration of mechanical system, to engage in lifelong learning for the advancement of technology and its adaptation in multi-disciplinary environments.
- PSO6 Implementation of professional engineering solutions for the betterment of society keeping the environmental context in mind, be aware of professional ethics and be able to communicate effectively.

Course Outcomes

SEMESTER – III

Environmental Engineering-ME-3011

- CO1- Convey a clear idea of the interdisciplinary nature of environmental and health risk assessment.
- CO2- The ability to apply the fundamental knowledge of science and engineering to assess environmental and health risk.
- CO3- Ability to understand environmental laws and regulations to develop guidelines, procedures and processes for health and safety issues.
- CO4- Understand what are professional ethics and how do ethics affect the outcomes of environmental laws and regulations

Strength of Materials-ME-3021

- CO1- Students will be able to predict mechanical behavior of the member by determining the stresses, strains and deflections produced by the loads up to the elastic limit.
- CO2- Students will be able to solve the stresses in determinate and indeterminate, homogeneous and composite bars under concentrated loads, self weight and thermal loads.
- CO3- Students will be proficient to construct Shear Force and Bending Moment diagrams for statically determinate beam due to concentrated load, uniformly distributed load, uniformly varying load and couple.

CO4- Students will be able to determine bending and shear stresses in machine elements.

Manufacturing Processes-ME-3031

CO1- select appropriate Manufacturing Processing to manufacture any component.

CO2- Interpret foundry practices like pattern making, mold making, Core making and Inspection of defects.

CO3- Differentiate various metal forming processes such as Hot and Cold Working, Rolling, Forging, Extrusion and Drawing Processes.

CO4- Classify different plastic molding processes, Extrusion of Plastic and Thermoforming.

Thermodynamics-ME-3041

CO1- Students will be able to explain the basic principles and applications of the thermodynamics to the various real life systems.

CO2- Students will be able to describe fundamental laws of thermodynamics.

CO3- Students will be able to apply the concepts such as Entropy, Energy Balance also the calculations of heat, work and other important thermodynamic properties for various ideal gas processes.

CO4- Students will be able to estimate performance of various thermodynamic gas power cycles and gas refrigeration cycle and availability in each case.

Value Education-ME-3051

CO1- Students will understand the importance of value based living.

CO2- Students will gain deeper understanding about the purpose of their life.

CO3- Students will understand and start applying the essential steps to become good leaders

CO4- Students will emerge as responsible citizens with clear conviction to practice values and ethics in life

Computer Aided Design- ME-3061

CO1- Ability to create fully constrained solid models that can be quickly modified using standard software tools.

CO2- Ability to use, identify and explain standard features in solid modeling including protrusions, revolutions, cutouts, and patterns

CO3- Ability to use standard software tools to create engineering drawings, or other documents, to fully describe the geometries and dimensions of parts, as well as to document assemblies according to standard practice

CO4- Ability to use standard software tools to create part assemblies and check for clearances.

Semester – IV

Engineering Mathematics-III-ME- 4011

CO1- Understand and apply knowledge of complex variables and integral for real life and engineering problems.

CO2- Able to apply the knowledge of various numerical methods and interpolation

CO3- Identify, formulate and solve engineering problems.

CO4- Understand and apply the concept of probability distribution to engineering problems

Kinematics and Theory Of Machines-ME- 4021

CO1- Kinematics& Theory of Machines I includes study of velocity, acceleration and force

analysis of different mechanisms, power transmitting elements.

- CO2- The students will be conversant with commonly used mechanism for industrial application.
- CO3- The students will get competency in drawing velocity and acceleration diagram for simple and complex mechanism
- CO4- Students will get analytical competency in solving kinematic problems using complex algebra method

Power Plant Engineering-ME-4031

- CO1- Students will be able to understand basic knowledge of Different types of Power Plants, site selection criteria of each one of them and understanding of Power Plant Economics, Energy Storage including compressed air energy and pumped hydro etc.
- CO2- Students will be able to select the suitability of site for a power plant and will be able to calculate load factor, capacity factor, average load and peak load on a power plant. Students will also be able to propose ash handling, coal handling method in a the thermal power plant.
- CO3- Students will be able to understand the working of Diesel & Gas Turbine Power plant.
- CO4- Students will be able to design the Power Plant Instrumentation and understand Environmental Impact

Material Science & Metallurgy-ME-4041

- CO1- Be able to apply core concepts in Materials Science and Metallurgy to solve engineering problems.
- CO2- Interpret about material fundamental and material processing.
- CO3- Figure out the different mechanical properties of material by studying different destructive and non- destructive testing.
- CO4- Acknowledge the importance of surface modification and study the different surface modification methods.

Machine Drawing-Me -4051

- CO1- Able to understand product symbols, weld symbols, pipe joints
- CO2- Understand orthographic projections of machine elements
- CO3- Understand isometric projections of machine elements
- CO4- Understand detailed assembly drawings of Plummer block, Tailstock, Welded joints, tool head of shaper.

Java Programming-ME -4061

- CO1- The student should be able to use an integrated development environment to write, compile, run, and test simple object-oriented Java programs.
- CO2 Read and make elementary modifications to Java programs that solve real-world problems.

Semester-V

Dynamics of Machinery-ME-5011

- CO1- The students will get competency in graphical and analytical method for solving problems in static and dynamic force analysis.
- CO2- The students will get competency in conducting laboratory experiments for finding moment of inertia of rigid bodies, verification of displacement relation for Hooke's joints,

to measure power transmitted and absorbed by dynamometer and brakes respectively.

- CO3- The students will be able to understand working principles of various types of brakes, clutches and dynamometers.
- CO4- This subject will help students to apply knowledge in the field of automobile, aerospace and Novel industries, where mechanisms and moving members force and kinematic analysis and Power transmitting elements play vital role.

Fluid Mechanics-ME- 5021

- CO 1- Use of various properties in solving the problems in fluids.
- CO 2- Use of Bernoulli's equation for solutions in fluids.
- CO 3- Determination of forces drags and lifts on immersed bodies.
- CO 4- Analyze the performance of hydraulic machines.
- CO 5- Analyze the performance of pumps.

Artificial Intelligence & Neural Network- ME- 5031(A)

- CO1. Learn about importance of AI techniques. Adoption of Artificial Intelligence (AI) technologies is widely expanding in our society.
- CO2. Know the exact application of AI Techniques. Such applications use AI techniques to interpret information from a wide variety of sources and use it to enable intelligent, goal-directed behavior.
- CO3. Understand the working of Modern AI based systems.
- CO4. Know about sensors used in AI based systems.

Mechatronics EngineeringME- 5041(A)

- CO 1 Identification of key elements of mechatronics system and its representation in terms of block diagram,
- CO 2 Understanding the concept of signal processing and use of interfacing systems such as ADC, DAC, digital
- CO3 Understand principles of sensors, its characteristics, interfacing with DAQ microcontrollerInterfacing of Sensors, Actuators using appropriate DAQ micro-controller
- CO 4 Outline appropriate sensors and actuators for an engineering application.

Semester –VI

Internal Combustion Engine-ME - 6011

- CO1- To teach students the operating characteristics and thermodynamic analysis of common internal combustion engine cycles
- CO2- To teach students to analyze the combustion process of common fuels.
- CO3- To make students aware of the roles of fluid flow and heat transfer in engine operation.
- CO4- To teach students the environmental, social, and technological issues related to the future widespread use of internal combustion engines

Machine Component Design-ME - 6021

- CO1- Helping the student in machine design their technical ideas
- CO2- Creating knowledge about the various practices with regard to the shaft, bearing and development of views
- CO3- Understanding the importance of the linking functional and visualization aspects in the

preparation of the part object

- CO4- Interpretation of machine Component that in turn help the students in the preparation of the production of machine Component.

Gas Dynamics- ME – 6031(A)

- CO1- Explain basic concepts of gas dynamics and describe the basic fundamental equations of one dimensional flow of compressible fluid and isentropic flow of an ideal gas.
- CO2- Analyze the steady one-dimensional is entropic flow; frictional flow and isothermal flow and express the concepts of steady one dimensional flow with heat transfer
- CO3- Discuss the effect of heat transfer on flow parameters.
- CO4- Describe the jet propulsion engines

Mechanical Measurement &Control- ME – 6041(A)

- CO1- Inspection of engineering parts with various precision instruments.
- CO2- Principles of measuring instruments and gauges and their uses.
- CO3- Understand the importance of material properties in designing the mechanical systems and identify the material for specific applications
- CO4- Understand the changes in phases of material with respect to time and temperature

Advance Workshop Practice-ME - 6061

- CO1 Student will be able to make various joints in the given object with the available work material
- CO2- Student will be able to know how much time a joint will take for the assessment of time

Minor Project-I-ME-6081

- CO1- Enable the Students to undertake short research projects in a team under the direction of members of the faculty
- CO2- To impart skills in preparing detailed report describing the project and results.
- CO3- To enable the students to undertake fabrication work of new experimental set up/devices or develop software packages
- CO4- effectively communicate by making an oral presentation before an evaluation committee

Semester – VII

Heat and Mass Transfer- ME - 7011

- CO1- Students will be able to comprehend the different mode of heat transfer
- CO2- Students are expected to formulate basic equations and Laws for heat transfer Problems
- CO3- To understand the knowledge of mass transfer by applying principles of diffusion, mass transfer coefficients and interphase mass transfer.
- CO4- Applications of heat transfer principles to design and calculate performance of thermal systems related to one dimensional, steady state and transient state for conduction and convection heat transfer

Mechanical System Design- ME – 7021(B)

- CO1- Apply different types power transmission drive in practical use.
- CO2- Understand different types of slip less drive like gears in different machinery.
- CO3- Undertake different types of internal combustion engines in vehicles and different drives.

CO4- Undertake project work different types of optimization techniques.

MECHANICAL VIBRATION ME-7021(C)

CO1- Ability enhancement in analyzing the machine vibrations in various degree of freedom systems.

CO2- Ability enhancement in determining the various causes of machine vibrations and noise and in applying the perfect compensatory system to control them.

CO3- Ability enhancement in practically using the different vibrations.

CO4- Noise measuring instruments to find the basic causes of machine failures.

RELIABILITY ENGINEERING-ME – 7031(A)

CO1-Students should be able to understand about what is Reliability.

CO2-Students should be able to understand how to allocate Reliability to each component

CO3-Students should be able to understand what is Maintainability.

CO4-Students should be able to understand about strength based reliability.

TRIBOLOGY-ME-7031(B)

CO1 Understand the wear theories, types of wear, mechanism, factors and selection of materials

CO2- To introduce students to the field of teratology.

CO3 To enhance students' awareness of tribological issues in the design of machine components, such as rolling element bearings, journal bearings, thrust bearings, seals, and braking systems.

CO4-Calculate surface topographical parameters of Surface properties & Friction

MATERIAL HANDLING EQUIPMENT-ME – 7021 (C)

CO1- To discuss the material handling equipment& their applications.

CO2- To study the construction, working & maintenance of traction type conveyors

CO3- To discuss the different components of material handling systems.

CO4- To study the mechanism used in material handling equipment in industries.

AUTOMOBILE ENGINEERING-ME – 7021 (A)

CO1- Identify the different parts of the automobile

CO2-Explain the working of various parts like engine, transmission, clutch, brakes

CO3-Understand the environmental implications of automobile emissions

CO4 Develop a strong base for understanding future developments in the automobile industry

MAJOR PROJECT-I-ME - 7061

CO1 Enable the Students to undertake short research projects in a team under the direction of members of the faculty

CO2 To impart skills in preparing detailed report describing the project and results.

CO3 To enable the students to undertake fabrication work of new experimental set up/devices or develop software packages

CO4 To effectively communicate by making an oral presentation before an evaluation committee

MECHANICAL SYSTEM DESIGN LA-ME – 7041(B)

- CO1- Apply knowledge of different types of machine parts and its attachments.
- CO2 Apply applications of computer programming of different linkage and its Factor of Safety.
- CO3- Understand different creep of creep and fatigue principles in machine parts.
- CO4- Understand different types of brakes and clutches in practical applications.

MECHANICAL VIBRATION LAB-ME – 7051

- CO1- Apply knowledge of different types Linkage mechanisms in kinematic chain.
- CO2- Apply applications of Governors and Gyroscope in different machinery.
- CO3- Understand different types of sliding and rolling element bearing.
- CO4- Understand the principle of Critical speed of shaft in single and multi rotor system.

MATERIAL HANDLING EQUIPMENT LAB ME – 7051

- CO1- Apply different types of material handling equipment in industry.
- CO2- Understand different types of material handling conveyers and Host in factory.
- CO3- Apply different types Hooks and cranes for small and heavy load in industry.
- CO4- Apply different types of AGV in different types of industry.

RELIABILITY ENGINEERING LAB -ME – 7051

- CO1- Students should be able to understand Addition of Probabilities.
- CO2- Students should be able to understand Mean Time to Failure (MTTF).
- CO3- Students should be able to understand Hazard Rate as conditional probability.
- CO4-Students should be able to understand Effects and Criticality analysis.

TRIBOLOGY LAB-ME-7051

- CO1- Understand the tribological systems and their characteristic features.
- CO2- To Understand mechanism of wear.
- CO3- To Understand the Lubrication System.
- CO4-To Understand Bearing Design Consideration & Characteristics.

Semester – VIII

REFRIGERATION & AIR CONDITIONIN- ME – 8011

- CO1- Students should be able to understand various refrigeration cycles and evaluate performance using Mollier charts and/ or refrigerant property tables
- CO2- Students should be able to illustrate the fundamental principles and applications of refrigeration and air conditioning system
- CO3- Students should be able to obtain cooling capacity and coefficient of performance by conducting test on vapor compression refrigeration systems
- CO4- Students should be able to present the properties, applications and environmental issues of different refrigerants

I.E. & OPERATION RESEARCH-ME - 8021

- CO1- Students will be proficient to recognize the importance and value of Operations Research and mathematical modeling in solving practical problems in industry by Linear programming problems
- CO2- Planning and controlling of production system and use of modern forecasting and management techniques for different types of industries

- CO3- Students will be able to understand queuing & sequencing models and apply them to real-life problems
- CO4- Students will be capable to understand the mathematical tools that are needed to formulate & solve transportation problems for optimization

ROBOTICS-ME - 8031

- CO1- Apply the basic concepts of Robots
- CO2- Apply and evaluate the concepts of Kinematics of Robotics.
- CO3- Apply the Motions, velocities and dynamic analysis of force.
- CO4- Apply and evaluate Motion planning

SIMULATION & MODELLING ME - 8031

- CO1 The student will be able to appreciate the utility of the tools like ANSYS or FLUENT in solving real time problems and day to day problems
- CO2 Use of these tools for any engineering and real time applications.
- CO3 Acquire knowledge on utilizing these tools for a better project in their curriculum as well as they will be prepared to handle industry problems with confidence when it matters to use these tools in their employment.

SOLAR ENERGY UTILISATION ME - 8031

- CO1 Gain an understanding of the available solar energy and the current solar energy conversion and utilization processes.
- CO2 Have a working knowledge of semiconductor physics, optical systems, photovoltaic engineering, load matching, and storage and grid connections.
- CO3 Be able to comprehend the challenges in sustainable energy processes, perform cost analysis, design photovoltaic systems for different applications meeting residential and industrial needs, predict and test performance,
- CO4 Understand the manufacturing processes involve, environmental challenges that need to be solved, economic aspects, and future potentials.

QUALITY MANAGEMENT & CONTROL ME - 8031

- CO1- To realize the importance of significance of quality
- CO2- Manage quality improvement teams
- CO3- Identify requirements of quality improvement programs

MAJOR PROJECT-ME - 8041

- CO1. Identify, analyze & define the problem.
- CO2. Generate alternative solutions to the problem identified.
- CO3. Compare & select feasible solutions from alternatives generated.
- CO4. Design, develop, manufacture & operate equipment/program.
- CO5. Acquire higher-level technical knowledge by studying recent development in Engineering field.
- CO6. Compare machines/devices/apparatus for performance practices.
- CO7. Work effectively in a team.

Masters of Technology (Production and Industrial Engineering)

Programme Outcome (POs)

- PO1- Graduates will have the ability to apply knowledge of Science, Engineering concepts and Management principles to solve the problems.
- PO2- Problem analysis will have the ability to identify, formulate and solve
- PO3- complex engineering problems using principles of mathematics, natural sciences, and engineering sciences in various design, manufacturing process and industrial management.
- PO4- Design/development of solutions will be able to design a system to meet desired needs within environmental, economic, political, ethical, health and safety, manufacturing and management knowledge and techniques.
- PO5- Conduct investigations of complex problems will be able to demonstrate the knowledge of design and conduct experiment as well as interpret data to find the solutions for industrial problems.
- PO6- Modern tool usage will have the ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.
- PO7- The engineer and society will have the knowledge of recent and advanced techniques related to social, health, and safety, legal and cultural issues in production field.
- PO8- Environment and sustainability Will have the ability to understand and analyze the impact of engineering applications in the global and societal context and need for sustainable development.
- PO9- Ethics Will be able to understand the professional and ethical responsibility in decision making in various engineering fields.
- PO10- Individual and team work: Graduates will be able to exhibit the skills of functioning individually as well as in the multidisciplinary teams.
- PO11- Communication Will be able to communicate effectively in the manufacturing field and at the societal level and make effective presentations with clear decisions to demonstrate the leadership quality.
- PO12- Life-long learning: Will have the ability to recognize the necessity and importance of knowledge and engage in life-long learning process to update with current scenario.

Program Specific Outcomes (PSOs):

- PSO1- Industrial & Production engineering graduates will be able to apply the knowledge acquired in the program about materials and finishing process.

PSO2- Industrial & Production engineering graduates will be able to design product based on Ergonomic Principles.

Course Outcomes:

Semester – I

Advanced Mathematics-MTPIE-1001

- CO1 Understand the basic principles of sets and operations in sets.
- CO2 Apply counting principles to determine probabilities.
- CO3 Demonstrate different traversal methods for trees and graphs.
- CO4 Write model problems in computer science using trees and graphs.

Advance Production Technology-MTPIE-1002

- CO1 To acquire abilities and capabilities in the areas of advanced manufacturing methods.
- CO2 Students will understand the fundamentals of control in automation as they apply to manufacturing.
- CO3 Design of Pneumatic Circuit for manufacturing application.
- CO4 Able to implement concepts of automation in machine tools and plant.

Production Planning & Inventory Control- MTPIE-1003

- CO1 Use and analysis of inventory models for independent and dependent demand (including determining and evaluating the impact of inventory holding, ordering, and setup costs).
- CO2 Application of various Materials Requirement Planning models (including the analysis of various lot-sizing techniques and order policies and the development and evaluation of Master Production Plans and Master Scheduling).
- CO3 Application and evaluation of scheduling and sequencing methodologies.
- CO4 The course enables the students to acquire the ability to design and conduct experiments, analyze and interpret data, develop implementation strategies, shape recommendations so that results will be achieved and findings will be communicated effectively.

Theory of Metal Cutting and Tool forming-MTPIE-1004

- CO1 Overview of Machining Technology.
- CO2 Theory of Chip Formation in Metal Machining.
- CO3 Force Relationships and the Merchant Equation.
- CO4 Power and Energy Relationships in Machining & Cutting Temperature.

Operation Research-MTPIE-1005

- CO1 Students will be able to apply and analyse mathematical optimization functions to various applications
- CO2 Students will be proficient to recognize the importance and value of Operations Research and mathematical modeling in solving practical problems in industry by

Linear programming problems

- CO3 Students will be capable to understand the mathematical tools that are needed to formulate & solve transportation problems for optimization.
- CO4 Students will be able to formulate & analyze a managerial decision problem into a mathematical model using game theory & investment analysis.

Lab-I Advanced Production Process-MTPIE-1006

- CO1 To study the basics of metal machining and mechanics of metal machining.
- CO2 To study the different cutting tool materials and types & geometry of cutting tools.
- CO3 To learn introductory concepts of various advanced machining processes.
- CO4 To learn introductory concepts of roughness during the metal cutting.

Lab-II Operation Research-MTPIE-1007

- CO1 Identify necessity and development of mathematical models for various industries.
- CO2 Describe basic optimization and simulation techniques applied to various industries.
- CO3 Carry out investment analysis and game theory.
- CO4 Determine and analyse the industrial systems under the conditions of certainty, uncertainty and risk.
- CO5 Construct and solve the network models and to understand the concept of reliability concept.

Semester – II

Reliability and Quality Management-MTPIE-2001

- CO1 Attain the basic techniques of quality improvement, fundamental knowledge of statistics and probability.
- CO2 Use control charts to analyze for improving the process quality.
- CO3 Describe different sampling plans.
- CO4 Acquire basic knowledge of total quality management.

Simulation Modeling Process-MTPIE-2002

- CO1 The student will be able to appreciate the utility of the tools like ANSYS or FLUENT in solving real time problems and day to day problems
- CO2 Use of these tools for any engineering and real time applications.
- CO3 Acquire knowledge on utilizing these tools for a better project in their curriculum as well as they will be prepared to handle industry problems with confidence when it matters to use these tools in their employment.

Metrology and Measurement- MTPIE-2003

- CO1- The student shall be measuring the various parameters like length, height, angle, displacement, flatness etc., by using various instruments like vernier calipers, micrometer, dial indicator, etc.
- CO2- The student shall be able to measure the threads, gear tooth profiles and surface roughness using appropriate instruments and analyze the data.
- CO3- The student shall be able to recognize various types of governors and gyroscopes, and improve their performance as per requirement.
- CO4- The student shall be able to determine the balancing forces, inertial forces of rotating and reciprocating components in real life problems.

Method Study&Engineering Economy-MTPIE-2004

- CO1 Evaluate the Economic Theories, Cost Concepts And Pricing Policies.
- CO2 Understand the Market Structures and Integration Concepts.
- CO3 understand the measures of national income, the functions of banks and concepts of globalization.
- CO4 Apply Quality Engineering Tools For Process Control And Improvement.

Machines of Dynamics-MTPIE-2005

- CO1 Students will be able to understand about control of sound.
- CO2 Students will be able to carry out noise analysis of simple systems.
- CO3 Students will be able to model and analyze simple, two degrees of freedom vibration systems.
- CO4 Students will be able to measure the vibration methods.

Lab-I Metrology and Measurement-MTPIE-2006

- CO1 Identify various gauges for measurement.
- CO2 Demonstrate linear and angular measurement using precision instruments.
- CO3 Apply the load cell to measure the force and torque
- CO4 Use thermocouple and vibrometer for taking measurement.
- CO5 Measure pressure and surface roughness
- CO6 Measure length and angles using line-graduated instruments, i. e. vernier callipers, micrometers, bevel protractor, sine bar and surface plates.

Lab-II Dynamics of Machine-MTPIE-2007

- CO1 Recall and study the motion of different mechanisms.
- CO2 Understand the working principle of governor and gyroscope and demonstrate the effect of forces and moments on their motion.
- CO3 Analyze the motion and the dynamical forces acting on cams.
- CO4 Demonstrate the effect of unbalances resulting from rotary motions.

Semester – III**Computer Integrated Manufacturing-MTPIE-3001 (A)**

- CO1 Students will be able to apply knowledge about Computer Aided Quality control and Process Planning Control.
- CO2 Students will be able to Design Flexible manufacturing cell after carrying out Group technology study and finally creating FMS
- CO3 Students will be able to apply knowledge about various methods of communication in CIMS.
- CO4 They will be able to apply data management and its importance for decision making in CIMS environment.

Entrepreneurship and Management Concepts-MTPIE-3001 (B)

- CO1 Understand the process of entrepreneurship and the institutional facilities available to an entrepreneur in India
- CO2 Know the process of starting a new venture and create their business plan.
- CO3 Know about valuation of business and the sources for financing new venture.

- CO4 Gain the knowledge on legal aspects and government policy relating to entrepreneurship.

Principals of Robotics and Applications-MTPIE-3001 (C)

- CO1 Identify a Robot for a specific application.
CO2 Interface various Servo and hardware components with Controller based projects.
CO3 Identify parameters required to be controlled in a Robot.
CO4 Develop small automatic / autotronics applications with the help of Robotics.

Industrial Training (6 month)- MTPIE-3002

- CO1 Recall recognizes, visualize, illustrate, demonstrate, criticize and appraise the aspects of Mechanical engineering systems.
CO2 Interaction among the industrial people.

Seminar-MTPIE-3003

- CO1 Recognize and examine ethical situations that affect engineering.
CO2 Identify and anticipate professional issues as a Production & Industrial engineer.
CO3 Prepare for management of ethical and legal issues that Production & Industrial engineers face as Professionals.
CO4 Understand the need to be knowledgeable of contemporary issues.

Dissertation Part- I (Literature Review/Problem Formulation/ Synopsis-MTPIE-3004

- CO1 To define and limit of the research.
CO2 To place your study in an historical perspective.
CO3 To avoid unnecessary duplication.
CO4 To evaluate promising research methods.

Semester – IV

Dissertation Part- II-MTPIE-4001

- CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Mechanical engineering where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/ M. Tech programme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the credit system.
CO2 Every candidate shall be required to submit the record of dissertation work at the end of fourth semester.

Masters of Technology (Thermal)

Programme Outcome (POs)

- PO1- Graduates will demonstrate sound domain knowledge on wider perspective to become successful professionals.
- PO2- Graduates will demonstrate an ability to identify, formulate and solve thermal engineering problems.
- PO3- Graduates will demonstrate an ability to conceptualise designs of thermal system or component and evaluate them to select optimal feasible solution considering safety, environment and other realistic constraints.
- PO4- Graduates will demonstrate skill of good researcher to work on a problem, starting from scratch, to research into literatures, methodologies, techniques, tools, and conduct experiments and interpret data.
- PO5- Graduates will demonstrate research skills to critically analyse complex thermal engineering problem for synthesizing new and existing information for their solutions.
- PO6- Graduates will demonstrate skills to use modern engineering tools, software and equipment to analyze and solve complex engineering problems.
- PO7- Graduates will exhibit the traits of professional integrity and ethics and demonstrate the responsibility to implement the research outcome for sustainable development of the society.
- PO8- Graduates will be able to communicate effectively to comprehend and write effective reports following engineering standards.
- PO9- Graduates will demonstrate skills of presenting their work unequivocally before scientific community, and give and take clear instructions.
- PO10- Graduate will demonstrate traits of manager in handling engineering projects and related finance, and coordinate workforce towards achieving their goals.
- PO11 Graduates will exhibit the traits of good academician and engage in independent and reflective lifelong learning.
- PO12 Graduates will demonstrate an ability to work on laboratory and multidisciplinary tasks.”

PROGRAMS SPECIFIC OUTCOMES (PSO's)

- PSO1- Able to function in various domains of Mechanical Engineering related with production engineering, thermal engineering and design engineering and do the analysis and design of basic mechanical system using relevant tools and techniques for applying the knowledge of mechanical engineering for the solution of industry problem.
- PSO2- Able to model, simulate, analyze and optimize mechanical systems / processes through application of software's.
- PSO3- Acquainted with the platforms, tools, research and development in mechanical engineering.

Course Outcomes

Semester – I

Advanced Mathematics-METH-1001

- CO1 Understand the basic principles of sets and operations in sets.
- CO2 Apply counting principles to determine probabilities.
- CO3 Demonstrate different traversal methods for trees and graphs.
- CO4 Write model problems in computer science using trees and graphs.

Advanced Thermodynamics -METH-1002

- CO1 To apply the knowledge of mathematics, science and engineering fundamentals to model the energy conversion phenomenon.
- CO2 To identify and formulate power production based on the fundamentals laws of thermal engineering.
- CO3 To investigate the effectiveness of energy conversion process in mechanical power generation for the benefit of mankind.
- CO4 To appreciate concepts learnt in fundamentals laws of thermodynamics from which learning ideas how to sustain in energy crisis and think beyond curriculum in the field of alternative and renewable sources of energy.

Heat & Mass Transfer -METH-1003

- CO1 Students will be able to comprehend the different mode of heat transfer.
- CO2 Students are expected to formulate basic equations and Laws for heat transfer problems.
- CO3 To understand the knowledge of mass transfer by applying principles of diffusion, mass transfer coefficients and inter phase mass transfer.
- CO4 Applications of heat transfer principles to design and calculate performance of thermal systems related to one dimensional, steady state and transient state for conduction and convection heat transfer

Gas Dynamics and Flow Through Turbo machines -METH-1004

- CO1 Give examples of the main applications of turbo machines.
- CO2 Determine the velocity triangles in turbo machinery stages operating at design and off design conditions.
- CO3 Explain the working principle of various types of hydro turbines and know their application range.
- CO4 Explain and understand how the flow varies downstream of a turbo machinery blade row.

Refrigeration & Air Conditioning -METH-1005

- CO1 Students should be able to understand various refrigeration cycles and evaluate performance using Mollier charts and/ or refrigerant property tables
- CO2 Students should be able to illustrate the fundamental principles and applications of refrigeration and air conditioning system
- CO3 Students should be able to obtain cooling capacity and coefficient of performance by conducting test on vapor compression refrigeration systems
- CO4 Students should be able to present the properties, applications and environmental issues of different refrigerants

THERMAL ENGINEERING LAB - I -METH-1006

- CO1: Measure the compressibility of real gases and dryness fraction of steam.
- CO2: Evaluate the performance of variable compression engines, air conditioning systems, heat pipe and refrigeration system.
- CO3: Analyze exhaust gases and test the evacuated tube concentrator.
- CO4: Determine overall heat transfer co-efficient for double pipe heat exchanger with parallel and counter flow.

REFRIGERATION & AIR CONDITIONING LAB - II -METH-1007

- CO1 Understand the principles and applications of refrigeration systems.
- CO2 Evaluate performance of Vapor compression refrigeration system.
- CO3 Apply working principle of VAR/VCR system to solve numerical based on VCR and VAR system.
- CO4 Understand basics of psychometric, air conditioning processes and different air conditioning systems.

Semester – II**INSTRUMENTATION AND CONTROL -METH-2001**

- CO1 Develop advanced skills of technical communication in English.
- CO2 Communicate confidently and competently in English language in all spheres.
- CO3 Develop writing competence- technical report, business letters, job applications etc.
- CO4 Develop reading comprehension skill through non-technical texts.
- CO5 Conduct conversation practice: face to face and via media.

THERMAL POWER PLANT ENGINEERING -METH-2002

- CO1 Describe the power generation scenario, the layout components of thermal power plant and analyze the improved Rankin cycle, Cogeneration cycle.
- CO2 Analyze the steam condensers; recognize environmental impacts of thermal power plant and method to control the same.
- CO3 Recognize the layout, component details of hydroelectric power plant and nuclear power plant.
- CO4 Realize the details of diesel power plant, gas power plant and analyze gas turbine power cycle.

Theory & Design of Heat Exchangers -METH-2003

- CO1 Students will be able to learn and explain various modes of heat and mass transfer operations in food processing industry.
- CO2 Understand the basic laws of heat transfer.
- CO3 Analyze heat exchanger performance by using the method of log mean temperature difference.
- CO4 Evaluate heat transfer coefficients for natural convection. 8. Evaluate heat transfer coefficients for forced convection inside ducts.

Design of Power Generation System -METH-2004

- CO1. Ability to understand the Layout of Various Generating Power Stations.

- CO2. Ability to Design Turbine Layout of Various Generating Stations.
- CO3. Ability to Determine design parameters required for Distribution systems.
- CO4. Ability to understand the features and need of various Compensation Systems.

WIND ENERGY AND ITS UTILIZATION -METH-2005

- CO1 Understand the need of energy conversion and the various methods of energy Storage.
- CO2 Identify Winds energy as alternate form of energy and to know how it can be tapped.
- CO3 Understand the Geothermal &Tidal energy, its mechanism of production and its applications.
- CO4 Illustrate the concepts of Direct Energy Conversion systems & their applications.

THERMAL ENGINEERING LAB - II -METH-2006

- CO1 Recognize typical designs of turbo machines and differentiate from positive displacement machines.
- CO2 Explain the working principles of turbo machines and apply it to various types of machines.
- CO3 Recognize relations between choices made early in the turbo machinery design process and the final components and operability.
- CO4 Recognize and discuss today's and tomorrow's use of turbo machines for enabling a sustainable society.

Semester – III

Theory & Design of Blowers &Compressors &Industrial Steam Turbines -METH-3001

- CO1 To have a concept of discrete probability density functions and probability distributions like Binomial Distribution and Poisson distribution.
- CO2 To have a concept of continuous probability density functions and probability distributions like Normal, Gamma and Exponential distribution.
- CO3 To understand the concept of Laplace and inverse Laplace transforms and apply them to solve ordinary differential equations.
- CO4 To use the iteration and interpolation methods to solve engineering problems.

Advance Vibration System-METH-3001

- CO1 Understand free and forced vibrations of single degree freedom systems Turbo Machinery.
- CO2 Explain Inertia force and inertia torque in reciprocating engine & Equivalent dynamical system
- CO3 Analyze balancing problems in rotating and reciprocating machinery.
- CO4 Characterize and design flywheels.
- CO5 Analyze and design centrifugal governors.

General Computers Aided Design -METH-3002

- CO1 Understand the concepts of CAD and CAD tools.
- CO2 Design and create Part Models and Assembly Models
- CO3 Understand concepts 3D Modelling: Concepts, Wireframe, Surface, and Solid Modeling
- CO4 Create Engineering Drawings by using Part and Assembly models

CONCURRENT ENGINEERING METH-3002

- CO1 The graduates shall have the ability to understand the importance of product design in leveraging both manufacturing cost and product lifecycle cost.
- CO2 The graduates shall have the ability to plan and implement a product development program.
- CO3 The graduates shall have the ability to participate in multi-discipline Integrated Product Development teams.
- CO4 The graduates shall have the ability to design and conduct experiments to ensure that the product design is robust and compatible with the capability of the manufacturing process.

Semester IV

PROJECT DISSERTATION – II METH-4001

- CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Computer Science where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/M.Techprogramme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the creditsystem.
- CO2 Every candidate shall be required to submit the record of dissertation work at the end of fourth semester.

Masters of Technology (Electrical Power System)

Programme Outcome (POs)

- PO1- Apply the knowledge of science and mathematics in designing, analyzing and using power converters for various industrial and domestic applications.
- PO2- Design the modern electric machines, drives, power converters, and control circuits

for specific application.

- PO3- Use modern tools, professional software platforms, embedded systems for the diversified applications.
- PO4- Solve the problems which need critical and independent thinking to show reflective learning.
- PO5- Explore ideas for inculcating research skills and undertake independent research in cutting edge technologies
- PO6- Visualize the larger picture and correlate the domain knowledge with the global industrial problems
- PO7- Understanding of the impact of electronics & communications in an economic, social and environment context.
- PO8- Understanding of intellectual property rights and overall professional & ethical responsibility.
- PO9- Ability to communicate effectively with a wide range of audience.
- PO10- Ability to learn independently and engage in life-long learning.

Program Specific Outcomes (PSO's):

- PSO1 Apply engineering fundamental knowledge to identify, formulate, design and investigate complex engineering problems of electric circuits, power electronics, electrical machines and power systems and to succeed in competitive exams.
(Engineering Knowledge and Analysis)
- PSO2 Apply appropriate techniques and modern engineering hardware and software tools in power systems and power electronics to meet desired needs within realistic constraints such as economical, environmental, social, political, ethical, health and safety, manufacturability and sustainability. (System Design & Professionalism)
- PSO3 Function effectively as an individual or a leader in a team to manage different projects in multidisciplinary environment and appreciate the need for, and an ability to engage in life-long learning

Course Outcomes:

Semester – I

Advanced Mathematics-MEPS-1001

- CO1 Find Laplace Transform of periodic functions, Dirac delta function and solve differential equations, integral equations and integro- differential equations and can analyze the solution.
- CO2 Analyse solutions of partial differential equations and solve the initial value and

boundary value problems. Extremis the functional and also solve Isoperimetric problems.

- CO3 Understand Principle of operation and characteristics, biasing arrangements of Field effect transistors and MOSFETs and types of oscillator.
- CO4 Evaluate conversion of numbers from one code to other code, logic gates and truth table of digital circuits.

Advance power system protection relays-MEPS -1002

- CO1 Ability to design and analyze the real time electrical transmission system with respect to various electrical parameters considering environmental and economic obligations.
- CO2 Develop the ability to implement the appropriate safety equipments for design of electrical power system with enhancing the efficiency of the transmission and distribution system with environment friendly technology.
- CO3 Ability to implement the knowledge of basic mathematical.
- CO4 Judge the suitability of installing overhead and underground power transmission strategies considering electrical, mechanical, environmental, performance, safety and economic constraints

Advance Course in Electrical Machines- MEPS—1003

- CO1 Students are able to understand modeling of power system .
- CO2 Able to know the various types of DC drive.
- CO3 Able to analyze and implementation of power electronics to different drives
- CO4 Able to analyze operation and control of induction motor and synchronous motor drives

PowerElectronics Applications in Power System- MEPS—1004

- CO1 Familiarize the power semiconductor switching devices for power conversion .
- CO2 Understand the principle of working of resonant converter and resonant switch converter.
- CO3 Understand different modulation techniques .
- CO4 Understand power electronic applications in FACTS.

Power System Dynamics Analysis and Control-MEPS-1005

- CO1 Proficiency in voltage & frequency control of modern power system
- CO2 Understand power system security concepts and study the methods to rank the contingencies.
- CO3 Capability to contrive Load Dispatch functions.
- CO4 Understand how to analyze various types of faults in power system

Semester – II

Reactive Power Control & Facts- MEPS—2001

- CO1 Analyze the power system data for load-flow and fault studies.
- CO2 An ability to apply knowledge of FACTS Controllers.
- CO3 A knowledge of recent trend in FACTS controllers and application of FACTS controllers.
- CO4 An ability to design a Compensators within realistic constraints.

Restructured Power Systems-MEPS—2002

- CO1 To provide in-depth understanding of operation of deregulated electricity market systems.

- CO2 To examine topical issues in electricity markets and how these are handled world-wide in various markets.
- CO3 To train the students to analyze various types of electricity market operational and control issues under congestion management.
- CO4 To learn different pricing mechanism and power trading in restructured power system.

Power System Transients- MEPS--2003

- CO To understand the concepts of switching transients and damping, transients in DC and three phase circuits
- CO2 To learn the origin and nature and type of power system transients
- CO3 To learn about the travelling surges, lighting phenomena and fast transients in transmission lines
- CO4 To understand the insulation coordination procedures for high voltages systems

Power Quality and Conditioning MEPS-2004

- CO1 Reliably identify the sources of various power quality problems.
- CO2 Explain about causes of harmonic and its distortion effect.
- CO3 Educate the harmful effects of poor power quality and harmonics.
- CO4 Decide the compensators and filters to keep the power quality indices within the standards.

Energy Conservation and Management -MEPS-2005

- CO1 Acquaintance with conservation of energy and its management, energy planning, and energy economics.
- CO2 Know-How of energy efficient machinery systems, energy losses and their management.
- CO3 Competency in Energy analysis techniques and methods & Energy conservation planning and practices.
- CO4 Know-How of Energy forecasting, Energy economics, Energy pricing and incentives for energy conservation.

Semester – III

Power System Instrumentation MEPS--3001 (A)

- CO1 To understand the concepts of Power System Instrumentation.
- CO2 To enable design an selection of Power System Instrumentation sub-systems.

DSP & its Application MTEPS-3001 (B)

- CO1 Thorough understanding of frequency domain analysis of discrete time signals.
- CO2 Ability to design & analyze DSP systems like FIR and IIR Filter etc
- CO3 Practical implementation issues such as computational complexity, hardware resource limitations as well as cost of DSP systems or DSP Processors.
- CO4 Understanding of spectral analysis of the signals.

Power Controller-MEPS--3001 (C)

- CO1 Competency in function of various power electronics devices

- CO2 Skill of analyzing power electronic devices.
- CO3 Capability in testing & harnessing typical characteristics of power semiconductor devices.
- CO4 Competency in Triggering & Protective mechanism of semiconductor devices.

Special Machines MEPS-3002 (A)

- CO1 To provide knowledge about the fundamentals of electrical machines by using transformation theory based mathematical modelling.
- CO2 To impart knowledge about principle of operation and performance of DC, Synchronous, Induction machines and transformers.
- CO3 Ability to analyze steady state & dynamic operation of induction machine
- CO4 Capability in contriving drive operation as per the industry requirements.

Advanced Electrical Drives-MEPS-3002 (B)

- CO1 Competency in developing Dynamic model of drive system.
- CO2 Fitness' in solving typical drive issues.
- CO3 Ability in control strategy of cycloconverter based Drives.
- CO4 Skill in Transient analysis of drive system.

SEMESTER – IV

Dissertation Part- II- MTEPS-4001

- CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Electrical engineering where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/ M. Tech programme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the credit system.
- CO2 Every candidate shall be required to submit the record of dissertation work at the end.

Masters of Technology (POWER ELECTRONICS)

Programme Outcome (POs)

- PO1 Acquire in-depth knowledge in the domain of power systems
- PO2 Ability to critically analyze various power system components, models and their operation
- PO3 Ability to apply fundamentals and concepts to analyze, formulate and solve complex problems of electrical power systems and its components.
- PO4 Apply advanced concepts of electrical power engineering to analyze, design and develop electrical components, apparatus and systems to put forward scientific findings at national and international levels.
- PO5 Ability to use advanced techniques, skills and modern scientific and engineering tools for professional practice.
- PO6 Ability to use advanced techniques, skills and modern scientific and engineering tools for professional practice.

- PO7 Demonstrate and apply knowledge and understanding of engineering principles for project management.
- PO8 To motivate exploring ideas and to encourage for independent, reflective and lifelong learning. To understand the impact of engineering solutions in a global, economic, environmental and societal context

PROGRAM SPECIFIC OUTCOMES (PSOs)

- PSO1 To apply state-of-the-art knowledge in analysis design and complex problem solving with effective implementation in the multidisciplinary area of Electrical Engineering with due regard to environment and social concerns.
- PSO2 To prepare graduates for continuous self learning to apply technical knowledge and pursue research in advanced areas in the field of Electrical Engineering for successful professional career to serve the society ethically.

Course Outcomes:

Semester – I

ADVANCED MATHAMETICS-MTPE-1001

- CO1 Knowledge gained: • Concept of group action and theorems about group actions. • Structure of permutation groups. • Polynomial rings, EDs, PIDs, & UFDs, and relations among them. • Universality of Polynomial rings
- CO2 Skills gained: • Solving problems using the powerful concept of group action. • Facility in understanding the structure of a problem where the problem involves a permutation group - e.g. nature of the roots of a polynomial equation. • Ability to understand a large class of commutative rings by regarding them as quotients of polynomial rings by suitable ideals.
- CO3 Knowledge gained: Topological spaces • Connectedness, compactness, separation axioms • Continuity • Metric spaces review • Fundamental groups • Covering spaces • Computations
- CO4 Skills gained: Generalization of concepts like continuity • Generalizations of theorems • Distinguishing spaces up to homeomorphisms
- CO5 Competency gained: • Understanding of topological spaces and having a grasp on basic results
- CO6 Knowledge gained: Metric spaces (in particular, the complex plane). • Analytic functions, Cauchy-Riemann differential equations, harmonic functions
- CO7 Competency developed: • Understanding of topological and geometric properties of the complex plane. • Differentiation and integration of functions on $\mathbb{C}$, with applications to problems from real analysis. • Viewing analytic functions as conformal mappings
- CO8 Knowledge gained: • Matrix theory, determinants and their application to systems of linear equations. • Eigenvalues, diagonalization of matrices and reduction of systems of linear equations into simpler systems of easily tractable nature. • Vector theory: subspace, basis, linear independence, inner product spaces etc. • Applications of matrix algebra.

ADVANCED CONTROL SYSTEM-MTPE-1002

- CO1 Linearize the non linear physical systems.

- CO2 Study the non linear system behavior by phase plane and describing function methods
- CO3 Study the stability of linear and nonlinear systems by Lyapunov method.
- CO4 Design IMC with Uncertainty and Disturbances
- CO5 Represent any system in any canonical form.
- CO6 Determine response of system
- CO7 Design Lead, Lag and Lead – lag compensator using frequency domain method or time domain method
- CO8 Design PID compensator

Forced Commutation Circuits-MTPE-1003

- CO1 Acquire knowledge about fundamental concepts and techniques used in power electronics.
- CO2 Ability to analyze various single phase and three phase power converter circuits and understand their applications
- CO3 Foster ability to identify basic requirements for power electronics based design application.
- CO4 To develop skills to build, and troubleshoot power electronics circuits
- CO5 Foster ability to understand the use of power converters in commercial and industrial applications
- CO6 To understand and acquire knowledge about various power semiconductor devices.
- CO7 To prepare the students to analyze and design different power converter circuits.
- CO8 AC Voltage Controllers: Principle of phase control, Principle of integral cycle control, single phase ac voltage controller with R load and RL load

Power Electronics Devices and Phase controlled circuits-MTPE-1004

- CO1 Relate basic semiconductor physics to properties of power devices, and combine circuit mathematics and characteristics of linear and non-linear devices.
- CO2 Describe basic operation and compare performance of various power semiconductor devices, passive components and switching circuits
- CO3 Design and Analyze power converter circuits and learn to select suitable power electronic devices by assessing the requirements of application fields
- CO4 Formulate and analyze a power electronic design at the system level and assess the performance.
- CO5 Identify the critical areas in application levels and derive typical alternative solutions, select suitable power converters to control Electrical Motors and other industry grade apparatus
- CO6 Recognize the role power electronics play in the improvement of energy usage efficiency and the applications of power electronics in emerging areas.

ELECTRICAL DRIV- MTPE-1005

- CO1 Examine various applications in industrial and domestic areas where use of electric drives are essential.
- CO2 Classify types of electric drives systems based on nature of loads, control objectives, performance and reliability
- CO3 Combine concepts of previously learnt courses such as, electrical machines, Control and power electronics to cater to the need of automations in industries
- CO4 Select most suitable type and specification of motor drive combination for efficient conversion and control of electric power

- CO5 Design and justify new control and power conversion schemes for implementing alternative solutions considering the critical and contemporary issues.
- CO6 Identify the critical areas in application levels, and derive typical solutions
- CO7 Effect of starting on Power supply, motor and load.Methods of stating of electric motors.Acceleration time Energy relation during stating, methods to reduce the Energy loss during starting.
- CO8 Types of braking, braking of DC motor, Induction motor and Synchronous motor, Energy loss during braking,

Semester-II

Power Electronic App. to Power System-MTPE-2001

- CO1 Acquire in-depth knowledge in the domain of power systems. To understand the impact of engineering solutions in a global, economic, environmental and societal context.
- CO2 Ability to critically analyze various power system components, models and their operation
- CO3 Ability to apply fundamentals and concepts to analyze, formulate and solve complex problems of electrical power systems and its components
- CO4 Apply advanced concepts of electrical power engineering to analyze, design and develop electrical components, apparatus and systems to put forward scientific findings at national and international levels
- CO5 Ability to use advanced techniques, skills and modern scientific and engineering tools for professional practice
- CO6 Preparedness to lead a multidisciplinary scientific research team and communicate effectively
- CO7 Demonstrate and apply knowledge and understanding of engineering principles for project management
- CO8 To motivate exploring ideas and to encourage for independent, reflective and lifelong learning

Solid State controllers of Drives-MTPE-2002

- CO1 understand the operation of the converter / chopper fed dc drive and to solve simple problems
- CO2 understand the operation of both classical and modern induction motor drives
- CO3 apply this skills to design the current and speed controllers for a closed loop solid-state DC motor drive
- CO4 understand the concept of AC AND DC drive system
- CO5 discriminates to drive the systems required for special machines
- CO6 To understand the stable steady-state operation and transient dynamics of a motorload system.
- CO7 To study and analyze the operation of the converter / chopper fed dc drive and to solve simple problems
- CO8 To understand the differences between synchronous motor drive and induction motor drive and to learn the basics of permanent magnet synchronous motor drives

Modeling and Simulation of Drives-MTPE-2003

- CO1 Exhibit knowledge of basic sciences and engineering and manufacturing processes.
- CO2 Demonstrate the ability to accomplish the integration of systems using appropriate analytical, computational and application practices and procedures.

- CO3 Demonstrate the ability to apply knowledge of probability and statistics, optimization techniques, simulation modeling, engineering economic analysis and cost control, and other technical sciences and specialties necessary in the field of industrial engineering and management
- CO4 Be able to identify, formulate, solve problems and implement solutions for engineering, managerial and societal requirements
- CO5 Possess skills related to design / re-design and conduct experiments, analyze and interpret data through systems thinking and modeling approaches
- CO6 Exhibit knowledge of values and professional ethics in their areas of work
- CO7 Develop an ability to adapt and continuously learn to pursue successful careers in chosen professional field
- CO8 Manage projects in various sectors of economy with a focus on conceptual, technical and human aspects

Advanced Microprocessor and Application-MTPE-2004

- CO1 Understand the necessity, features and architecture of 8086.
- CO2 Analyse the addressing modes and understand the functions of 8086 instructions
- CO3 Write simple assembly language programs
- CO4 Understand the need and handling of interrupts in 8086 and features of peripheral IC
- CO5 Explain the architecture of generic advanced microprocessor and features of advanced microprocessors
- CO6 Understand the need and features of bus standards.
- CO7 Teacher is expected to observe and record the progress of students' activities; Assessment shall be made based on the following rubrics table
- CO8 Know the features of advanced microprocessors.

Power Quality and Conditioning-MTPE-2005

- CO1 Explain the importance of Power Quality
- CO2 Describe about power quality problems, categories, causes and its effects
- CO3 Interpret the role of power quality standards and charts
- CO4 Demonstrate the various types of linear and nonlinear loads
- CO5 Summarize Power Conditioning devices and Power Quality Monitoring systems.
- CO6 According to load aspect, it is defined as the power supplied for satisfactory performance of all equipment i.e., all sensitive equipment.
- CO7 In IEEE dictionary, power quality is defined as “the concept of powering and grounding sensitive equipment in a manner that is suitable to the operation of that equipment
- CO8 This depends upon the end user. According to end user point of view, it is defined as, any power problem manifested in voltage, current, or frequency deviations that result in failure or misoperation of customer equipment

Semester – III

Micro Controllers and Control -MTPE-3001

- CO1 Explain the internal organization and operation of microprocessors/microcontrollers.
- CO2 Program 8086 Microprocessor, 8051 and PIC Microcontrollers for application specific solution
- CO3 Design microprocessors/microcontrollers-based systems
- CO4 Implement and develop new experiments on microprocessor/microcontroller based

systems.

- CO5 Understand the need and features of bus standards.
- CO6 Explain microcontroller application and basic architecture of PIC,ARM and ATMEGA processors.
- CO7 Know the internal organization, addressing modes and instruction sets of 8085 processor
- CO8 Understand embedded C and assembly language program by using 8051 Instruction sets and addressing modes

EHV AC and DC Transmission-MTPE-3002

- CO1 Develop the knowledge of HVDC transmission and HVDC converters and the applicability and advantage of HVDC transmission over conventional AC transmission.
- CO2 Formulate and solve mathematical problems related to rectifier and inverter control methods and learn about different control schemes as well as starting and stopping of DC links
- CO3 Analyze the different harmonics generated by the converters and their variation with the change in firing angles.
- CO4 Develop harmonic models and use the knowledge of circuit theory to develop filters and assess the requirement and type of protection for the filters.
- CO5 Study and understand the nature of faults happening on both the AC and DC sides of the converters and formulate protection schemes for the same.
- CO6 Review the existing HVDC systems along with MTDC systems and their controls and recognize the need to follow the advancements in both the existing systems and HVDC systems and determine the most economic coexistence of both.
- CO7 Introduction of DC power transmission technology
- CO8 Generation of harmonics by converters, characteristics of harmonics on DC side, characteristics of current harmonics

SEMESTER – IV

Dissertation Part- II- MTPE -4001

- CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Electronics engineering where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/ M. Tech programme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the credit system.
- CO2 Every candidate shall be required to submit the record of dissertation work at the end.

Masters of Technology (Power System)

Programme Outcome (POs)

- PO1- Apply the knowledge of science and mathematics in designing, analyzing and using power converters for various industrial and domestic applications.
- PO2- Design the modern electric machines, drives, power converters, and control circuits for specific application.
- PO3- Use modern tools, professional software platforms, embedded systems for the diversified applications.
- PO4- Solve the problems which need critical and independent thinking to show reflective learning.
- PO5- Explore ideas for inculcating research skills and undertake independent research in cutting edge technologies
- PO6- Visualize the larger picture and correlate the domain knowledge with the global industrial problems
- PO7- Understanding of the impact of electronics & communications in an economic, social and environment context.
- PO8- Understanding of intellectual property rights and overall professional & ethical responsibility.
- PO9- Ability to communicate effectively with a wide range of audience.
- PO10- Ability to learn independently and engage in life-long learning.

Program Specific Outcomes (PSO's):

- PSO1 Apply engineering fundamental knowledge to identify, formulate, design and investigate complex engineering problems of electric circuits, power electronics, electrical machines and power systems and to succeed in competitive exams.
(Engineering Knowledge and Analysis)
- PSO2 Apply appropriate techniques and modern engineering hardware and software tools in

- power systems and power electronics to meet desired needs within realistic constraints such as economical, environmental, social, political, ethical, health and safety, manufacturability and sustainability. (System Design & Professionalism)
- PSO3 Function effectively as an individual or a leader in a team to manage different projects in multidisciplinary environment and appreciate the need for, and an ability to engage in life-long learning

Course Outcomes:

Semester – I

Advanced Mathematics-MEPS-1001

- CO1 Find Laplace Transform of periodic functions, Dirac delta function and solve differential equations, integral equations and integro- differential equations and can analyze the solution.
- CO2 Analyse solutions of partial differential equations and solve the initial value and boundary value problems. Extremis the functional and also solve Isoperimetric problems.
- CO3 Understand Principle of operation and characteristics, biasing arrangements of Field effect transistors and MOSFETs and types of oscillator.
- CO4 Evaluate conversion of numbers from one code to other code, logic gates and truth table of digital circuits.

Advance power system protection relays-MEPS -1002

- CO1 Ability to design and analyze the real time electrical transmission system with respect to various electrical parameters considering environmental and economic obligations.
- CO2 Develop the ability to implement the appropriate safety equipments for design of electrical power system with enhancing the efficiency of the transmission and distribution system with environment friendly technology.
- CO3 Ability to implement the knowledge of basic mathematical.
- CO4 Judge the suitability of installing overhead and underground power transmission strategies considering electrical, mechanical, environmental, performance, safety and economic constraints

Advance Course InElectrical Machines- MEPS--1003

- CO1 Students are able to understand modeling of power system .
- CO2 Able to know the various types of DC drive.
- CO3 Able to analyze and implementation of power electronics to different drives
- CO4 Able to analyze operation and control of induction motor and synchronous motor drives

Power Electronics Applications in Power System- MEPS--1004

- CO1 Familiarize the power semiconductor switching devices for power conversion.
- CO2 Understand the principle of working of resonant converter and resonant switch converter.
- CO3 Understand different modulation techniques.
- CO4 Understand power electronic applications in FACTS.

Power System Dynamics Analysis and Control-MEPS-1005

- CO1 Proficiency in voltage & frequency control of modern power system
- CO2 Understand power system security concepts and study the methods to rank the contingencies.
- CO3 Capability to contrive Load Dispatch functions.
- CO4 Understand how to analyze various types of faults in power system

Semester – II

Reactive Power Control & Facts- MEPS--2001

- CO1 Analyze the power system data for load-flow and fault studies.
- CO2 An ability to apply knowledge of FACTS Controllers.
- CO3 A knowledge of recent trend in FACTS controllers and application of FACTS controllers.
- CO4 An ability to design a Compensators within realistic constraints.

Restructured Power Systems-MEPS--2002

- CO1 To provide in-depth understanding of operation of deregulated electricity market systems.
- CO2 To examine topical issues in electricity markets and how these are handled world-wide in various markets.
- CO3 To train the students to analyze various types of electricity market operational and control issues under congestion management.
- CO4 To learn different pricing mechanism and power trading in restructured power system.

Power System Transients- MEPS--2003

- CO1 To understand the concepts of switching transients and damping, transients in DC and three phase circuits
- CO2 To learn the origin and nature and type of power system transients
- CO3 To learn about the travelling surges, lightning phenomena and fast transients in transmission lines
- CO4 To understand the insulation coordination procedures for high voltages systems

Power Quality and Conditioning MEPS-2004

- CO1 Reliably identify the sources of various power quality problems.
- CO2 Explain about causes of harmonic and its distortion effect.
- CO3 Educate the harmful effects of poor power quality and harmonics.
- CO4 Decide the compensators and filters to keep the power quality indices within the standards.

Energy Conservation and Management -MEPS-2005

- CO1 Acquaintance with conservation of energy and its management, energy planning, and energy economics.
- CO2 Know-How of energy efficient machinery systems, energy losses and their management.
- CO3 Competency in Energy analysis techniques and methods & Energy conservation planning and practices.
- CO4 Know-How of Energy forecasting, Energy economics, Energy pricing and incentives for energy conservation.

Semester – III

Power System Instrumentation MEPS--3001 (A)

- CO1 To understand the concepts of Power System Instrumentation.
- CO2 To enable design an selection of Power System Instrumentation sub-systems.

DSP & its Application MTEPS-3001 (B)

- CO1 Thorough understanding of frequency domain analysis of discrete time signals.
- CO2 Ability to design & analyze DSP systems like FIR and IIR Filter etc
- CO3 Practical implementation issues such as computational complexity, hardware resource limitations as well as cost of DSP systems or DSP Processors.
- CO4 Understanding of spectral analysis of the signals.

Power Controller-MEPS--3001 (C)

- CO1- Competency in function of various power electronics devices
- CO2- Skill of analyzing power electronic devices.
- CO3 Capability in testing & harnessing typical characteristics of power semiconductor devices.
- CO4 Competency in Triggering & Protective mechanism of semiconductor devices.

Special Machines MEPS-3002 (A)

- CO1 To provide knowledge about the fundamentals of electrical machines by using transformation theory based mathematical modelling.
- CO2 To impart knowledge about principle of operation and performance of DC, Synchronous, Induction machines and transformers.
- CO3 Ability to analyze steady state & dynamic operation of induction machine
- CO4 Capability in contriving drive operation as per the industry requirements.

Advanced Electrical Drives-MEPS-3002 (B)

- CO1 Competency in developing Dynamic model of drive system.
- CO2 Fitness' in solving typical drive issues.
- CO3 Ability in control strategy of cycloconverter based Drives.
- CO4 Skill in Transient analysis of drive system.

SEMESTER – IV

Dissertation Part- II- MTEPS-4001

- CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Electrical engineering where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/ M. Tech programme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the credit system.
- CO2 Every candidate shall be required to submit the record of dissertation work at the end.

Masters of Technology (Digital Communication)

Programme Outcome (POs)

- PO1 Research Skills** → The approach in this course has been to provide a strong exposure to fundamentals with full mathematical rigor in Signal Processing, Communications in state of art in wireless, wire line and optical communications. This provides a strong background to engage in developments in these communication systems.
- PO2 Problem Solving** → The course involves mathematical modeling of communication events including noise, devices and systems that are different across various channels and hence is intensively problem oriented.
- PO3 Usage of Modern Tools** → The student is exposed to Numerical and Algorithmic procedures in the theoretical courses with a strong lab component using Matlab/SCILAB Environment.
- PO4** Encoding and Decoding, Modulation and Demodulation, Radio Frequency Conversion, Channel Transmission, and performance extraction.
- PO5 Critical Thinking** → The course involves understanding of the physical issues in communications and its abstraction to mathematical models, followed by engineering approximation leading to a viable algorithm.
- PO6 Ethical Practice and Social Responsibility** → In the preparation of reports and in the process of accessing and understanding journal papers, the student gets to understand the importance of acknowledging the works of prior state of the art and peer reviews. This ethical practice is built at this stage.
- PO7 Communication Skills** → As a part of progress reports on M.Tech Projects the student is expected to develop his skills in written and oral presentation of the work that he has accomplished.
- PO8 Life Long Learning** → Exposure to prerequisite Mathematics a rigorous approach to communication theory will provide him with all the necessary background to pursue a career in any field of communications going forward in his career.
- PO9 Project management and finance** → As a part M.Tech Project the student learns the value of simulation and emulation, thereby understanding that a cost effective software or hardware realization is possible only through simulations
- PO10 Independent and Reflective Learning** → In the individual lab assignments and Project Work, the student is exposed to thought provoking issues in communication system practice that need association of theoretical learning with real issues in a communication environment.

Program Specific Outcomes (PSOs)

- PSO1** Use the knowledge of, Communications, Networks, Electromagnetics and signal processing to simulate algorithms in virtual environments and implement them on embedded platforms.
- PSO2** Critically and systematically integrate knowledge to analyze, estimate solve complex problems and meet the challenges in the communication domain
- PSO3** Apply concepts of Statistics, Linear Algebra and Residue Calculus in Communication, Signal processing and Electromagnetics domain
- PSO4** Solve issues in real world communication sectors, and develop feasible and viable communication systems.
- PSO5** Inculcate effective communication skills, practice effective team work, professional ethics and pursue research.

Course Outcomes

I Semester

Advanced Mathematics - MTDC 1001

- CO1 Perform basic algebraic manipulation with complex numbers
- CO2 Understand the geometric interpretation of complex numbers
- CO3 Know methods of finding the nth roots of complex numbers and the solutions of simple polynomial equations.
- CO4 Develop a working knowledge of concepts and methods related to designing and managing operations and supply chains.
- CO5 Develop a skill set for quality and process improvement

Micro Controller System Design -MTDC 1002

- CO1 Fundamental principles of 8 bit, 16 bit Microprocessor Controller design and layout and its interfacing techniques.
- CO2 Have an overview of single chip microcontrollers. its memory mapping, addressing modes, etc
- CO3 Learn about software development modular approach and Object oriented interfacing and programming.
- CO4 Get Knowledge of Design and application of Micro-Controller in Data acquisition, Embedded controllers, Process control,etc
- CO5 Have hands-on design experience on DSP Processor architecture and sample design using TI – DSP.

DSP Application-MTDC 1003

- CO1 Fundamental principles of Discrete time signals and systems.
- CO2 Understand the need and basics Z Transform – Properties, ROC, Stability, Causality, and Criterion. Inverse Z Transform, Recursive and Non Recursive systems.
- CO3 Become familiar with types of DFT- its Properties, Linear and Circular convolution, Discrete Cosine Transform, Relationship between DFT and DCT.
- CO4 Understand the Utility of FIR and IIR systems, its Basic structure, Bilinear Transformation, Designing of Discrete time IIR filter-Butterworth ,Chebychev , Inverse Chebyshev Elliptic etc.

- CO5 Application of MATLAB for Design of Digital filters. Effect of Finite register length in filter Design.
- CO6 Understand the concept of Discrete time Random signals, energy signals, Discrete time random process, Spectrum Representation of finite energy signals and response of linear systems to random signals.
- CO7 Basic principles of spectrum estimation, estimate of auto con variance, power spectrum, cross con variance and cross spectrum.
- CO8 Learn the Utility of Advance signal processing technique and transforms, introduction to discrete Hilberts Transform, Wavelet Transform, Haar Transform etc.

VLSI Design-MTDC 1004

- CO1 Basic concept of integrated circuits and manufacturing, Design fundamental for digital MOS circuits, Design Abstraction and circuit Validation.
- CO2 Understand the basics of CMOS circuit and Logic Design, CMOS Logic gate design, Basic Physical design and Power and Delay considerations.
- CO3 Understand the Utility of System Design, CMOS Chip Design, standard cells, Programmable gate array, Design Capture, Simulation and Verification
- CO4 Get Knowledge of Subsystem Design, Data Operation, CMOS Sub System Design, Memory and Control Strategies, PLA and ROM Implementation
- CO5 Understand the concept of CAD system and Algorithms, Layout Analysis, Placement and Routing Algorithms, Timing Analysis, Optimization, Logic Synthesis and Simulations.

Data Communication and Computer Network - MTDC 1005

- CO1 Understand basics of synchronous and asynchronous transmission, circuit switching, message switching, packet switching and their comparison, RS-232 C and X.21 standards, modern operation, null model.
- CO2 Learn the concept of various communication protocols like Data link control, point-to-point, etc , various ARQ technique for error control and their comparison and performance analysis, HDLC as a bit oriented link control protocol.Replacement and Game Theory.
- CO3 Get knowledge of Communication Network, Virtual circuit and datagram, routing algorithm, Dijkstra and Bellman ford least cost, algorithm, various routing protocol, congestion control technique, deadlock and its avoidance.
- CO4 Get a comprehensive knowledge of various LAN topologies and medium access control schemes such as contention, polling, token parsing and performance analysis, various IEEE standards for LAN, UBS LANs, FDDI.
- CO5 Get Understanding of WAN packet switching technologies such as ATM and Frame relay. Introduction to TCP / IP protocols.

II Semester

System Programming- MTDC 2001

- CO1 Get the Concepts of Fundamental of programming, steps in problem solving with digitalcomputer algorithm, flow chart and textual representation, primitive actions, control construct like conditional, iteration, conditional repetition, recursion, programming with Pascal of C

- CO2 Get knowledge of Data types, data representation, data structure array-various operations with array, concept of pointers and pointers manipulations, pointers for data structures and functions, static and dynamic allocations, implementations with arrays and pointers, various operations like searching, appending, insertion & deletion in lists, doubly linked list and their implementations.
- CO3 Will understand the concept of Searching and sorting, linear, binary and Hash search, minimum and maximum selection, divide and conquer, sorting, insertion sort, bubble sort, quick sort & heap sort, matrix operations, dynamic programming.
- CO4 Understand Overview of system programs, Assembler, interpreter, compiler, Editor and operating system.

Modeling and Simulation of Computer- MTDC 2002

- CO1 Learn basics of Discrete event system simulation, its applications, Discrete and Homogeneous system, modeling of system and type of models, Various steps in simulation, General concept in discrete event simulation.
- CO2 Get familiar with Practical models in simulation and its concepts, useful statistical models, discrete distributions, continuous distributions, Poisson process and empirical distribution .
- CO3 Get Knowledge of Queuing model and its Characteristics , measures of performance using queuing systems property.
- CO4 Learn the concept of Random number and its generation, Properties of random numbers, distribution of pseudo random no, test for random no., Random variant Distribution, inverse transform technique, Direct transformation for normal distribution, Acceptance and rejection technique.
- CO5 Understand the applications of Modeling, Data Collection, identifying the distribution with data, parameter variation, etc.
- CO6 Get Introduction and validation of simulation models, output analysis for single model, nature of output data, types of simulation with respect to output analysis and types of performance and their estimation.

Network Design Technology - MTDC 2003

- CO1 Understand basic concepts of Layering and Layered models- OSI & TCP/IP LAN Technology, transmission Medium, Topology, Medium Access Control (MAC) Techniques including MAC & LLC sub layers.
- CO2 Learn the concept of LAN system, Ethernet system, Fast Ethernet & Gigabit Ethernet, Token Ring, FDDI Internet working with TCP/IP, Internet Protocol (IP) Suite including IP V4, IP V6 Transport Protocols, TCP and UDP.
- CO3 Get knowledge of IP routing, various interior gateways protocols like RIP, OSPF and exterior gateway protocols like BGP
- CO4 Get a comprehensive knowledge of label Switching and MPLS WAN technology, WAN Vs LAN, Circuit switching mechanism and network design, packet switched networking including routing and traffic control, X.25 ISDN and Broadband ISDN, High Speed network frame relay, frame relay protocols, services and congestion control,
- CO5 ATM: ATM adaptation layer (AAL), ATM traffic and congestion control ATM LAN, ATM LAN emulation and multi protocols over ATM (MPOA)

Optical Network - MTDC 2004

- CO1 Basic concept of optical network, first generation optical network, multiplexing technique, second generation optical network, virtual circuit services and data gram.
- CO2 Understand the basics of Network components like couplers, Isolators, Circulators, Multiplexer, filter, fiber Bragg gratings as ADD/Drop multiplexers, acoustics optical tunable filters, characterization of switches, mechanical, electro-optic, thermo-optic, and SOA switches, switching architecture.
- CO3 Understand the concept of First generation of optical network: SONET, SDH, goals of SONET design, Multiplexing in SONET, elements of SONET/SDH infrastructure, SONET physical layer, FDDI, ATM, IP layered architecture, physical layer, data link layer, network layer, transport layer.
- CO4 Get Knowledge of network topologies for broadcast networks like bus topology, star topology, media access control (MAC) protocols, throughput calculation, synchronization, aloha and slotted ALOHA.
- CO5 Understand the concept of Wavelength routing network, optical layer, wavelength cross connect, wavelength reuse reliability, virtual topology and circuit switching and node design, degree of wavelength conversion, network design and operation traffic models, and performance criteria, static and reconfigurable network, classification of light paths.
- CO6 Learn about Photonic packet switching, optical time domain multiplexing (OTDM), Method of multiplexing and demultiplexing, OTDM network, bit interleaving and packet interleaving, etc.

Mobile and Satellite Communication: - MTDC 205

- CO1 Get Knowledge of wireless and cellular radio communication: The cellular concept, system design fundamentals, frequency reuse, reused distance, cluster size, channel assignment strategies, handoff strategies, co-channel interference and system capacity, trunking and grade of service.
- CO2 Learn about Speech coding for wireless system applications and broadcast systems, coding techniques for audio and voice and popular speech codes. Brief introduction to radio channel characterization, multi-path propagation, co channel interference, exponential power delay profile, propagation effects, scattering, ground reflection, fading, long normal shadowing, coherence bandwidth
- CO3 Learn concepts of Modulation techniques for mobile and satellite communication, their generation and detection, performance of spectral and power efficiency. Physical layer technique, diversity, spread, spectrum, frequency hopping, direct sequence, adaptive equalization, Orthogonal Frequency Division Multiplexing (OFDM)
- CO4 Get Knowledge of MAC Protocols; 802.11 and its variants, ETSI-HILARAN type 1 MAC protocol, multiple access with collision avoidance.
- CO5 Get the concepts of GEO, MEO and LEO satellite systems, Antenna positioning in GEO and Link calculations, wideband CDMA concepts principles.

III Semester

Information Theory and Coding - MTDC 3001(A)

- CO1 Learn basics of uncertainty, information, entropy and its properties, entropy of binary memory less source and its extension to discrete memory less source, coding theorem, data compression, prefix coding, HUFFMAN coding, Lempel-Ziv Coding.
- CO2 Understand Discrete memory less channels, Binary symmetric channel, mutual

information & its properties, channel capacity, channel coding theorem, and its application to BSC, Shannon's theorem on channel capacity, capacity of channel of infinite bandwidth, Bandwidth signal to noise Trade off, Practical communication system in light of Shannon's theorem, Fading Channel.

- CO3 Learn about Group and field of Binary system Galois field and its construction in GF(2) and its basic properties, vector spaces and matrices in GF(2), Linear Block Codes, Systematic codes, and its encoding circuits, syndrome and error detection ,minimum distance, error detecting and correcting capabilities of block code, Decoding circuits, Probability of undetected error for linear block code in BSC ,Hamming code and their applications.
- CO4 Learn basics of Cyclic codes and its basic properties, Generator & parity check matrix of cyclic codes, encoding & decoding circuits, syndrome computation & error detection, cyclic Hamming codes.
- CO5 Get Introduction to BCH codes, its encoding & decoding, error location & correction.
- CO6 Get knowledge about convolution codes, its construction & viterbi algorithm for maximum likelihood decoding.

Advanced Digital Communication - MTDC 3002(A)

- CO1 Learn about digital modulation technique and their spectral characteristics, optimum receivers for signals corrupted by AWGN and their performance for memory less channel, optimum receivers for PCM, regenerative repeaters and link budget analysis.
- CO2 Get Knowledge about Estimation of signal parameters, carrier phase and symbol timings. Signal design band limited channels and their characterization, probability of error in detection PAM with zero ISI, modulation codes for spectrum spacing.
- CO3 Get basics of Optimum receivers for channels with ISI and AWGN, linear equalization and decision feedback equalization, adaptive linear and adaptive decision feedback equalizer.
- CO4 Become familiar with Multi channel and multi carrier systems, spread spectrum signals for digital communication, direct sequence spread spectrum signals and frequency hopped spread spectrum signals and their performances, OFDM.
- CO5 Learn concepts of fading multi path channels, frequency non-selective slowly fading channels, diversity techniques for fading multi path channels, coded waveform for fading channels and their application

Optical Instrumentation and Measurement - MTDC 3002(B)

- CO1 Understand basics of Optical Time Domain Reflector, Optical low Coherence Reflect meter, Optical Spectrum Analyzer, Optical power and energy meter, Monochromator, CCD, Ellipsometer, transducer, etc.
- CO2 Learn about Fiber Optics Component and Devices like Direction Couplers, beam splitters, switches, modulators, connectors, couplers, polarizer, polarization controllers, amplifiers, fiber laser, reflector, wavelength filters, polarizing beam splitter, wavelength division multiplexes, fiber optic isolator etc.
- CO3 Get knowledge of Fiber optic sensors, sensors based on characteristics like intensity, phase, polarization, frequency and wavelength of light wave.
- CO4 Get Knowledge of Fiber optic Measurements and its techniques.
- CO5 Multimode Fiber: Refractive Index Profile, Geometric Measurement, Numerical

- Aperture, Total Attenuation, Scattering Loss and differential mode loss, Non destructive loss Measurement (OTDR), Transmission, Bandwidth and dispersion, Bandwidth of Jointed fiber, Differential Mode Delay (DMD)
- CO6 Understand concept of Single Mode Fiber Attenuation, Refractive Index Profile (RIP), Mode Field Diameter, Equivalent step Index (EXI) Profile, Mode Cut off Wave length and the Single Mode operating regime, Dispersion, Birefringence Measurement, Measurement of the Propagation constant of fiber mode.

SEMESTER – IV

Dissertation Part- II- MTDC -4001

- CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Electronics engineering where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/ M. Tech programme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the credit system.
- CO2 Every candidate shall be required to submit the record of dissertation work at the end.

Masters of Technology (VLSI)

Programme Outcome (POs)

- PO1** Ability to understand social, ethical, global, environmental responsibilities & contemporary issues for the benefit of the society and possess critical thinking and innovative skills.
- PO2:-** The course involves ability to attain, identify and apply knowledge of mathematics, science & engineering in Electronics VLSI design discipline to the solution of various engineering problems and develops ability to design a component or a product applying all the relevant standards and with realistic constraints.
- PO3:-** The course involves ability to develop the design capability among students so that they have the ability to participate in creative, synthetic and integrative activities of the relevant branch of engineering and has clear understanding of professional and ethical responsibility.
- PO4:-** The approach in this course has been to provide a strong exposure to fundamentals with full mathematical rigor in Signal Processing, Communications in state of art in wireless, wire line and optical communications. This provides a strong background to engage in developments in these communication systems.
- PO5:-** Ability to use techniques and modern CAD Tools so as to implement them in engineering practice to develop professional skills that will prepare the students for immediate employment in the relevant branch of engineering in industry and design thinking capability.
- PO6:-** Ability to communicate effectively in multi-disciplinary teams on various issues & engage in lifelong learning and will design and perform experimentations for academic and industrial needs. Will be able to solve the engineering problems for economically and environmentally viable solution.
- PO7:-** Ability to design and conduct experiments, as well as to analyze and interpret data and will use techniques, skills and modern engineering tools necessary for engineering practice.
- PO8:-** Ability to apply mathematics and science in engineering applications. Sense-Making Skills of creating unique insights in what is being seen or observed.
- PO9:-** The course develops computational thinking i.e. ability to translate vast data in to abstract concepts and to understand database reasoning, Problem solving ability-solving social issues and engineering problems.

Program Specific Outcomes (PSOs)

- PSO1** Be technically competent in design, development and implementation of electronics and VLSI design and extends into applications in the different thrust areas.
- PSO2** Possess suitable knowledge for analyzing, modeling, and evaluating the research problems in major thrust areas of VLSI design.
- PSO3** Possess interpersonal skills, team work capabilities, communication skills, leadership and awareness of the social, ethical and legal responsibilities leading to lifelong learning and career development.
- PSO4** Be successfully employed in electronics profession in industry/ research organization and to have entrepreneurial skill.
- PSO5** Acquire competency in areas of VLSI and Embedded Systems, IC Fabrication, Design, Testing, Verification and prototype development focusing on applications.

- PSO6** Integrate multiple sub-systems to develop System on Chip, optimize its performance and excel in industry sectors related to VLSI / Embedded domain.
- PSO7** Apply advanced concepts in Physics of semiconductor devices to design VLSI systems and design ASIC and FPGA based systems using industry standard tools.

Course Outcomes

I Semester

Solid State Device Modeling and Simulation:- MTVD-1001

- CO1 Learn Quantum Mechanical Concepts, Carrier Concentration, Transport Equation, Band gap, Mobility and Resistivity, Carrier Generation and Recombination, Avalanche Process, Noise Sources.
- CO2 Get Knowledge of Injection and Transport Model, Continuity Equation, Diode Small Signal and Large Signal (Charge Control Model), Transistor Models: Eberly - Moll's and Gummel Port Model, Mextram model, SPICE modeling temperature and area effects.
- CO3 Learn about MOS Transistor Current, Threshold Voltage, Temperature Short Channel and Narrow Width Effect, Models for Enhancement, Depletion Type MOSFET, CMOS Models in SPICE.
- CO4 Learn about Specific Bipolar Measurement, Depletion Capacitance, Series Resistances, Early Effect, Gummel Plots, etc.
- CO5 Understand Static and Dynamic Models, Rate Equations, Numerical Technique, Equivalent Circuits, Modeling of LEDs, Laser Diode and Photo-detectors.

CMOS Analog Circuit Design- MTVD-1002

- CO1 Learn about analog and digital signals, analog sampled-data or discrete time signal, circuit design, analysis of a circuit, synthesis of a circuit, differences between integrated and discrete analog circuit design, design process of an analog integrated circuit.
- CO2 Get basic concepts of MOS fabrication processes, PN junction, MOS transistor, passive components, lateral and substrate BJT and latchup.
- CO3 Get Knowledge of Large-signal and small-signal model for the MOS transistor, computer simulation models, simulation of MOS circuits using SPICE
- CO4 Learn basics of MOS switch, MOS diode/ active resistor, current sinks and sources, current mirrors, current and voltage references, bandgap reference, simulation of CMOS sub circuits using SPICE.
- CO5 Know about Common-Source stage (with resistive load, diode connected load, current-source load, triode load, source degeneration), source follower, common-gate stage, cascode stage, folded cascode stage, simulation of CMOS amplifiers using SPICE.
- CO6 Attain Knowledge of Single-ended operation, differential operation, basic differential pair, large-signal and small-signal behavior, common-mode response, differential pair with MOS loads, simulation of differential amplifiers using SPICE.
- CO7 Get the concept of op-amp, two-stage CMOS op-amp, cascode op-amps, 741 op-amp circuit, dc & small-signal analysis of 741 op-amp, simulation of 741 op-amp using SPICE, macro models for op-amps. Frequency response of CS stage, CD stage, CG stage, cascode stage, differential pair, two-stage CMOS opamp, folded cascode op-amp and 741 op-amp.

- CO8 Learn about Noise characteristics in the frequency and time domains, thermal noise, shot noise, flicker noise, popcorn noise, noise models of IC components, representation of noise in circuits, noise in single-stage amplifiers (CS, CD and CG stages), noise in differential pairs, noise bandwidth, noise figure, noise temperature

Digital Signal Processing Structures for VLSI- MTVD-1003

- CO1 **Introduction to Digital Signal Processing**
Linear System Theory- Convolution- Correlation - DFT- FFT- Basic concepts in FIR Filters and IIR Filters - Filter Realizations. Representation of DSP Algorithms - Block diagram-SFG-DFG.
- CO2 **Iteration Bound, Pipelining and Parallel Processing of FIR Filter**
Iteration Bound: Data-Flow Graph Representations- Loop Bound and Iteration Bound- Algorithms for Computing Iteration Bound-LPM Algorithm. Pipelining and Parallel Processing: Pipelining of FIR Digital Filters- Parallel Processing- Pipelining and Parallel Processing for Low Power.
- CO3 **Fast Convolution and Arithmetic Strength Reduction in Filters**
Fast Convolution: Cook-Toom Algorithm- Modified Cook-Toom Algorithm. Design of Fast Convolution, Algorithm by Inspection. Parallel FIR filters-Fast FIR algorithms- Two parallel and three parallel. Parallel architectures for Rank Order filters-Odd Even Merge sort architecture-Rank Order filter architecture-Parallel Rank Order filters-Running Order Merge Order Sorter-Low power Rank Order filter.
- CO4 **Pipelined and Parallel Recursive Filters**
Pipelined and Parallel Recursive Filters : Pipeline Interleaving in Digital Filters- Pipelining in 1st Order IIR, Digital Filters- Pipelining in Higher- Order IIR Filters- Clustered Look ahead and Stable Clustered Look ahead- Parallel Processing for IIR Filters and Problems.
- CO5 **Scaling and Round off Noise**
Scaling and Round off Noise: Scaling and Round off Noise- State Variable Description of Digital Filters, Scaling and Round off Noise Computation.

Low Power VLSI Design- MTVD-104

- CO1 **Introduction to low power VLSI design and analysis**
Introduction to low power VLSI design-Need for low power-CMOS leakage current-static current-Basic principles of low power design-probabilistic power analysis-random logic signal-probability and frequency power analysis techniques-signal entropy.
- CO2 **Circuit level and logic level design techniques**
Circuit - transistor and gate sizing - pin ordering - network restructuring and reorganization – adjustable threshold voltages - logic-signal gating - logic encoding. Pre-computation logic.
- CO3 **Special low power VLSI design techniques – I**
Power reduction in clock networks - CMOS floating node - low power bus - delay balancing - SRAM.
- CO4 **Special low power VLSI design techniques – II**
Switching activity reduction - parallel voltage reduction - operator reduction -Adiabatic computation – pass transistor logic
- CO5 **Software design and power estimation**

Low power circuit design style - Software power estimation – co-design.

Graph Theory and Optimization Techniques - MTVD-105

CO1 Paths and Circuits

Graphs – Data structures for graphs – Subgraphs – Operations on Graphs Connectivity – Networks and the maximum flow – Minimum cut theorem – Trees – Spanning trees – Rooted trees – Matrix representation of graphs.

CO2 Graphs & Graph Colorings

Eulerian graphs and Hamiltonian graphs – Standard theorems – Planar graphs – Euler's formula Five colour theorem – Coloring of graphs – Chromatic number (vertex and edge) properties and examples – Directed graphs

CO3 Applications

Computer Representation of graphs – Basic graph algorithms – Minimal spanning tree algorithm – Kruskal and Prim's algorithm – Shortest path algorithms – Dijkstra's algorithm – DFS and BFS algorithms.

CO4 Optimization Techniques

Single variable and multivariable optimization – Lagrangian method – Kuhn-Tucker conditions – Random pattern and Random search methods

CO5 Unconstrained Minimization Methods

Classification – Techniques of unconstrained minimization – Scaling of Design variables – Direct search methods – Random jumping method – Grid search method – Univariate method – Hook and Jeeve's method – Newton – Raphson method – Fibonacci method.

II Semester

Testing of VLSI Circuits - MTVD-2001

CO1 Basics of Testing and Fault modeling

Introduction - Need for testing - VLSI Testing Process And Test Equipment - Types of testing - ATE ADVANTEST model T6682-Block Diagram and Specification– Electrical Parametric testing – AC and DC Test Economics - Fault Modeling-Stuck at faults. Calculation of DCR and ECR.

CO2 Test generation for combinational circuits

Faults in Digital circuits-failures and faults. Modeling of faults.Temporary faults. Test generation for combinational logic circuits-combinational ATPG-Boolean Difference Method-D-Algorithm-PODEM-FAN. Algorithm.Testable combinational logic circuit design.Algorithm for true value simulation and for fault simulation.

CO3 Test generation and Testability of sequential circuits

Test generation for sequential circuits. Design of testable sequential circuits-sequential ATPG Implementation and complexity- Time Frame Expansion-simulation method. DSP based testing - Static ADC & DAC testing methods- Testable memory design – Reduced Functional Faults-MARCH and MAT+ algorithm. Analog and Mixed signal tests.

CO4 Memory, Delay fault and IDDQ Testing

Delay test - Path delay test and fault models - Transition faults - delay test methodologies – practical consideration - IDDQ testing - Testing methods - Limitations of IDDQ testing - DFT IDDQ.

CO5 Built-in Self-Test.

DFT - Scan Design - Partial scan design - BIST- TPG for BIST - output response analysis, BIST Architectures- Random Logic BIST - Memory and delay fault BIST - JTAG - System test and core based design.

Digital VLSI Design- MTVD-2002

CO1 Introduction to MOS Device

MOS Transistor-First Glance at the MOS device MOS Transistor under static conditions-threshold voltage, Resistive operation-saturation region -channel length modulation-velocity saturation-Hot carrier effect-drain current V_s voltage charts - sub threshold conduction - equivalent resistance-MOS structure capacitance, Design logic gates using NMOS and PMOS and CMOS devices-Stick Diagram.

CO2 MOS Transistor Device Modeling

Modeling of MOS Transistor using modeling technique-various model CMOS Inverter. Performance of propagation delay sizing inverter consumption-static consumption PSPICE-Introduction – Basic Concepts-LEVEL1-LEVEL2-LEVEL3 comparison. Static CMOS inverter, Evaluating the Robustness of CMOS inverter: Dynamic Behavior-computing the capacitance-for performance-sizing a chain of invertors-Dynamic power

CO3 CMOS combinational logic design

Static CMOS design-complementary CMOS - static properties- complementary CMOS design-Power consumption in CMOS logic gates-dynamic or glitching transitions - Design techniques to reduce switching activity - Radioed logic-DC VSL - pass transistor logic - Differential pass transistor logic - Sizing of level restorer-Sizing in pass transistor-Dynamic CMOS design-Basic principles - Domino logic-optimization of Domino logic-NPCMOS-logic style selection -Designing logic for reduced supply voltages.

CO4 CMOS sequential logic design

Timing metrics for sequential circuit -latches V_s registers -static latches and registers - Bistability principle multiplexer based latches-master slave edge triggered registers-non-ideal clock signals-low voltage static latches-static SR flip flop - Dynamic latches and registers-C2MOS register - Dual edge registers-True single phase clocked registers-pipelining to optimize sequential circuit latch V_s register based pipelines-non, Bistable sequential circuit-Schmitt trigger-mono stable –Astable -sequential circuit - choosing a clocking strategy.

CO5 CMOS subsystem design

Data Path Operations: Addition/Subtraction - Comparators- Zero/One Detectors- Binary Counters- ALUs Multiplication- Shifters- Memory elements- control: Finite-State Machines.

DSP Architectures and Applications - MTVD-2003

CO1 Overview of Digital Signal Processing

Advantages of DSP over analog systems, salient features and characteristics of DSP systems, applications of DSP systems, Introduction to DSP Processors: Common features of DSP processors, numeric representations in DSP processor, data path of a DSP processor, memory structures in DSP processors, VLIW architecture, special addressing modes in DSP processors, pipelining concepts, on-chip peripherals found in DSP processors.

CO2 TMS320C5X Processors

Architecture of TMS320C5X Processors- Assembly Instructions- Addressing Modes- Pipelining and Peripherals- lab exercise.

CO3 TMS320C3X Processors

Architecture of TMS320C3X- Instruction Set- Addressing Modes- Data Formats- Floating Point Operation, Pipelining and Peripherals- lab exercise.

CO4 Black fin Processor

Introduction to Black fin processor- Architecture overview-processor core-addressing modes-instruction sets, Targeted applications- lab exercise.

CO5 SHARC Processor

VLIW Architecture- SHARC- SIMD- MIMD Architectures- Application: Adaptive filters-DSP based biometry receiver-speech processing-position control system for hard disk drive-DSP based power meter.

System on Chip Design - MTVD-2004

CO1 SOC fundamentals

Essential issues of SoC design – A SoC for Digital still camera – multimedia IP development : Image and video codecs.

CO2 SOC software and energy management

SoC embedded software – energy management techniques for SoC design.

CO3 On-chip networking System design and methodology

Design methodology for NOC based systems – Mapping concurrent application onto architectural platforms.

CO4 Hardware and basic infrastructure

Packet switched network for on-chip communication – energy reliability tradeoff for NoC's – clocking strategies – parallel computer as a NoC's region.

CO5 Software and application interfaces

MP-SoC from software to hardware – NoC APIs – multilevel software validation for NoC – Software for network on chip

Fundamentals and Application of Mems - MTVD-2005

CO1 Overview of MEMS and Microsystems

MEMS and Microsystems, Microsystems and microelectronics, Microsystems and miniaturization, Working principle of micro system – Micro sensors, Micro actuators, MEMS with Micro actuators.

CO2 Materials For MEMS

Substrate and Wafer, Silicon as a Substrate Material, Silicon Compound, Silicon Piezo-Resistors, Gallium Arsenide, Quartz, Piezoelectric Crystals, Polymers and Packaging Materials.

CO3 Microsystems Fabrication Process

Fabrication Process – Photolithography, Ion implantation, Oxidation, Chemical vapor deposition (CVD), Physical vapor deposition, Deposition by Epitaxy, Etching. Manufacturing Process – Bulk Micromachining, Surface Micromachining, LIGA Process.

CO4 Microsystems Design, Assembly and Packaging

Micro system Design – Design consideration, process design, Mechanical design,

Mechanical design using MEMS, Mechanical packaging of Microsystems, Microsystems packaging, interfacing in Microsystems packaging, packaging technology, selection of packaging materials, signal mapping and transduction.

CO5 Case Study of MEMS Devices

Case study on strain sensors, Temperature sensors, Pressure sensors, Humidity sensors, Accelerometers, Gyroscopes , RF MEMS Switch, phase shifter, and smart sensors. Case study of MEMS pressure sensor Packaging.

III Semester

ASIC Design - MTVD-3001(A)

- CO1 Types of ASICs – Design flow – CMOS transistors CMOS Design rules – Combinational Logic Cell Sequential logic cell – Data path logic cell – Transistors as Resistors – Transistor Parasitic Capacitance Logical effort – Library cell design – Library architecture.
- CO2 Anti fuse – static RAM – EPROM and EEPROM technology – PREP benchmarks – Actel ACT – Xilinx, LCA – Altera FLEX – Altera Max DC & AC inputs and outputs – Clock & Power inputs – Xilinx I/O blocks.
- CO3 Actel ACT –Xilinx LCA – Xilinx EPLD – Altera MAX 5000 and 7000 – Altera MAX 9000 – Altera FLEX- Design systems – Logic Synthesis – Half gate ASIC –Schematic entry – Low level design language – PLA tools – EDIF- CFI design representation.
- CO4 ASIC Construction: Physical Design- System Partitioning- FPGA Partitioning- Partitioning Methods. Floor planning and Placement: Floor planning- Placement- Physical Design Flow. Routing: Global Routing –Detailed Routing- special Routing. Design checks
- CO5 Planar subset problem(PSP) –single layer global routing single layer detailed routing wire length and bend minimization technique-over the cell(OTC) Routing-multichip modules(MCM)-Programmable logic arrays, Transistor chaining-Weinberger Arrays- Gate Matrix Layout-ID Compaction-2D compaction

Programming in HDL - MTVD-3002(A)

- CO1 **Basic concepts:** Operators, Data types, Number specification, System tasks and compiler directives, Modules and ports, Gate-level Modeling, Dataflow Modeling, Behavioral Modeling-example for each modeling –test bench- lab exercise.
- CO2 **Tasks and Functions**-example-useful modeling techniques-Timing and delays-Switch level modeling-user defined primitives- lab exercise.
- CO3 **Basic Concepts:** Data Objects, Data Types, Operators, Concurrent and Sequential Assignment Statements, Different Styles of Modeling, Simple Examples-test bench- lab exercise.
- CO4 **Procedure and functions** - examples-packages - Generic constants and statements - examples. Component and configuration- lab exercise.
- CO5 **Introduction to synthesis** - Verilog synthesis-modeling tips for Verilog logic synthesis-combinational and sequential logic synthesis using VHDL-VHDL modeling restrictions- lab exercise.

Embedded Wireless Sensor Networks - MTVD-3002(B)

CO1 Introduction

Embedded network systems – representation of signals – signal propagation – sensor principles.

CO2 Communication

Source detection and identification – digital communications – multiple source estimation and multiple access communications.

CO3 Networking

Networking – network position and synchronization services.

CO4 Network management

Energy management – data management – articulation mobility and infrastructure.

CO5 Nodes, data and application

Node architecture – network data integrity – experimental system design.

SEMESTER – IV

Dissertation Part- II- MTDC -4001

CO1 The programme of instruction will consist of advanced subjects of the respective specialization. The complete programme is distributed over four semesters with two semesters per academic year. Course work is offered in the first two semesters (except for PG programme in Electronics engineering where it is extended up to third semester) and the dissertation work will be carried out during third and fourth semesters. Every branch of M.E/ M. Tech programme will have a curriculum and syllabi for the courses recommended by the board of studies and approved by the academic council. The academic programmes of the Institute follow the credit system.

CO2 Every candidate shall be required to submit the record of dissertation work at the end.

Masters of Technology (Computer Science & Engineering)

Programme Outcome (POs)

- PO1- An ability to independently carry out research / investigation and development work to solve practical problems.
- PO2- An ability to write and present a substantial technical report / document.
- PO3- Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program.
- PO4- Apply the knowledge of engineering principles to develop software systems, products and processes thus to solve real world multifaceted problems.
- PO5- Ability to design and conduct experiments, procedures and technical skills necessary for engineering exploration to solve societal problems and environmental contexts for sustainable development.
- PO6- Recognize the need to engage in self-governing and life-long learning by making use of professional and ethical principles.

Program Specific Outcomes

- PSO1- Analyze software products, processes in a systematic way by applying problem solving skills and employable in product oriented Industry.
- PSO2- Ability to take up higher studies, Research & Development and Entrepreneurships in the modern computing environment.
- PSO3- Model computational problems by applying mathematical concepts and design solutions using suitable data structures and algorithmic techniques
- PSO4- Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms.
- PSO5- Develop system solutions involving both hardware and software modules.

Course Outcomes:

Semester –I

Mathematical Foundations of Computer Science- MTCS-1001

- CO1 Knowledge gained: Concept of group action and theorems about group actions. Structure of permutation groups. Polynomial rings, EDs, PIDs, & UFDs, and relations among them. Universality of Polynomial rings
- CO2 Skills gained: Solving problems using the powerful concept of group action. Facility in understanding the structure of a problem where the problem involves a permutation group - e.g. nature of the roots of a polynomial equation. Ability to understand a large class of commutative rings by regarding them as quotients of polynomial rings by suitable ideals.
- CO3 Knowledge gained: Topological spaces Connectedness, compactness, separation axioms Continuity Metric spaces review Fundamental groups Covering spaces Computations
- CO4 Skills gained: Generalization of concepts like continuity Generalizations of theorems Distinguishing spaces up to homeomorphisms
- CO5 Competency gained: Understanding of topological spaces and having a grasp on basic results

Pattern Recognition-MTCS-1002

- CO1 Students will learn the fundamentals of pattern recognition and its relevance to classical

and modern problems.

- CO2 A main objective is to be able to identify where, when and how pattern recognition can be applied.
- CO3 Students will learn the sufficient background necessary to read more advance texts as well as journal articles on the field.
- CO4 Several applications of pattern recognition on classical computer and electrical engineering problems (e.g. word/sentence-based searches, signal analysis, speech and visual processing, engineering system design, medical diagnosis, etc.) will show the student how to use pattern recognition in real settings.
- CO5 The student will also be introduced to more recent applications of pattern recognition, such as cognitive neuroscience and bioinformatics. Students will work on a selected project.

High Performance Computer Architecture-MTCS-1003

- CO1 You will gain basic knowledge required to design an analyze high performance computer systems.
- CO2 You will become aware of parallel architecture in modern intel i3, i5 and ARM based computer systems.
- CO3 You will know how parallelism is achieved using various pipelining techniques in ARM and Intel high performance systems.
- CO4 You will learn how to evaluate and analyze cost and performance of multi processor systems.
- CO5 You will learn various type of interconnection networks used to achieve high performance in modern systems
- CO6 You will learn how various type of memories are used in parallel architecture to achieve data parallelism. 8. You will acquire the background for understanding next-generation GPUs using CUDA.
- CO7 You will learn how to achieve parallelism using pipelining and memory organization in CUDA architecture using GPUs
- CO8 You will increase your knowledge about various parallel computing model in modern systems.

Advance Algorithm Design & Analysis-MTCS-1004

- CO1 Discuss advanced algorithm design techniques
- CO2 Discuss Algorithm Design Techniques, including – greedy algorithms, divide-and-conquer algorithms, Dynamic programming, Randomised Algorithms and Back-tracking.
- CO3 Discuss the fundamentals of Data Compression, including Huffman coding and decoding algorithm.
- CO4 Discuss string and pattern matching techniques – brute-force method, Knuth-Morris-Pratt method
- CO5 Discuss dynamic programming algorithms
- CO6 Understand the fundamentals of data structures
- CO7 Describe advanced data structures such as binomial heaps, fibonacci heaps
- CO8 Study advanced graph problems such as all-pairs shortest paths, min-flow max-flow problem, graph partitioning

Distributed Systems-MTCS-1005

- CO1** Upon Completion of the course, the students will be able to List the principles of distributed systems and describe the problems and challenges associated with these principles.
- CO2** Understand Distributed Computing techniques, Synchronous and Processes. Apply Shared Data access and Files concepts.
- CO3** Design a distributed system that fulfills requirements with regards to key distributed systems properties
- CO4** Understand Distributed File Systems and Distributed Shared Memory.
- CO5** Apply Distributed web-based system. Understand the importance of security in distributed systems

Semester-II

Advanced Computer Networking-MTCS-2001

- CO1 Recognize the technological trends of Computer Networking.
- CO2 Discuss the key technological components of the Network.
- CO3 Evaluate the challenges in building networks and solutions to those

Real Time Systems-MTCS-2002

- CO1 Explain the fundamentals of interaction of OS with a computer and User computation
- CO2 Explain the fundamental concepts of how process are created and controlled with OS
- CO3 Describe the programming logic of modeling Process based on range of OS features.
- CO4 Develop the target system by porting RTOS
- CO5 Compare types and Functionalities in commercial OS.
- CO6 Application development using RTOS

Advance Soft Computing-MTCS-2003

- CO1 Apply soft computing techniques to solve engineering problems.
- CO2 Handle multi-objective optimization problems.
- CO3 Apply advanced AI techniques of swarm intelligence , particle swarm optimization, ant Colony optimization and patients.
- CO4 Apply rough set theory and granular computing to solve process control applications

Distributed And Parallel Databases-MTCS-2004

- CO1 The need for parallelism in data processing.
- CO2 The different types of parallelism
- CO3 The different parallel processing architecture.
- CO4 How to measure the performance of parallel processing.
- CO5 Basic parallel query processing.

Advance Network Security-MTCS-2005

- CO1 At the end of the program, graduates will be able to get insights into various fields of information security with a deep understanding of theoretical aspects of security and related analysis.
- CO2 Graduates should also get a broader understanding of various security systems, protocols, complexities, standards, practical applicability, and their limitations.
- CO3 During the course, students should enhance their inquisitiveness to ever-evolving domain of information security and apply their knowledge to solve problems.
- CO4 With a focused one year research leading to a thesis, students should be able to understand the "art" and "science" of research and should be capable enough to apply this training to newer/other fields and problems.
- CO5 Students should also be able to solve security issues with understanding of system security and cryptographic attributes with a relevance to standards

Semester-III

Fault Tolerant Computing Systems-MTCS-3001

- CO1 Evaluate reliability of systems with permanent and temporary faults
- CO2 Carry out assessment of the relationship between software testing, residual defects and security vulnerabilities.
- CO3 Understand cost-dependability trade-offs and the limits of computer system dependability

Statistical Data Mining-MTCS-3002

- CO1 Describe and discuss the key terminology, concepts tools and techniques used in business statistical analysis
- CO2 Critically evaluate the underlying assumptions of analysis tools
- CO3 Understand and critically discuss the issues surrounding sampling and significance
- CO4 Discuss critically the uses and limitations of statistical analysis
- CO5 Solve a range of problems using the techniques covered
- CO6 Conduct basic statistical analysis of data.